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Client Credentials Check TASK 46-13-00-860-803			213	PTF ALL
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ONBOARD NETWORK SYSTEM - MAINTENANCE PRACTICES

1. General

A. This section gives maintenance tasks that support the Onboard Network System (ONS) and the Mass Storage Device (MSD).

B. This procedure has the tasks that follow:

(1) Onboard Network System Maintenance Laptop Connection.

(2) Onboard Network System Multifunction Display (MFD).

(3) Onboard Network System Airplane Credentials Check.

(4) Onboard Network System 46 NFS OAS AIRPLANE KEYS UMS Installation.

(5) Onboard Network System Server Credentials Check.

(6) Onboard Network System Client Credentials Check.

(7) Onboard Network System Airplane Identification Check.

(8) Onboard Network System Input Monitoring Check.

(9) Onboard Network System Configuration Check.

(10) Onboard Network System LRU Software Parts Installation.

(11) Mass Storage Device Software Parts Check.

(12) Mass Storage Device Software Parts Removal.

(13) Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD).

(14) Download Files using the Software Management Tool (SMT).

(15) Stage Software Parts on a Mass Storage Device via Ground Network or Software Management Tool.

TASK 46-13-00-480-801

2. Maintenance Laptop Connection

A. General

(1) This task gives the steps to connect the universal maintenance device, SPL-11321, to the Onboard Network System (ONS) and to start an ONS maintenance browser session.

(2) You can connect a laptop using an Ethernet cable, or with a wireless connection to the optional Crew Wireless LAN Unit (CWLU).

(a) To connect with an Ethernet cable, do the Connection Procedure with Ethernet Cable below.

(b) To connect wirelessly, do the Connection Procedure with the Optional CWLU below.

NOTE: Wireless access requires installation of the CWLU hardware. If your airplane does not have a CWLU installed, the wireless task is not applicable and should not be performed.

1) To determine whether the optional CWLU is installed, run a system configuration for ATA “46 - Information Systems”. Do this task: System Configuration Check, TASK 46-13-00-710-802.

NOTE: “Wireless Access Point 1” will show on the system configuration if the CWLU is installed.

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2) To verify the required UMS parts, run a system configuration for ATA “46 - Information Systems”. Do this task: System Configuration Check, TASK 46-13-00-710-802.

NOTE: Serviceable user-modifiable software (UMS) parts must be installed on the Network File Server (NFS) for the CWLU to work. The required UMS parts are:

- 46 NFS OAS AIRPLANE KEYS UMS
- 46 NFS EXCOMM UMS
- 46 NFS OAS AIRLINE CERTS UMS
- 46 NFS OAS AIRLINE CRL UMS

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205
STD-9080	Cable - Ethernet (Cat 5E, RJ-45 Connectors)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

D. Connection Procedure with Ethernet Cable

SUBTASK 46-13-00-480-001

(1) Connect one end of the RJ-45 Ethernet cable, STD-9080, to the RJ45 port of the universal maintenance device, SPL-11321.

(2) Connect the opposite end to one of these data ports:

- Flight Compartment - On the P61 Panel, find the RJ45 ONS port on the Maintenance BITE Panel.
- E/E Bay - On the E4-2 rack, find the RJ45 port behind the door on the Network File Server (NFS) front panel.

SUBTASK 46-13-00-860-001

(3) Energize the universal maintenance device, SPL-11321.

(a) When it shows Welcome to Windows, then continue.

SUBTASK 46-13-00-860-002

(4) Log-in as follows:

(a) Push the CTRL, ALT and DELETE keys at the same time.

1) Make sure it shows: Log On to Windows.

(b) Enter the User Name and Password and then select OK.

SUBTASK 46-13-00-860-003

(5) Do these steps to launch the ONS maintenance application:

(a) Double-click on the ONS maintenance browser desktop icon.

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1) If the ONS maintenance application fails to connect, then refer to the IFIM and do the applicable procedure(s).

E. Connection Procedure using the Optional Crew Wireless LAN Unit

SUBTASK 46-13-00-860-004

(1) Get the crew wireless access code:

(a) Use one of the large inboard displays (Captain's or First Officer's Multi Function Display (MFD)).

(b) On the P61 Maintenance BITE Panel or on one of the two forward console locations, set the outer MAINT SELECTOR switch to the L(eft) or to the R(ight) MFD position.

(c) Turn the inner SEL knob until you see MAINT on the MFD.

1) If the MAINT button does not show on the MFD, push the SYS key on the Multi-Function Panel (P9-82).

2) If the MAINT button still does not show on the MFD, then refer to the IFIM and do the applicable procedure(s).

(d) Set the cursor on the word MAINT and push the SEL knob.

(e) Select ONBD MAINT.

1) If the MFD shows “ONBD MAINT NOT AVAILABLE” in amber or white color text, then refer to the IFIM and do the applicable procedure(s) based on the text color.

(f) From the Onboard Maintenance menu, make the LINE MAINT selection.

(g) On the LINE MAINT menu, make the CREW WIRELESS ACCESS CODE selection.

NOTE: It may take up to one minute for the CREW WIRELESS ACCESS CODE selection to show after selecting LINE MAINT.

1) If the crew wireless access code shows, then make a record of the access code and the expiration.

NOTE: The time that shows is in minutes and seconds. The access code is valid for five minutes.

2) If the crew wireless access code does not show and the word “Expired” shows, then make the “NEW ACCESS CODE” selection.

a) Make sure that a new crew wireless access code shows. Make a record of the access code and the expiration.

SUBTASK 46-13-00-860-005

(2) Use a Wi-Fi enabled, universal maintenance device, SPL-11321, to do the steps that follow.

(a) Energize the laptop and show the desktop.

(b) With the cursor, find and select the Network icon.

1) You can left-click on the network icon, in the Windows task tray.

2) You can get access using the Windows Start icon, and control panel selection.

(c) Select the correct wireless network, and click on the word Connect.

1) The wireless network name is: Maintenance-[tail ID].

2) Examine the signal strength bars that show in the task tray to make sure the network connection is serviceable.

SUBTASK 46-13-00-860-006

(3) Do these steps to launch the ONS maintenance application:

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(a) Double-click on the ONS Maintenance desktop icon.

1) If the ONS Maintenance desktop icon does not open, then refer to the IFIM and do the applicable procedure(s).

(b) Enter the Crew Wireless Access Code that you recorded from the steps above and click on SAVE.

NOTE: A keyboard is necessary to enter a code. If the universal maintenance device, SPL-11321, does not have a keyboard which shows automatically when you click to enter the code, you need to go into Windows and manually show the keyboard.

(c) Make sure the ONS Maintenance main menu shows.

——— END OF TASK ———

TASK 46-13-00-480-803

3. Multi Function Display Connection

A. General

(1) This task gives the steps to access the Onboard Network System (ONS) using the Multi Function Display (MFD) and to start an ONS maintenance browser session.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

C. Connection Procedure with Multifunction Display

SUBTASK 46-13-00-860-037

(1) Do these steps to access the ONS using the MFD:

(a) Use one of the large inboard displays (Captain or First Officer's MFD).

(b) On the P61 Maintenance BITE Panel or on one of the two forward console locations, set the outer MAIN SELECTOR switch to the L (left) or to the R (right) MFD position.

NOTE: If the cursor gets stuck on the screen, rotate the Rotary Selector knob to reset the cursor to the upper left-hand section of the screen.

1) To navigate/tab through the selectable elements on the MFD, turn the Rotary Selector knob on the P9-82 panel or the MAINT SELECTOR knob on the P61-4 panel clockwise or counter-clockwise until the desired element is highlighted.

2) To make a selection on the MFD, push the center selection button “SEL” of the Rotary Selector knob on the P9-82 panel or the MAINT SELECTOR knob on the P61-4 panel.

(c) Turn the inner SEL knob until you see MAINT on the MFD.

1) If the MAINT button does not show on the MFD, push the SYS key on the Multi-Function Panel (P9-82).

2) If the MAINT button still does not show on the MFD, then refer to the IFIM and do the applicable procedure(s).

(d) Set the cursor on the word MAINT and push the SEL knob.

(e) Select the ONBD MAINT button from the Maintenance Menu bar.

NOTE: Not all functionality on the UMD is available on the MFD. Whether a function is available on the MFD or not will be noted as needed.

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1) If the MFD shows “ONBD MAINT NOT AVAILABLE” in amber or white color text, then refer to the IFIM and do the applicable procedure(s) based on the text color.

——— END OF TASK ———

TASK 46-13-00-860-801

4. Airplane Credentials Check

(Figure 201)

A. General

(1) This task contains two sections.

(a) There are steps to do a check of the airplane credential installed in the Network File Server (NFS).

(b) There are steps to make (or, generate) a Certificate Signing Request (CSR).

NOTE: The Boeing Company recommends that you make a CSR only when specified by a maintenance order.

(2) There are different procedures to transmit the CSR. Contact your Engineering department for the preferred method.

- Manual - use an approved data loader or maintenance device.
- Automatic - use Boeing Electronic Distribution of Software (BEDS).

NOTE: For the automatic method, after CSR generation, there are two options: download CSR via the Universal Maintenance Device (UMD) and send it manually to operator's network, or wait for the CSR to downlink from the NFS queue to operator's network. The NFS downlink queue offboards data via priority within 168 hours. If the data is not offboarded within the aloted time, it will be stored to the Download Manager.

(3) The aircraft must be on the ground to do this task.

(4) This task is available when the Onboard Network System (ONS) is accessed with the UMD.

B. References

Reference	Title
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Airplane Credentials Check

SUBTASK 46-13-00-480-017

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

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SUBTASK 46-13-00-700-007

(2) Make sure that the Airplane Identification information is completely populated (TASK 46-13-00-750-801).

F. Airplane Credentials Check

SUBTASK 46-13-00-860-007

(1) To do the Airplane Credentials Check, do these steps:

(a) Select LINE MAINTENANCE > Security > AIRPLANE CREDENTIALS.

(b) Examine the results in the box below Credential Status.

1) If the result shows “A valid, CA-signed Airplane Certificate exists”, then the credential is serviceable. Go to step “Put the Airplane Back to Its Usual Condition”

2) If the result shows “No Valid Airplane Certificate exists”, then continue to the next step.

NOTE: If a valid CSR previously existed, and this result is unexpected, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

G. Certificate Signing Request

SUBTASK 46-13-00-470-001

(1) To make a new CSR, do the steps that follow:

NOTE: The Boeing Company recommends that you make a CSR only when specified by a maintenance order.

(a) Select the GENERATE button.

(b) When the GENERATE AIRPLANE CREDENTIALS box shows, make sure that all fields are populated and correct and then press CONTINUE (Figure 201).

NOTE: If the data fields are not populated and new Client Credentials were recently created, wait up to two (2) minutes and try again.

NOTE: System Date, System Time, Tail Number, and Airplane ID are automatically populated by the Display Processing Computer (DPC). The Aircraft Type must be set manually (TASK 46-13-00-750-801).

(c) Wait for the new credentials to be created.

NOTE: In the Credentials Generation Status box, the progress indicator will turn until the operation is complete.

(d) Make sure that the Credential Status box shows that a new self-signed Airplane Certificate was generated and downlinked for signing on the current date and time.

(e) Make sure that the Credentials Generation Status box shows a message that credentials were successfully created, with a file name.

(f) Make a record of the file name and file location.

NOTE: The file name is necessary to find and upload the file in the next step.

SUBTASK 46-13-00-470-002

(2) To download the CSR file from the NFS to the UMD using the Download Manager, do these steps:

(a) Select LINE MAINTENANCE > Download Manager.

(b) On the Select Application menu, select the NFS button.

(c) On the Select Type menu, select “NFS:OAS” filter.

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(d) Select CONTINUE.

(e) On the FILES tab page, select the most recent file with “CSR” in the file name.

(f) Select the REPORT icon.

(g) Make sure that the destination “Maintenance Device” is selected.

(h) Select the SEND button.

(i) When the Windows popup shows, select where the data will be downloaded on the UMD.

NOTE: The Boeing Company recommends to use the auto-created file name. The file is named as follows: airplane identification/tail ID, filter type (NFS-OAS), date and time.

(j) Select the CLOSE button.

SUBTASK 46-13-00-470-019

(3) To download the CSR file from the NFS to the UMD with the Software Management Tool (SMT), do this step:

(a) Make sure that the date and time information of the CSR is known.

(b) Do this task: Download Files using the Software Management Tool (SMT), TASK 46-13-00-470-803.

SUBTASK 46-13-00-470-003

(4) Send the file to the engineering department, or network operator for processing.

SUBTASK 46-13-00-070-001

(5) If there are one or more obsolete CSR files on the NFS, remove them.

NOTE: Do not erase the last CSR file that was sent to engineering.

(a) Click once on the CSR file to highlight.

(b) Click once on the ERASE button.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-001

(1) Make sure that there are no active maintenance messages for ATA “46 - Information Systems” (TASK 46-13-02-700-805).

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

SUBTASK 46-13-00-020-002

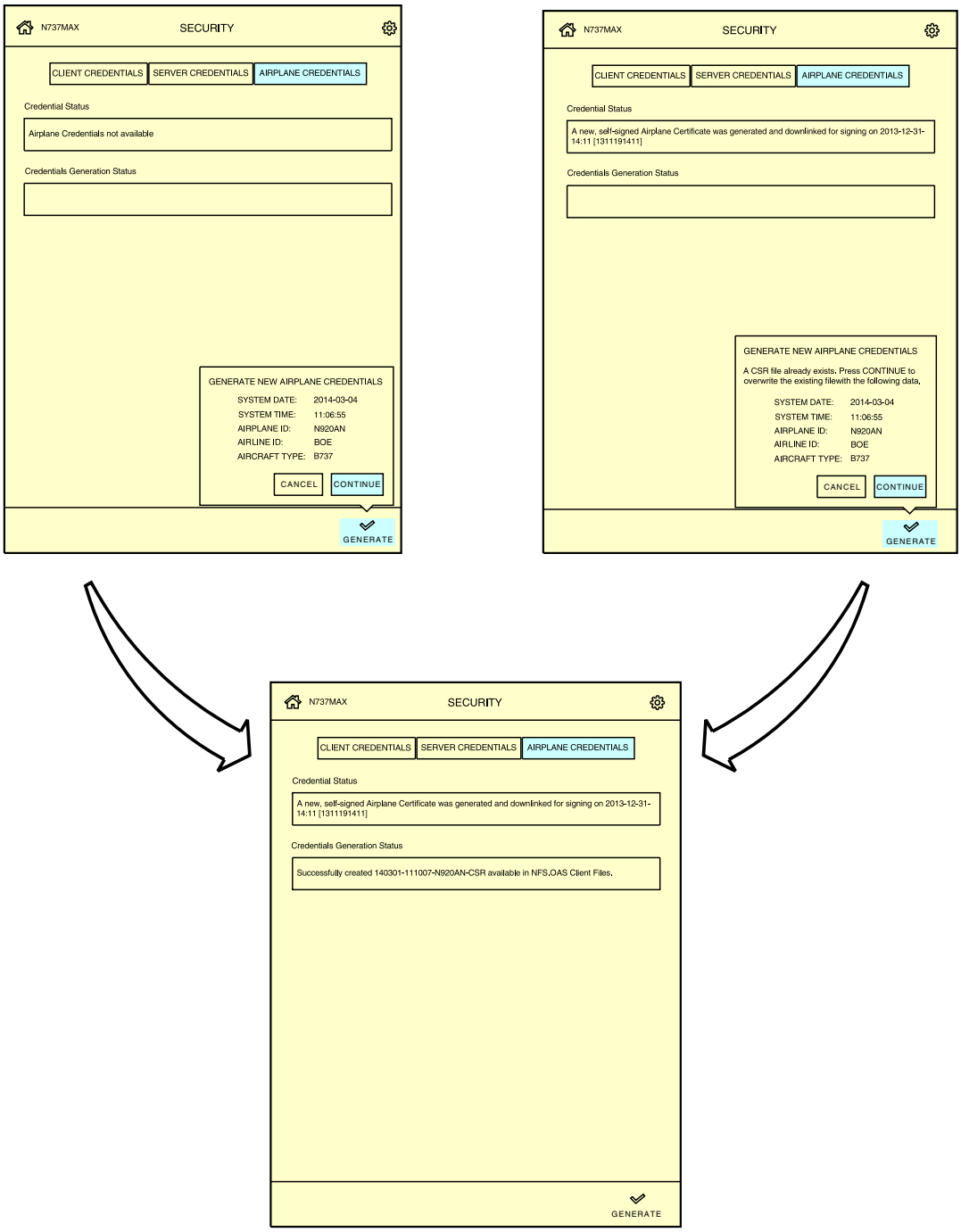
(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

SUBTASK 46-13-00-470-046

(3) If the 46 NFS OAS AIRPLANE KEYS UMS needs to be installed, then continue to this task: 46 NFS OAS AIRPLANE KEYS UMS Installation, TASK 46-13-00-860-805.

——— END OF TASK ———

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NOTE:

THE DATA THAT SHOWS ON THE DISPLAY IS ONLY AN EXAMPLE.

2475689 S0000580813_V1

Airplane Credentials Check
Figure 201/46-13-00-990-810

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TASK 46-13-00-860-805

5. 46 NFS OAS AIRPLANE KEYS UMS Installation

A. General

(1) This task contains two sections.

(a) There are steps to install the 46 NFS OAS AIRPLANE KEYS UMS into the Network File Server (NFS) operating system.

(b) There are steps to verify if the 46 NFS OAS AIRPLANE KEYS UMS were installed successfully.

(2) Contact your Engineering department to generate the 46 NFS OAS AIRPLANE KEYS UMS.

(3) The aircraft must be on the ground to do this task.

(4) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Prepare for the Airplane Keys UMS Installation

SUBTASK 46-13-00-480-020

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

E. Airplane Keys Installation

SUBTASK 46-13-00-470-043

(1) Get the new Loadable Software Airplane Part (LSAP) 46 NFS OAS AIRPLANE KEYS UMS. Stage the part from the UMD to the Mass Storage Device (MSD). Do this task: Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD), TASK 46-13-00-470-802.

NOTE: This step is not applicable if BEDS is serviceable and a network administrator sends the 46 NFS OAS AIRPLANE KEYS UMS LSAP to the MSD automatically.

SUBTASK 46-13-00-470-044

(2) Dataload the 46 NFS OAS AIRPLANE KEYS UMS to the NFS. Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

F. Installation Test

SUBTASK 46-13-00-860-041

(1) On the ONS maintenance browser menu bar:

(a) Select LINE MAINTENANCE > Security > AIRPLANE CREDENTIALS.

(b) Examine the result in the box below Airplane Credentials Status.

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1) Make sure that the result shows: A valid, CA-signed Airplane Certificate exists.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-014

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems”. Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-00-860-802

6. Server Credentials Check

(Figure 202)

A. General

(1) This task gives the steps to do a check, and if necessary, to make new server credentials.

NOTE: When a new server credential is made, all client credentials that show “No credentials exist” must be made again (TASK 46-13-00-860-803).

(2) The Boeing Company recommends to make a new server credential only when specified by a maintenance order.

(3) The aircraft must be on the ground to do this task.

(4) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Check

SUBTASK 46-13-00-480-003

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

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F. Server Credentials Check

SUBTASK 46-13-00-470-034

(1) To do the Server Credentials Check, do these steps:

(a) Select LINE MAINTENANCE > Security > SERVER CREDENTIALS.

(b) Examine the Credential Status box:

1) Make sure that the data shows a valid time-date stamp when the last credentials were made (or, were “last generated”).

NOTE: A valid time-date stamp is any date between the year 2017 and the current year.

a) If a valid time-stamp exists, then no action is required for the remainder of this task.

b) If an invalid time-date stamp shows, then continue.

NOTE: If the time-date stamp shows the year 1999 or anything beyond 2038, do not continue and instead refer to the IFIM and do the applicable procedure(s).

SUBTASK 46-13-00-470-006

(2) To make a new server credential, do these steps:

NOTE: The Boeing Company recommends that you do this only when specified by a maintenance order.

(a) Click once on the GENERATE button.

(b) When the confirmation box shows, click once on CONTINUE.

(c) In the Credentials Generation Status box, make sure that the progress indicator turns and wait for the result.

NOTE: It is necessary to wait 10 minutes after the credential generation is initiated.

(d) Examine the results to make sure that the process is complete:

1) The Credentials Status box shows “Credentials last generated on <current date / time stamp of the credential>”.

2) The Credentials Generation Status box shows “generated on board certificate authority”.

SUBTASK 46-13-00-470-037

(3) If a new server credential was made, make sure that all LRUs have valid client credentials. Do this task: Client Credentials Check, TASK 46-13-00-860-803.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-002

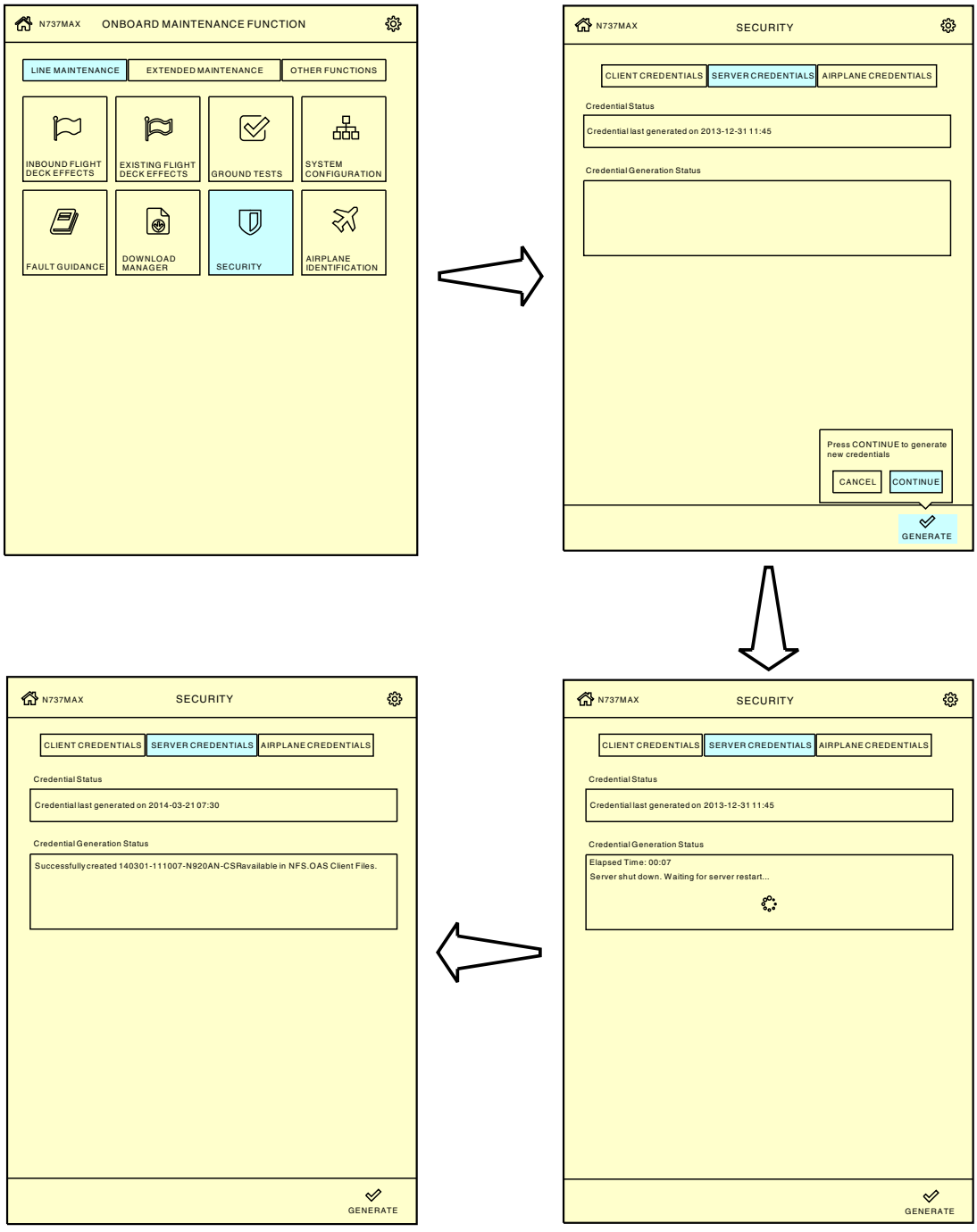
(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

END OF TASK

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NOTE:

THE DATA THAT SHOWS ON THE DISPLAY IS ONLY AN EXAMPLE.

2475691 S0000580814_V1

Onboard Network System Server Credentials Check
Figure 202/46-13-00-990-809

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TASK 46-13-00-860-803

7. Client Credentials Check

(Figure 203)

A. General

(1) This task gives the steps to do a check, and if necessary, to make and install a new client credential.

NOTE: Each Line Replaceable Unit (LRU) must have a client credential to communicate with the Network File Server (NFS).

(2) The client LRUs are:

- Aircraft Interface Device (AID)
- DFDAU
- EEC-1 (two channels)
- EEC-2 (two channels)
- NFS

(3) The aircraft must be on the ground to do this task.

(4) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-01-000-801	Network File Server Removal (P/B 401)
46-13-01-020-801	Network File Server SD Media Card Removal (P/B 201)
46-13-01-400-801	Network File Server Installation (P/B 401)
46-13-01-420-801	Network File Server SD Media Card Installation (P/B 201)
46-13-01-470-811	Network File Server OS Upgrade OS7.1 to OS9.1 (P/B 201)
46-13-01-470-812	Network File Server OS Downgrade OS9.1 to OS7.1 (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)
73-21-05-970-802-G00	Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Client Credentials Check

SUBTASK 46-13-00-480-004

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

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F. Client Credentials Check

SUBTASK 46-13-00-860-010

(1) To do the Client Credentials Check, do these steps:

(a) Make sure that power is supplied to any LRU(s) that require a credential check for at least 3 minutes.

NOTE: Power must be supplied to the AID for at least 7 minutes.

(b) Select LINE MAINTENANCE > Security > CLIENT CREDENTIALS.

(c) Wait for the page to show “Select the client(s) to generate new credentials”.

(d) Examine the page to find each client, and the condition of its credential.

NOTE: The credentials status should show the words “Last generated” with a date and time stamp.

1) Make sure that each credential status does not show “No credentials exist”.

2) If you find a missing credential, then continue.

SUBTASK 46-13-00-470-039

(2) To prepare the Electronic Engine Control (EEC)s for the client credentials, do these steps:

NOTE: For ONS OS9.1, There are three methods to generate EEC client credentials. Method 1 or method 2 is used on an EEC that does not have preloaded credentials. Method 3 is used on an EEC that has preloaded credentials.

- METHOD 1- NFS Downgrade/Upgrade: Downgrade NFS to OS7.1 > Generate EEC Client Credentials > Upgrade NFS to OS9.1. This process requires a reimage, which will wipe all data off the NFS. Please ensure all desired data is downloaded from the NFS before conducting the below method. If necessary, save a system configuration report (System Configuration Check, TASK 46-13-00-710-802) and a mass storage device report (Mass Storage Device Software Parts Check, TASK 46-13-00-860-804).
- METHOD 2- Preloaded OS7.1 NFS Swap: Remove OS9.1 NFS > Install OS7.1 NFS > Generate EEC Credentials > Remove OS7.1 NFS > Install OS9.1 NFS. Do NOT change the airplane ID during this procedure.
- METHOD 3- Server Credential Generation and EEC Power Cycle.

(a) METHOD 1: NFS DOWNGRADE/UPGRADE

1) Downgrade the NFS from OS9.1 to OS7.1, do this task: Network File Server OS Downgrade OS9.1 to OS7.1, TASK 46-13-01-470-812.

a) Ensure no SD card is installed in the NFS before conducting the reimage step, do this task: Network File Server SD Media Card Removal, TASK 46-13-01-020-801.

NOTE: Do not conduct steps 4-9 of SUBTASK 46-13-01-470-057.

2) Power on the EECs.

a) If the EECs are not powered, do these steps:

<1> On the Multi Function Display (MFD), select MAINT CTRL PGS.

<2> Select MISC SYSTEM CTRLS (MSC).

<3> In the EEC MAINT POWER box, power on ENG 1 EECs.

<a> Select TEST under ENG 1 to power on ENG 1 EECs.

<b> Verify ENG 1 STATUS shows PWR after 30 seconds.

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<4> In the EEC MAINT POWER box, power on ENG 2 EECs.

<a> Select TEST under ENG 2 to power on ENG 2 EECs.

<b> Verify ENG 2 STATUS shows PWR after 30 seconds.

b) If the EECs are already powered, conduct a power cycle of the EECs.

<1> On the MFD, select MAINT CTRL PGS.

<2> Select MISC SYSTEM CTRLS (MSC).

<3> In the EEC MAINT POWER box, if ENG 1 STATUS shows PWR:

<a> Select NORM under ENG 1 to power off ENG 1 EECs.

<b> Verify ENG 1 STATUS shows OFF after 15 seconds.

<4> In the EEC MAINT POWER box, if ENG 2 STATUS shows PWR:

<a> Select NORM under ENG 2 to power off ENG 2 EECs.

<b> Verify ENG 2 STATUS shows OFF after 15 seconds.

<5> In the EEC MAINT POWER box, power on ENG 1 EECs.

<a> Select TEST under ENG 1 to power on ENG 1 EECs.

<b> Verify ENG 1 STATUS shows PWR after 30 seconds.

<6> In the EEC MAINT POWER box, power on ENG 2 EECs.

<a> Select TEST under ENG 2 to power on ENG 2 EECs.

<b> Verify ENG 2 STATUS shows PWR after 30 seconds.

3) Generate EEC client credentials for EEC-1A, EEC-1B, EEC-2A, and EEC-2B.

a) On the ONS browser, select LINE MAINTENANCE > SECURITY > CLIENT CREDENTIALS.

b) Select EEC-1A, EEC-1B, EEC-2A, and EEC-2B.

c) Select GENERATE > CONTINUE.

d) Verify each EEC credential status shows the words “Last generated” with a current date and time stamp.

4) Upgrade the NFS from OS7.1 to OS9.1, do this task: Network File Server OS Upgrade OS7.1 to OS9.1, TASK 46-13-01-470-811.

a) Ensure no SD card is installed in the NFS before conducting the reimage step, do this task: Network File Server SD Media Card Removal, TASK 46-13-01-020-801.

5) Make sure that power is supplied to the DFDAU and any installed WWU or AID LRUs.

6) Cycle power on the EECs:

a) On the MFD, select MAINT CTRL PGS.

b) Select MISC SYSTEM CTRLS (MSC).

c) In the EEC MAINT POWER box, if ENG 1 STATUS shows PWR:

<1> Select NORM under ENG 1 to power off ENG 1 EECs.

<2> Verify ENG 1 STATUS shows OFF after 15 seconds.

d) In the EEC MAINT POWER box, if ENG 2 STATUS shows PWR:

<1> Select NORM under ENG 2 to power off ENG 2 EECs.

<2> Verify ENG 2 STATUS shows OFF after 15 seconds.

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e) In the EEC MAINT POWER box, power on ENG 1 EECs.
<1> Select TEST under ENG 1 to power on ENG 1 EECs.
<2> Verify ENG 1 STATUS shows PWR after 30 seconds.
f) In the EEC MAINT POWER box, power on ENG 2 EECs.
<1> Select TEST under ENG 2 to power on ENG 2 EECs.
<2> Verify ENG 2 STATUS shows PWR after 30 seconds.

7) Generate new Server Credentials, do this task: Server Credentials Check, TASK 46-13-00-860-802.

8) Upon completion of the Server Credential generation (after approximately 10 minutes), do these steps to cycle power on the EECs:
a) On the MFD, select MAINT CTRL PGS.
b) Select MISC SYSTEM CTRLS (MSC).
c) In the EEC MAINT POWER box, if ENG 1 STATUS shows PWR:
<1> Select NORM under ENG 1 to power off ENG 1 EECs.
<2> Verify ENG 1 STATUS shows OFF after 15 seconds.
d) In the EEC MAINT POWER box, if ENG 2 STATUS shows PWR:
<1> Select NORM under ENG 2 to power off ENG 2 EECs.
<2> Verify ENG 2 STATUS shows OFF after 15 seconds.
e) In the EEC MAINT POWER box, power on ENG 1 EECs.
<1> Select TEST under ENG 1 to power on ENG 1 EECs.
<2> Verify ENG 1 STATUS shows PWR after 30 seconds.
f) In the EEC MAINT POWER box, power on ENG 2 EECs.
<1> Select TEST under ENG 2 to power on ENG 2 EECs.
<2> Verify ENG 2 STATUS shows PWR after 30 seconds.

9) On the ONS Maintenance main menu, select LINE MAINTENANCE > Security.
NOTE: The Client Credentials page will not be updated automatically. The Security page must be left and then returned to.

10) Wait for the result to show “Select the client(s) to generate new credentials”.

11) Examine the record to find each EEC client, and the condition of its credential.
a) Make sure that the credentials status shows “Last generated” with a date and time stamp matching time of the steps above.

12) Stage all missing Mass Storage Device software parts to put the Mass Storage Device back to normal configuration, do this task: Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD), TASK 46-13-00-470-802.

13) If applicable, install the SD Media Card into the NFS, do this task: Network File Server SD Media Card Installation, TASK 46-13-01-420-801.

14) If applicable, generate a CSR and dataload a KEYS UMS, do this task: Airplane Credentials Check, TASK 46-13-00-860-801.

15) If this task was not performed immediately after an EEC replacement, any data present on the EEC before the credential generation must be downloaded to the NFS via the EEC Data Download page, do this task: Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS, TASK 73-21-05-970-802-G00.

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(b) METHOD 2: PRELOADED OS7.1 NFS SWAP

1) Remove the OS9.1 NFS, do this task: Network File Server Removal, TASK 46-13-01-000-801.
2) Install the OS7.1 NFS, do this task: Network File Server Installation, TASK 46-13-01-400-801.
3) Power on the EECs.
   a) If the EECs are not powered, do these steps:
      <1> On the MFD, select MAINT CTRL PGS.
      <2> Select MISC SYSTEM CTRLS (MSC).
      <3> In the EEC MAINT POWER box, power on ENG 1 EECs.
         <a> Select TEST under ENG 1 to power on ENG 1 EECs.
         <b> Verify ENG 1 STATUS shows PWR after 30 seconds.
      <4> In the EEC MAINT POWER box, power on ENG 2 EECs.
         <a> Select TEST under ENG 2 to power on ENG 2 EECs.
         <b> Verify ENG 2 STATUS shows PWR after 30 seconds.
   b) If the EECs are already powered, conduct a power cycle of the EECs.
      <1> On the MFD, select MAINT CTRL PGS.
      <2> Select MISC SYSTEM CTRLS (MSC).
      <3> In the EEC MAINT POWER box, if ENG 1 STATUS shows PWR:
         <a> Select NORM under ENG 1 to power off ENG 1 EECs.
         <b> Verify ENG 1 STATUS shows OFF after 15 seconds.
      <4> In the EEC MAINT POWER box, if ENG 2 STATUS shows PWR:
         <a> Select NORM under ENG 2 to power off ENG 2 EECs.
         <b> Verify ENG 2 STATUS shows OFF after 15 seconds.
      <5> In the EEC MAINT POWER box, power on ENG 1 EECs.
         <a> Select TEST under ENG 1 to power on ENG 1 EECs.
         <b> Verify ENG 1 STATUS shows PWR after 30 seconds.
      <6> In the EEC MAINT POWER box, power on ENG 2 EECs.
         <a> Select TEST under ENG 2 to power on ENG 2 EECs.
         <b> Verify ENG 2 STATUS shows PWR after 30 seconds.
4) Generate EEC client credentials for EEC-1A, EEC-1B, EEC-2A, and EEC-2B.
   a) On the ONS browser, select LINE MAINTENANCE > SECURITY > CLIENT CREDENTIALS.
   b) Select EEC-1A, EEC-1B, EEC-2A, and EEC-2B.
   c) Select GENERATE > CONTINUE.
   d) Verify each EEC credential status shows the words “Last generated” with a current date and time stamp.
5) Remove the OS7.1 NFS, do this task: Network File Server Removal, TASK 46-13-01-000-801.
6) Install the OS9.1 NFS, do this task: Network File Server Installation, TASK 46-13-01-400-801.

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7) Make sure that power is supplied to the DFDAU and any installed WWU or AID LRUs.

8) Cycle power on the EECs:
   a) On the MFD, select MAINT CTRL PGS.
   b) Select MISC SYSTEM CTRLS (MSC).
   c) In the EEC MAINT POWER box, if ENG 1 STATUS shows PWR:
      <1> Select NORM under ENG 1 to power off ENG 1 EECs.
      <2> Verify ENG 1 STATUS shows OFF after 15 seconds.
   d) In the EEC MAINT POWER box, if ENG 2 STATUS shows PWR:
      <1> Select NORM under ENG 2 to power off ENG 2 EECs.
      <2> Verify ENG 2 STATUS shows OFF after 15 seconds.
   e) In the EEC MAINT POWER box, power on ENG 1 EECs.
      <1> Select TEST under ENG 1 to power on ENG 1 EECs.
      <2> Verify ENG 1 STATUS shows PWR after 30 seconds.
   f) In the EEC MAINT POWER box, power on ENG 2 EECs.
      <1> Select TEST under ENG 2 to power on ENG 2 EECs.
      <2> Verify ENG 2 STATUS shows PWR after 30 seconds.

9) Generate new Server Credentials, do this task: Server Credentials Check, TASK 46-13-00-860-802.

10) Upon completion of the Server Credential generation (after approximately 10 minutes), do these steps to cycle power on the EECs:
   a) On the MFD, select MAINT CTRL PGS.
   b) Select MISC SYSTEM CTRLS (MSC).
   c) In the EEC MAINT POWER box, if ENG 1 STATUS shows PWR:
      <1> Select NORM under ENG 1 to power off ENG 1 EECs.
      <2> Verify ENG 1 STATUS shows OFF after 15 seconds.
   d) In the EEC MAINT POWER box, if ENG 2 STATUS shows PWR:
      <1> Select NORM under ENG 2 to power off ENG 2 EECs.
      <2> Verify ENG 2 STATUS shows OFF after 15 seconds.
   e) In the EEC MAINT POWER box, power on ENG 1 EECs.
      <1> Select TEST under ENG 1 to power on ENG 1 EECs.
      <2> Verify ENG 1 STATUS shows PWR after 30 seconds.
   f) In the EEC MAINT POWER box, power on ENG 2 EECs.
      <1> Select TEST under ENG 2 to power on ENG 2 EECs.
      <2> Verify ENG 2 STATUS shows PWR after 30 seconds.

11) On the ONS Maintenance main menu, select LINE MAINTENANCE > Security.

   NOTE: The Client Credentials page will not be updated automatically. The Security page must be left and then returned to.

12) Wait for the result to show “Select the client(s) to generate new credentials”.

13) Examine the record to find each EEC client, and the condition of its credential.

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a) Make sure that the credentials status shows “Last generated” with a date and time stamp matching time of the steps above.

14) If this task was not performed immediately after an EEC replacement, any data present on the EEC before the credential generation must be downloaded to the NFS via the EEC Data Download page, do this task: Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS, TASK 73-21-05-970-802-G00.

(c) METHOD 3: SERVER CREDENTIAL GENERATION AND EEC POWER CYCLE

1) Make sure that power is supplied to the DFDAU and any installed WWU or AID LRUs.

2) Cycle power on the EECs:

a) On the MFD, select MAINT CTRL PGS.

b) Select MISC SYSTEM CTRLS (MSC).

c) In the EEC MAINT POWER box, if ENG 1 STATUS shows PWR:

<1> Select NORM under ENG 1 to power off ENG 1 EECs.

<2> Verify ENG 1 STATUS shows OFF after 15 seconds.

d) In the EEC MAINT POWER box, if ENG 2 STATUS shows PWR:

<1> Select NORM under ENG 2 to power off ENG 2 EECs.

<2> Verify ENG 2 STATUS shows OFF after 15 seconds.

e) In the EEC MAINT POWER box, power on ENG 1 EECs.

<1> Select TEST under ENG 1 to power on ENG 1 EECs.

<2> Verify ENG 1 STATUS shows PWR after 30 seconds.

f) In the EEC MAINT POWER box, power on ENG 2 EECs.

<1> Select TEST under ENG 2 to power on ENG 2 EECs.

<2> Verify ENG 2 STATUS shows PWR after 30 seconds.

3) Generate new Server Credentials, do this task: Server Credentials Check, TASK 46-13-00-860-802.

4) Upon completion of the Server Credential generation (after approximately 10 minutes), do these steps to cycle power on the EECs:

a) On the MFD, select MAINT CTRL PGS.

b) Select MISC SYSTEM CTRLS (MSC).

c) In the EEC MAINT POWER box, if ENG 1 STATUS shows PWR:

<1> Select NORM under ENG 1 to power off ENG 1 EECs.

<2> Verify ENG 1 STATUS shows OFF after 15 seconds.

d) In the EEC MAINT POWER box, if ENG 2 STATUS shows PWR:

<1> Select NORM under ENG 2 to power off ENG 2 EECs.

<2> Verify ENG 2 STATUS shows OFF after 15 seconds.

e) In the EEC MAINT POWER box, power on ENG 1 EECs.

<1> Select TEST under ENG 1 to power on ENG 1 EECs.

<2> Verify ENG 1 STATUS shows PWR after 30 seconds.

f) In the EEC MAINT POWER box, power on ENG 2 EECs.

<1> Select TEST under ENG 2 to power on ENG 2 EECs.

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<2> Verify ENG 2 STATUS shows PWR after 30 seconds.

5) On the ONS Maintenance main menu in the upper left-hand corner, select LINE MAINTENANCE > Security.

NOTE: The Client Credentials page will not be updated automatically. The Security page must be left and then returned to.

6) Wait for the page to show “Select the client(s) to generate new credentials”.

7) Examine the record to find each EEC client, and the condition of its credential.

a) Make sure that the credentials status shows “Last generated” with a date and time stamp matching time of the steps above.

SUBTASK 46-13-00-470-007

(3) To make one or more new client credentials, do these steps:

NOTE: Each client (or LRU) must have a unique credential.

(a) Make sure that power is supplied to any LRU(s) that require new client credentials for at least three minutes.

(b) Click once on the box adjacent to the client name(s) where “No credentials exist” is shown.

1) Make sure that check mark(s) show in the box.

2) Do not select the EEC boxes.

NOTE: For OS9.1, client credentials for the EEC must be generated with METHOD 1, 2, or 3.

(c) When all selections are complete, select GENERATE.

(d) In the confirmation box, select CONTINUE.

(e) Wait for the credentials to be created.

NOTE: In the Credentials Generation Status box, the progress indicator will turn until the operation is complete.

(f) Examine the results in the Credentials Generation Status box.

1) Make sure that the Overall Status shows “All keys were successfully generated”.

NOTE: If the Overall Status shows “Keys were not generated for one or more clients”, examine the results to determine which LRU(s) did not generate a client credential and take maintenance action accordingly.

2) If necessary, use the scroll bar to see all of the clients and the status.

(g) Wait for at least five minutes before making the next credential or other ONS maintenance task.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-003

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems".

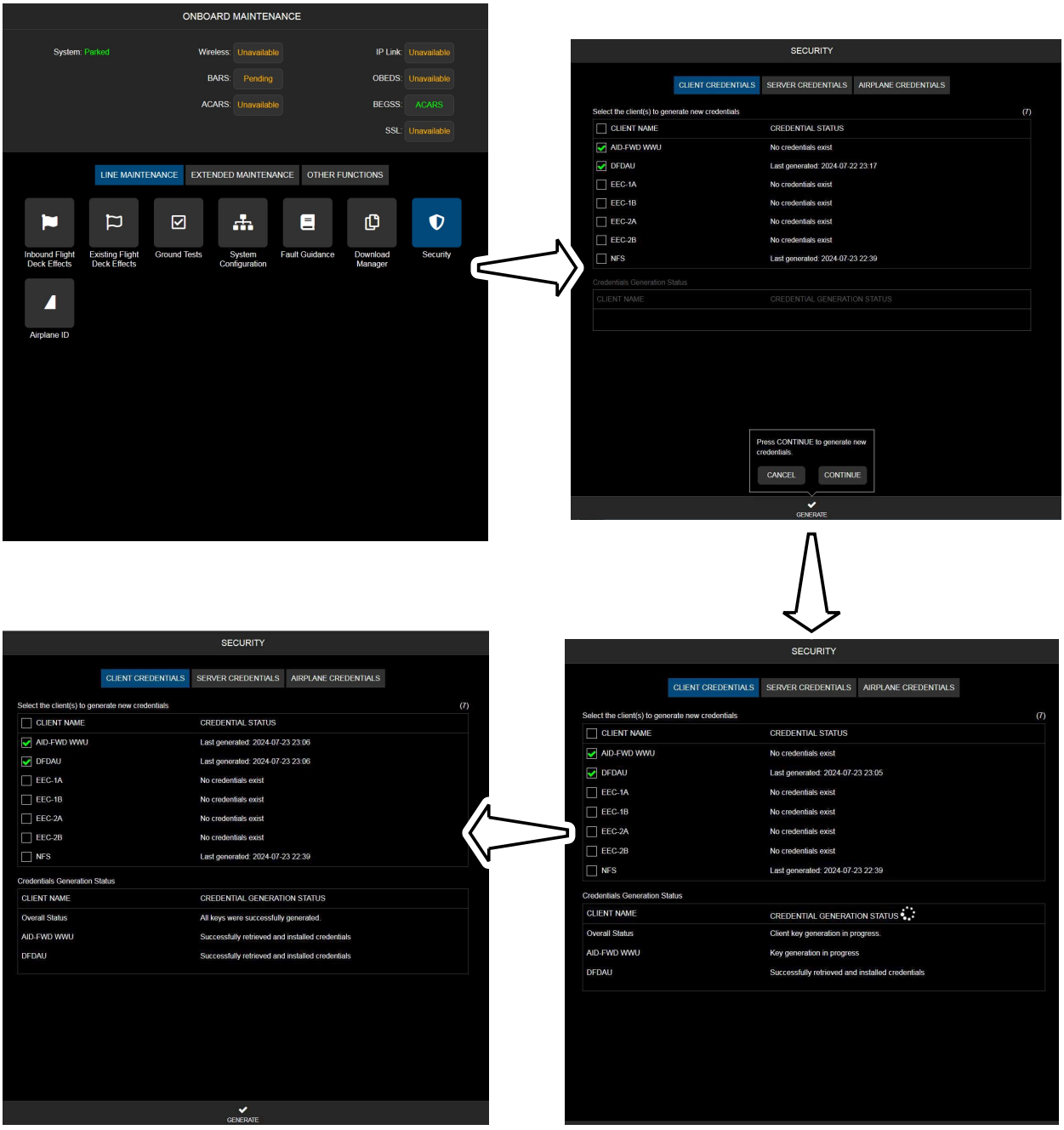
(a) Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(b) If there is an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

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NOTE:
THE DATA THAT SHOWS ON THE DISPLAY IS ONLY AN EXAMPLE.

2475702 S0000580815_V2

Onboard Network System Client Credentials
Figure 203/46-13-00-990-804

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TASK 46-13-00-750-801

8. Airplane Identification Check

(Figure 204)

A. General

(1) This task gives the steps to do a check, and if necessary, to update the airplane identification in the Network File Server (NFS).

(a) If the Onboard Network System (ONS) does not find Airplane Identification data, a pop-up message shows when the ONS maintenance browser is accessed.

(b) When there is no Airplane Identification data, some ONS and Onboard Maintenance Function (OMF) are not serviceable.

(2) It is necessary that the tail ID on the ONS is the same as the tail ID on the auxiliary outboard displays.

(a) The auxiliary outboard displays show on the outboard part of the LEFT and RIGHT outboard display units.

(3) The aircraft must be on the ground to do this task.

(4) This task is available when the ONS is accessed with the Universal Maintenance Device (UMD).

(5) The Airplane Identification Data can also be accessed and updated with the Multi Function Display (MFD) (TASK 31-65-00-970-804).

B. References

Reference	Title
31-65-00-970-804	Airplane Config Data (ACD) Page Function (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Check

SUBTASK 46-13-00-860-012

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

F. Airplane Identification Check

SUBTASK 46-13-00-970-001

(1) On the auxiliary outboard displays, find and make a record of the aircraft registration number (tail number).

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SUBTASK 46-13-00-860-013

(2) On the ONS maintenance browser:

(a) Select LINE MAINTENANCE > Airplane Identification.

(b) Compare the data on the page with the auxiliary outboard displays.

1) If Airline ID, Aircraft Type, and Airplane ID all agree, skip to step “Put the Airplane Back to Its Usual Condition”.

2) If one or more data records do not agree, then continue.

G. Airplane Identification Changes

SUBTASK 46-13-00-970-002

(1) Enter the correct data.

(a) Select the correct Aircraft Type from the Airline ID drop-down menu.

AIRCRAFT	DROPDOWN LIST AIRCRAFT TYPE
BOEING 737-600	B736 (737-600)
BOEING 737-700	B737 (737-700)
BOEING 737-800	B738 (737-800)
BOEING 737-900	B739 (737-900)
BOEING 737 MAX 7	B37M (737 MAX 7)
BOEING 737 MAX 8	B38M (737 MAX 8)
BOEING 737 MAX 200	B38M (737 MAX 200)
BOEING 737 MAX 9	B39M (737 MAX 9)
BOEING 737 MAX 10	B3XM (737 MAX 10)

(b) Using the MFD, enter the correct Airplane ID. Do this task: Airplane Config Data (ACD) Page Function, TASK 31-65-00-970-804.

NOTE: The airplane ID is called “TAIL NUMBER” on the MFD.

NOTE: If the DPCs are off or unavailable, the Airplane ID can be manually entered on the ONS Airplane Identification page. Once the Airplane ID is entered on the Airplane Identification page, select SAVE.

NOTE: When you change the Airplane ID, this clears the OMF fault history.

NOTE: After saving a new Airplane ID, wait 2 minutes and then power cycle the NFS. The NFS will take approximately 10 minutes to reboot.

(c) Using the MFD, enter the correct Airline ID. Do this task: Airplane Config Data (ACD) Page Function, TASK 31-65-00-970-804.

NOTE: If the DPCs are off or unavailable, the Airline ID can be manually entered on the ONS Airplane Identification page. Once the Airline ID is entered on the Airplane Identification page, select SAVE.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-004

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

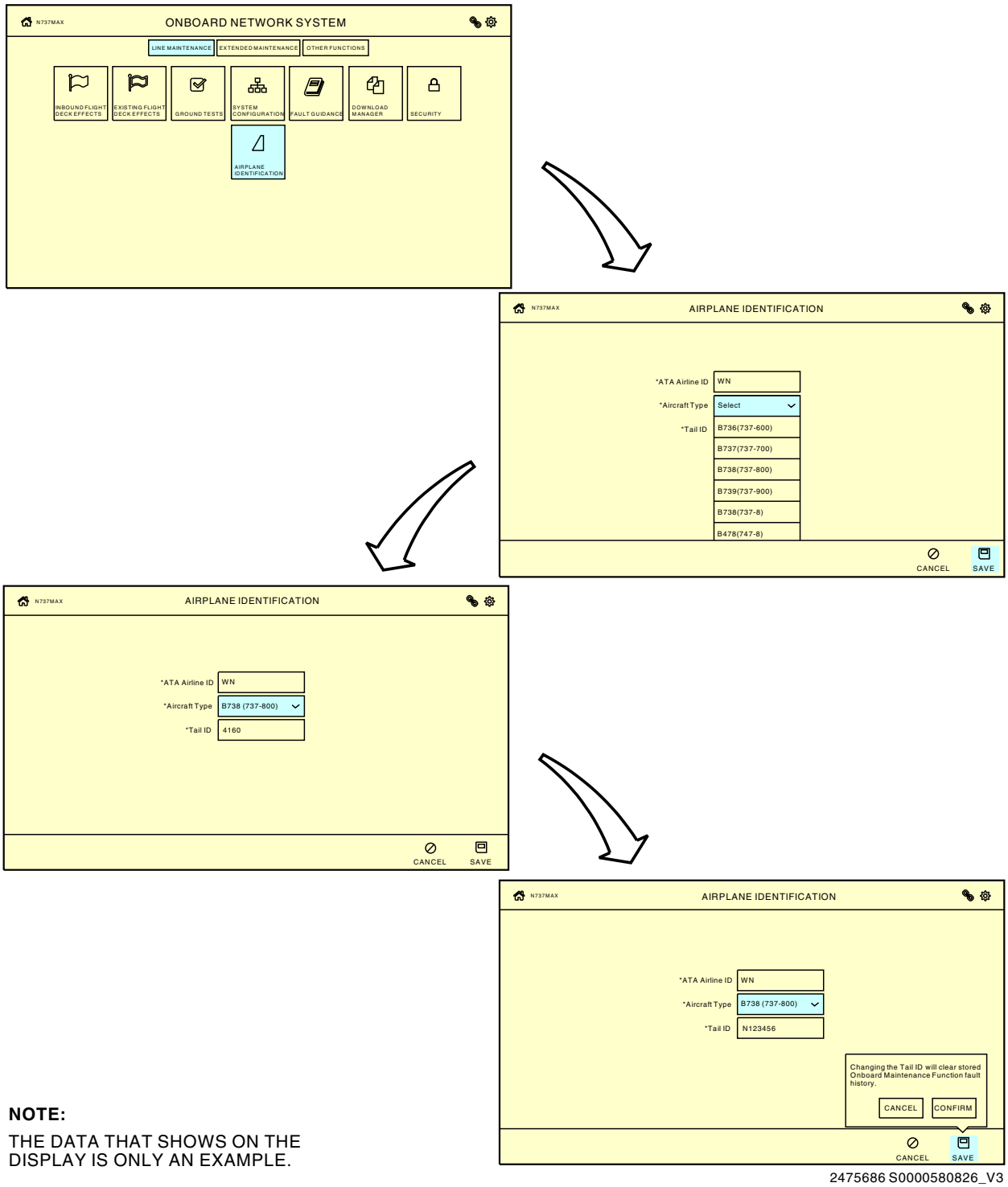
(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

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(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

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NOTE:

THE DATA THAT SHOWS ON THE DISPLAY IS ONLY AN EXAMPLE.

2475686 S0000580826_V3

Onboard Network System Airplane Identification
Figure 204/46-13-00-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-00-710-801

9. Input Monitoring Check

(Figure 205)

A. General

(1) This task gives the steps to do a check of the analog and digital inputs of the Network File Server (NFS), M2898.

(2) This task gives the condition of the communication channels that follow, for the applicable unit:

- ETHERNET
- 429 RECEIVE
- 429 TRANSMIT
- INPUT DISCRETES
- OUTPUT DISCRETES
- QAR RECEIVE

(3) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

(4) If a system is installed and enabled but input monitoring shows the bus is unserviceable, maintenance action is necessary.

B. References

Reference	Title
31-31-22-860-802	Digital ACMS Recorder (DAR) 717 Activation (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Check

SUBTASK 46-13-00-860-014

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

F. ONS Input Monitoring Check- NFS

SUBTASK 46-13-00-860-015

(1) On the ONS Maintenance Browser:

(a) Select EXTENDED MAINTENANCE > Input Monitoring.

(b) Select NFS.

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1) Make sure that the result shows the correct LRU and that these buttons show on the bottom of the display:
   - ETHERNET
   - 429 RECEIVE
   - 429 TRANSMIT
   - INPUT DISCRETES
   - OUTPUT DISCRETES
   - QAR RECEIVE

SUBTASK 46-13-00-700-001

(2) Click once on the ETHERNET button.

(a) Examine the status for each port that is configured.

Ethernet Input Monitoring

PORT	END CONNECTION*[1]
NP1	
NP2	
NP3	FLIGHT DECK CREW PORT
NP4	
NP5	
1	RESERVED
2	MMR-L
3	MMR-R
4	AID-FWD
5	WAP-1
6	DFDAU
7	DPC-1
8	DPC-2
9	EEC-1A
10	EEC-1B
11	EEC-2A
12	EEC-2B
13	WWU
14	SDU-L
15	EGPWS
16	

*[1] If Unconfigured is displayed on a port, the airplane is not configured for this option.

NOTE: The END CONNECTION ports will only show systems that are configured for the airplane being checked. If the system is not monitored, UNCONFIGURED is displayed in the END CONNECTION column of input monitoring table.

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1) If the status shows Up, the NFS senses the ethernet interface is operational.
2) If the status shows Down, the NFS senses the ethernet interface is non-operational.
3) If the status shows Unavailable, the NFS cannot currently provide status for the interface.
4) If the status shows Admin Down, the interface is not configured.
5) Use the scroll bar to show more rows of data.

SUBTASK 46-13-00-700-002

(3) Click once on the 429 RECEIVE button.

(a) Examine the result for each bus that shows.

ARINC 429 Receive Port

BUS ID	DISPLAY NAME*[1]	NOTES
1	DATA LOAD IN #1	Operational and shows High during data load of target LRU
2	Unconfigured	
3	W-FMC-1 Relay FMC-02	Shows Low when FMC-1 is operational and transmitting on the ARINC 429 bus
4	W-FMC-1 Relay FMC-09	Shows High when FMC-1 is operational and transmitting on the ARINC 429 bus
5	W-ADIRU-L ADR-4	Shows Low when ADIRU is operational and transmitting on the ARINC 429 bus
6	W-ADIRU-L IR-3	Shows High when ADIRU is operational and transmitting on the ARINC 429 bus
7	W-DPC-1 MDS-GP 1	Shows High when DPC-1 is powered up and transmitting on the ARINC 429 bus
8	W-DPC-2 MDS-GP 1	Shows High when DPC-2 is powered up and transmitting on the ARINC 429 bus
9	W-CMU OUTPUT BUS 3	Shows Low when CMU is powered up and transmitting on the ARINC 429 bus
10	Unconfigured	
11	W-MMR-1 GNSS Data 3	Shows High when MMR-1 is powered up and transmitting on the ARINC 429 bus
12	W-ATC-1 General Output 1	Shows High when ATC-1 is powered up and transmitting on the ARINC 429 bus

*[1] If Unconfigured is displayed on a Bus, the airplane is not configured for this option.

1) If the status shows Low, the low speed ARINC 429 bus is operational.
2) If the status shows High, the high speed ARINC 429 bus is operational.
3) If the status shows Disabled, the bus is not configured.
4) If the status shows Failed, the NFS is not sensing ARINC 429 bus activity.
5) If the status shows Unavailable, the NFS cannot currently provide status for ARINC 429 bus.
6) Use the scroll bar to show more rows of data.

SUBTASK 46-13-00-700-003

(4) Click once on the 429 TRANSMIT button.

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(a) Examine the result for each bus that shows.

ARINC 429 Transmit Port

BUS ID	DISPLAY NAME*[1]	NOTES
1	DATA LOAD OUT	Operational and shows High during data load of target LRU
2	BITE	Shows Low when NFS is sending ARINC 429 data to the FMC 1 and FMC 2
3	Unconfigured	
4	CABIN TERMINAL 1 TO CMU	Shows Low when NFS is sending ARINC 429 data to the CMU
5	Unconfigured	
6	Unconfigured	

*[1] If Unconfigured is displayed on a Bus, the airplane is not configured for this option.

1) If the status shows Low, the low speed ARINC 429 bus is operational.
2) If the status shows High, the high speed ARINC 429 bus is operational.
3) If the status shows Disabled, the bus is not configured.
4) If the status shows Failed, the NFS is not sensing ARINC 429 bus activity.
5) If the status shows Unavailable, the NFS cannot currently provide status for ARINC 429 bus.
6) Use the scroll bar to show more rows of data.

SUBTASK 46-13-00-700-004

(5) Click once on the INPUT DISCRETES button.

(a) Examine the result for each input discrete that shows.

Input Discrete

DISCRETE ID	DISPLAY NAME*[1]	NOTES
1	Unconfigured	
2	Unconfigured	
3	Unconfigured	
4	Unconfigured	
5	Unconfigured	
6	W-Air/Ground Relay	Status is Ground when relay in set on-ground and Open when relay is in-air

*[1] If Unconfigured is displayed on a Discrete ID, the airplane is not configured for this option.

1) If the status shows Open, the discrete signal is sensed as open.
2) If the status shows Ground, the discrete signal is sensed as grounded.
3) If the status shows Unavailable, the discrete is unconfigured or status is not available.
4) Use the scroll bar to show more rows of data.

SUBTASK 46-13-00-700-005

(6) Click once on the OUTPUT DISCRETES button.

(a) Examine the result for each output discrete that shows.

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Output Discrete

DISCRETE ID	DISPLAY NAME*[1]	NOTES
1	Unconfigured	
2	WWU RF Enable	Ground when WWU RF is enabled. Open when WWU RF is not enabled
3	DL FUNC DISC 1	Grounded as needed to support data load to target LRU
4	DL FUNC DISC 2	Grounded as needed to support data load to target LRU
5	DL FUNC DISC 3	Grounded as needed to support data load to target LRU
6	WAP-1 Power Enable	Ground when WAP-1 power is enabled. Open when WAP-1 power is not enabled
7	WAP RF Enable	Ground when WAP RF is enabled. Open when WAP RF is not enabled
8	DL RELAY CONTROL	Ground when Data Load from NFS to target LRU in progress
9	WAP-2 Power Enable	Ground when WAP-2 power is enabled. Open when WAP-2 power is not enabled
10	Unconfigured	
11	Unconfigured	
12	Unconfigured	

*[1] If Unconfigured is displayed on a Discrete ID, the airplane is not configured for this option.

1) If the status shows Open, the discrete signal is sensed as open.
2) If the status shows Ground, the discrete signal is sensed as grounded.
3) If the status shows Unavailable, the discrete is unconfigured or status is not available.
4) Use the scroll bar to show more rows of data.

SUBTASK 46-13-00-860-039

(7) Click once on the QAR RECEIVE button.

(a) Examine the result for QAR receive bus that shows.

QAR Receive

BUS	END CONNECTION	NOTES
1	DAR	When engines are not running, status will show Inactive
2	RS422	When engines are not running, status will show Inactive

1) If the status shows Active, the interface is operational.
2) If the status shows Inactive, there is no interface activity or engines are OFF.
a) To set QAR Receive to Active, do this task: Digital ACMS Recorder (DAR) 717 Activation, TASK 31-31-22-860-802.

G. ONS Input Monitoring Check- Avionics Parameter

SUBTASK 46-13-00-860-040

(1) Select the Avionic Parameters radio button at the top of the page.

(a) Make sure that the result shows the correct parameters on the display:

- Airline ID

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- Aircraft Tail ID
- Aircraft Type
- ICAO Address
- Time
- Date
- Origin
- Destination
- Weight on Wheels

NOTE: Airline ID and Aircraft Tail ID are populated by the DPCs. If the DPCs are off or unavailable, the Airline ID and Aircraft Tail ID can be manually entered on the Airplane ID page. This is the task: Airplane Identification Check, TASK 46-13-00-750-801.

NOTE: ICAO address is populated by the ATC Transponder.

NOTE: Aircraft Type must be manually entered on the Airplane ID page. This is the task: Airplane Identification Check, TASK 46-13-00-750-801.

NOTE: Origin and Destination are populated via the FMC flight plan. If the airplane has not flown since NFS installation or reimage, a flight plan can be manually entered into the FMC and then loaded to the NFS.

NOTE: Time and Date are populated by the DPCs. If the DPCs are off or unavailable, time and date are populated by the MMRs. If the MMRs are not available, but the DPCs are available, Time and Date can be manually entered to the DPCs.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-005

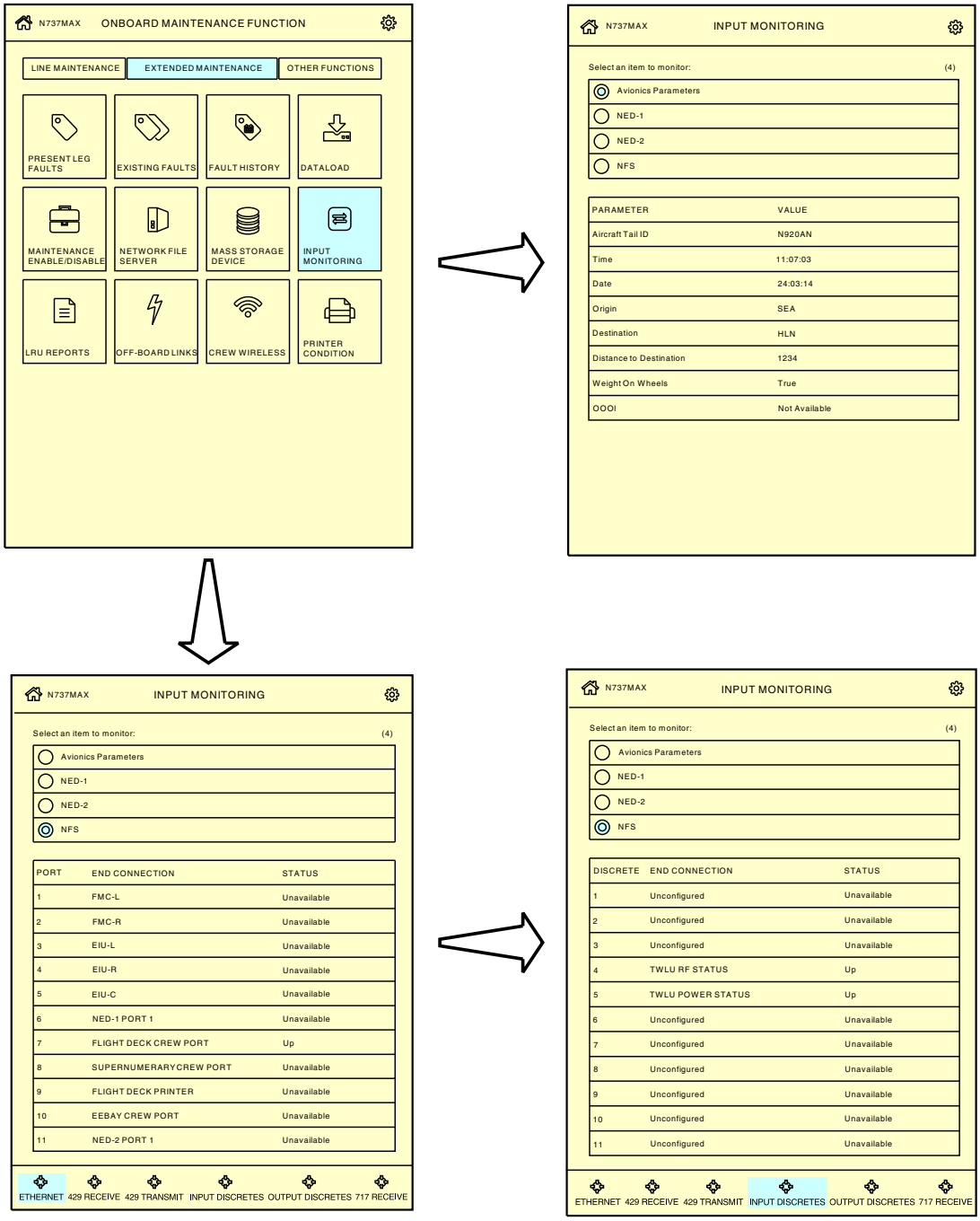
(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

THE DATA SHOWN HERE ON THE DISPLAY IS ONLY AN EXAMPLE.

2475683 S0000580676_V1

Onboard Network System Input Monitoring Check
Figure 205/46-13-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-00-710-802

10. System Configuration Check

(Figure 206)

A. General

(1) This task gives the serial number, the hardware part number, and software part number(s) for the applicable Line Replaceable Unit (LRU).

(2) This task can be used to determine the OS version of the Network File Server (NFS) by viewing the 46 NFS Boot OS software part number in the NFS system configuration. The following 46 NFS Boot OS software part numbers are tied to the following OS versions:

Boot OS Software Part Number	OS Version
BCG57-U000-02E3	OS7.0
BCG2A-U000-0458	OS7.1
BCG27-U000-0895	OS9.1
BCG24-U000-0CL8	OS9.1

(3) This procedure is applicable to the units that follow:

- Integrated Air System Controller (IASC)
- Spoiler Control Electronics (SCE)
- Display Processing Computer (DPC)
- Proximity Sensing Electronics Unit (PSEU)
- Tire/Brake Monitoring Unit (TBMU) (Optional)
- Multi-Mode Receiver (MMR)
- NFS
- Auxiliary Power Unit (APU) Electronic Control Unit (ECU)
- Electronic Engine Control (EEC)
- Wireless Access Point (WAP)
- Aircraft Interface Device (AID)

(4) See the software installation task for the specific LRU for additional steps and checks that may be required to perform this task.

(5) The aircraft must be on the ground to do this task.

(6) To do this task, more than ten minutes must have passed since an NFS power up was initiated.

(7) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

(8) The DATA REPORTS software parts are installed on the Mass Storage Device (MSD), but not dataloaded, so they will not show on a system configuration report. To do a check of the DATA REPORTS software parts, do this task: Mass Storage Device Software Parts Check, TASK 46-13-00-860-804. This task must be done with a UMD.

B. References

Reference	Title
46-13-02-700-805	Existing Faults Check (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. ONS Configuration Check

SUBTASK 46-13-00-860-016

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

(a) Select LINE MAINTENANCE > System Configuration.

(b) To see the system configuration report for a specific ATA, do the steps that follow:

1) Select an ATA system.

2) Select the CONTINUE button.

3) Make sure that the LRU(s) show for the ATA selected and the data shows hardware part number(s), hardware serial number(s), software location description(s), and software part numbers.

(c) To see the configuration of all ATA systems, do the steps that follow:

1) Select the ALL SYSTEMS button.

2) Make sure that all the LRUs show for all ATAs selected and the data shows hardware part numbers, hardware serial numbers, software location descriptions, and software part numbers.

NOTE: It can take approximately five minutes before the REPORT button is enabled (turns from blue to white).

(d) If it is necessary, make a record of the data for your target LRU(s)

1) To download a complete record of the data that shows, select REPORT, and follow the on-screen commands for the applicable destination.

2) Make sure that the data agrees with the requirement specified by your airline or operator.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-006

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

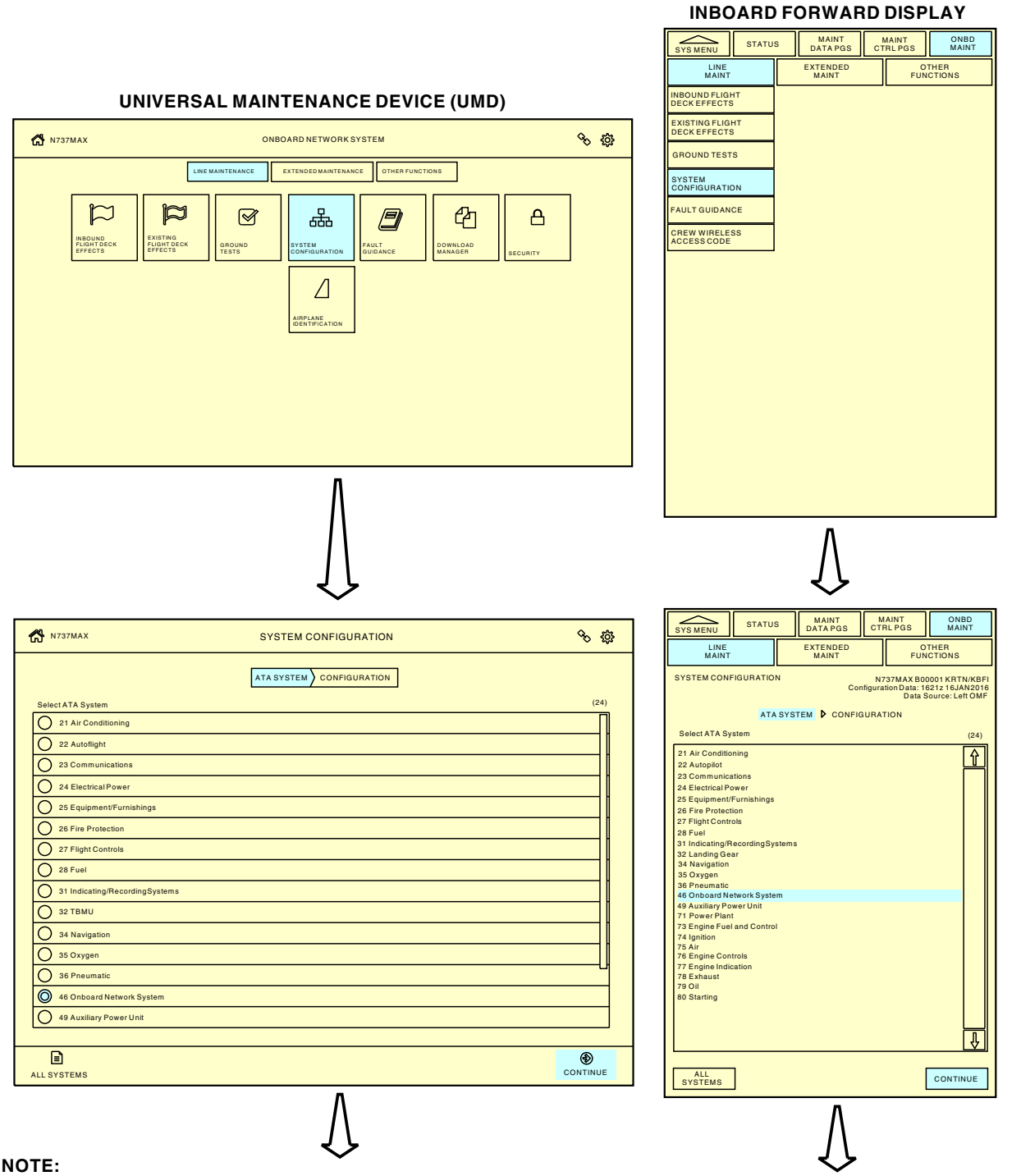
(a) If you find an active maintenance message, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

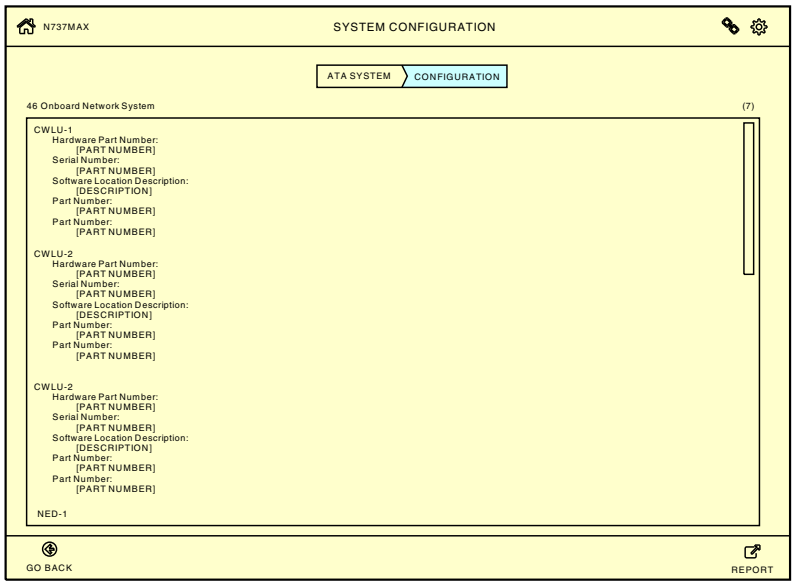
THE DATA SHOWN HERE ON THE DISPLAY IS ONLY AN EXAMPLE.

2476293 S0000580590_V3

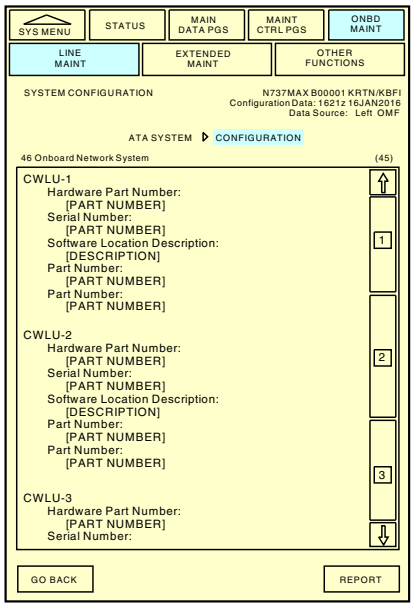
Onboard Network System (ONS) System Configuration Check
Figure 206/46-13-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

UNIVERSAL MAINTENANCE DEVICE (UMD)



INBOARD FORWARD DISPLAY



NOTE:

THE DATA SHOWN HERE ON THE DISPLAY IS ONLY AN EXAMPLE.

2476296 S0000580591_V3

Onboard Network System (ONS) System Configuration Check
Figure 206/46-13-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-00-470-801

11. LRU Software Parts Installation

(Figure 207)

A. General

(1) This procedure gives the steps to install software into Line Replaceable Unit (LRU) with the Onboard Network System (ONS).

(2) This procedure is applicable to the LRU that follow:

- Flight Control Computer (FCC)
- ACARS Communications Management Unit (CMU)
- SATCOM Satellite Data Unit (SDU)
- Forward Attendant Control Panel (ACP)
- Spoiler Control Electronics (SCE)
- Digital Flight Data Acquisition Unit (DFDAU)
- Airplane Condition And Monitoring System (ACMS)
- Display Processing Computer (DPC)
- Control Display Unit (CDU)
- Flight Management Computer (FMC)
- Enhanced Ground Proximity Warning Computer (EGPWC)
- Multimode Receiver (MMR)
- Auxiliary Power Unit Electronic Control Unit (APU ECU)
- Wireless Access Point (WAP)
- Network File Server (NFS)
- Electronic Engine Control (EEC)
- Aircraft Interface Device (AID)

(3) Confirm that the Loadable Software Airplane Part (LSAP) to be installed is saved on the NFS Mass Storage Device (MSD) (TASK 46-13-00-860-804).

(a) If the LSAP is not saved, install those parts to the NFS MSD (TASK 46-13-00-470-802).

(4) See the software installation task for your specific LRU for additional steps and checks that may be required to perform this task.

(5) The aircraft must be on the ground to do this task.

(6) This task is available when the ONS is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. OMF Data Load Operation - LRU Software Installation via the UMD

SUBTASK 46-13-00-470-030

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

SUBTASK 46-13-00-470-033

(2) Select EXTENDED MAINTENANCE > Data Load.

(a) Make sure that the DATA LOAD screen shows.

NOTE: On the DATA LOAD screen, look for these three tabs or function descriptions: LOAD QUEUE (which is highlighted), PRECONDITIONS, and STATUS.

(b) Select the applicable ATA and LRU.

NOTE: Both the ATA number and the applicable LRU are selected as one option in the left-hand column of the page. This column is labeled “Select LRU Destination.”

ONS Dataload - ATA and LRU Mapping

ATA	LRU
22	FCC-L, FCC-R
23	ACP*[1]
23	CMU
23	SDU-L
27	SCE-Lane-1, SCE-Lane-2, SCE-Lane-3
31	ACMS, DFDAU
	DPC-1, DPC-2
34	EGPWS
	FMC-L, FMC-R
	MCDU-L, MCDU-R
	MMR-L, MMR-R
46	NFS
	WAP-1, WAP-2
46	AID-FWD
49	APU-ECU
73	EEC-1A, EEC-1B, EEC-2A, EEC-2B

*[1] The AFT Attendant Control Panel (ACP) is also software loadable, but only when you use the ACP Crossload function to install software from the FWD ACP to the AFT ACP.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Find the correct software and select one or more parts to install by clicking once on the part(s).

NOTE: The correct software is located under the LOAD QUEUE tab on the right-hand column on the page. This column is labeled “Select Part Number.” When parts are selected, they will show up at the bottom of the page in the “Load Queue” box.

(d) Select the CONTINUE button at the bottom right of the screen.

(e) Make sure that the highlighted PRECONDITIONS tab shows.

(f) Follow the instructions that show on the PRECONDITIONS tab.

(g) Select CONTINUE at the bottom right of the screen.

(h) Make sure that the highlighted STATUS tab shows.

(i) Select START LOAD at the bottom of the screen.

NOTE: Do not pull the Network File Server circuit breaker while a dataload is in progress.

(j) If software is installed into the Rockwell CMU, do these steps after START LOAD is selected:

1) Select DATALOAD* LSK on the CDU.
2) Select PROCEED WITH DATALOAD LSK on the CDU.
3) Make sure that the message SCREEN WILL BLANK WHILE DATALOAD IS IN PROGESS appears on CDU, then the MENU shows without the ACARS prompt.

(k) Wait for the load progress indication on the START LOAD tab to show that the data transfer is complete.

NOTE: The percent for progress status comes from the LRU. The information that shows and the update frequency will be different for different system LRUs.

(l) Make sure that all software parts show Success in the Status field.

(m) Do the steps that follow to install a different part:

1) Select GO BACK at the bottom left of the screen and start again at this step above:
   - “Select the applicable ATA and LRU.”

(n) If software installation is complete, go to the next step.

NOTE: If AID Data-Load fails and prompts with a message "System Busy Generating", try another Data-Load.

NOTE: The LRUs listed in the Table 201 must reboot before you continue.

Table 201/46-13-00-993-801 Minimum Reboot Time for the LRUs

LRU	Minimum Reboot Time (after successful Data Load Message)
Aircraft Interface Device (AID)	9 minutes

F. OMF Data Load Operation - LRU Software Installation via the MFD

SUBTASK 46-13-00-470-040

(1) If not previously completed, check if the software is in the MSD (TASK 46-13-00-860-804). This task must be done with a UMD.

SUBTASK 46-13-00-480-019

(2) To access the ONS, do this task: Multi Function Display Connection, TASK 46-13-00-480-803.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 46-13-00-470-041

(3) Select EXTENDED MAINT > DATA LOAD.

SUBTASK 46-13-00-470-042

(4) Install the software into the LRU, these are the steps:

(a) Make sure that the DATA LOAD screen shows.

NOTE: On the DATA LOAD screen, look for these five tabs or function descriptions: DESTINATION (which is highlighted), LSP, SUMMARY, PRECONDITIONS and LOAD.

(b) On the DESTINATION function tab, select the applicable ATA system.

ONS Dataload - ATA System LRU Mapping

ATA System	LRU
22 - AUTO FLIGHT	FCC-L, FCC-R
23 - COMMUNICATIONS	ACP*[1]
23 - COMMUNICATIONS	CMU
23 - COMMUNICATIONS	SDU-L
27 - FLIGHT CONTROLS	SCE-Lane-1, SCE-Lane-2, SCE-Lane-3
31 - INDICATING/RECORDING SYSTEMS	ACMS, DFDAU
	DPC-1, DPC-2
34 - NAVIGATION	EGPWS
	FMC-L, FMC-R
	MCDU-L, MCDU-R
	MMR-L, MMR-R
46 - INFORMATION SYSTEMS	NFS
	WAP-1, WAP-2
46 - INFORMATION SYSTEMS	AID-FWD
49 - AIRBORNE AUXILIARY POWER	APU-ECU
73 - ELECTRONIC ENGINE CONTROL	EEC-1A, EEC-1B, EEC-2A, EEC-2B

*[1] The AFT Attendant Control Panel (ACP) is also software loadable, but only when the ACP Crossload function is used to install software from the FWD ACP to the AFT ACP.

(c) Select the applicable LRU.

(d) Select the CONTINUE button at the bottom right of the screen.

1) Make sure that the LSP (loadable software part) function tab is highlighted and that the necessary software part shows.

(e) On the LSP tab, select one or more software parts to install.

1) Click once to select the part(s).

(f) Select CONTINUE at the bottom right of the screen.

1) Make sure that SUMMARY is highlighted.

(g) On the SUMMARY tab, make sure that only the selected part to be installed shows.

(h) Select CONTINUE at the bottom of the screen.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the highlighted PRECONDITIONS tab shows.

(i) Follow the instructions that show on the PRECONDITIONS tab.

(j) Select CONTINUE at the bottom of the screen.

1) Make sure that the highlighted LOAD tab shows.

(k) Select START LOAD at the bottom of the screen.

(l) If software is installed into the Rockwell CMU, do these steps after START LOAD is selected:

1) Select DATALOAD* LSK on the CDU.

2) Select PROCEED WITH DATALOAD LSK on the CDU.

3) Make sure that the message SCREEN WILL BLANK WHILE DATALOAD IS IN PROGESS appears on CDU, then the MENU shows without the ACARS prompt.

(m) Wait for the load progress indication on the LOAD tab to show that the data transfer is complete.

NOTE: The percent for progress status comes from the LRU. The information that shows and the update frequency will be different for different system LRUs.

(n) Make sure that all software parts show Success in the Status field.

1) If necessary, clear the modal that lists the load result.

(o) If it is necessary, do the steps that follow to install a different part:

1) Start again at this step above:

- Select EXTENDED MAINT > DATA LOAD.

(p) If software installation is complete, go to the next step.

NOTE: The LRUs listed in the Table 202 must reboot before you continue.

Table 202/46-13-00-993-802 Minimum Reboot Time for the LRUs

LRU	Minimum Reboot Time (after successful Data Load Message)
AID	9 minutes

G. LRU Software Installation Test

SUBTASK 46-13-00-470-012

(1) To do a software configuration check, return to the software installation procedure for your applicable LRU.

NOTE: The software installation task for the applicable LRU will give you specific steps to do the software installation check.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-007

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems", do this task: Existing Faults Check, TASK 46-13-02-700-805.

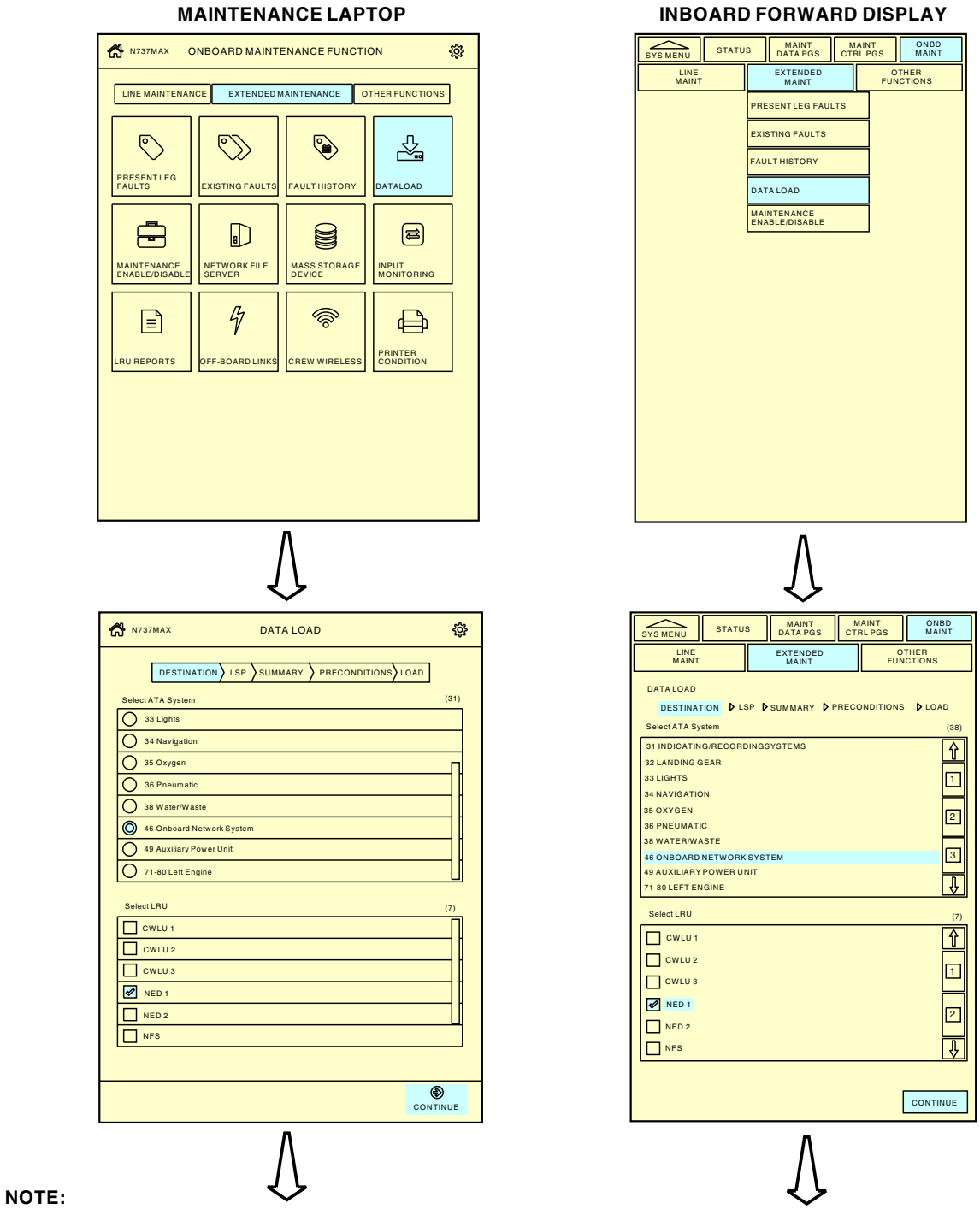
(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

SUBTASK 46-13-00-020-001

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

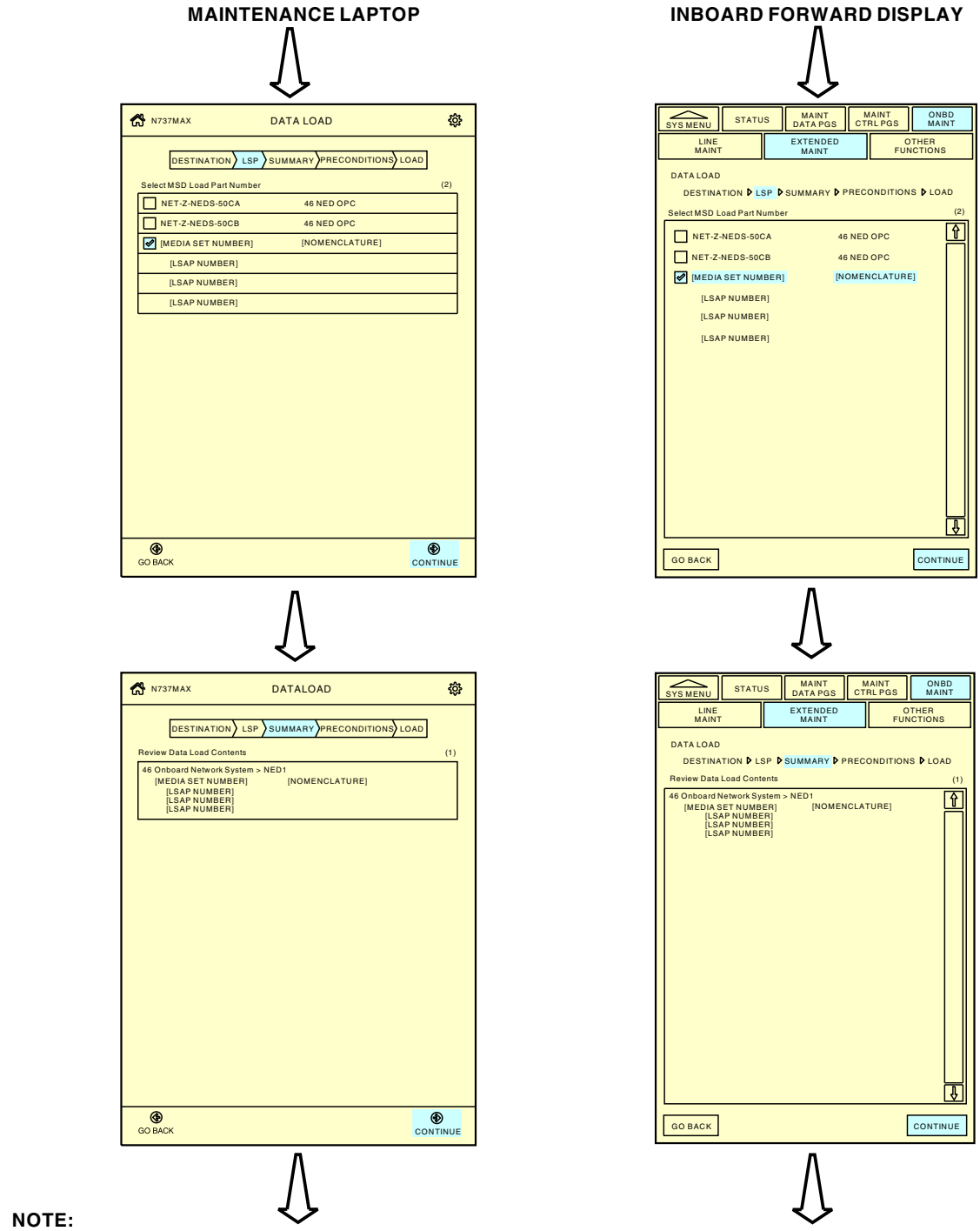


NOTE:
THE DATA ON THIS DISPLAY IS ONLY AN EXAMPLE.

2476933 S0000581167_V1

LRU Software Installation
Figure 207/46-13-00-990-807 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



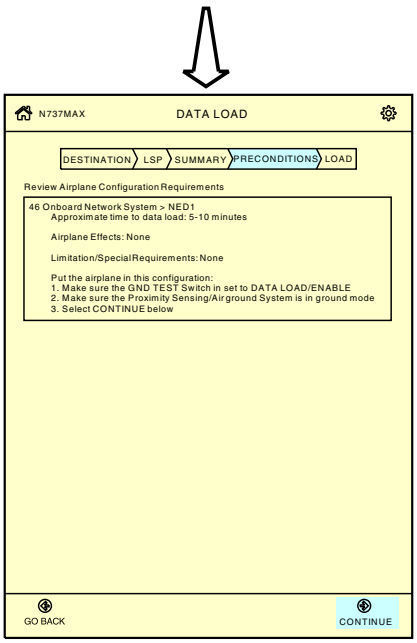
NOTE:
THE DATA ON THIS DISPLAY IS ONLY AN EXAMPLE.

2476935 S0000581170_V1

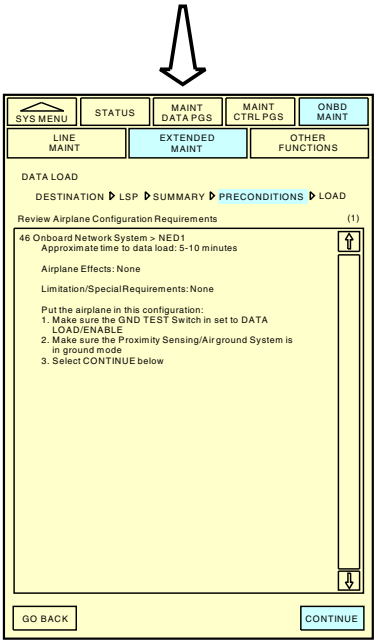
LRU Software Installation
Figure 207/46-13-00-990-807 (Sheet 2 of 4)

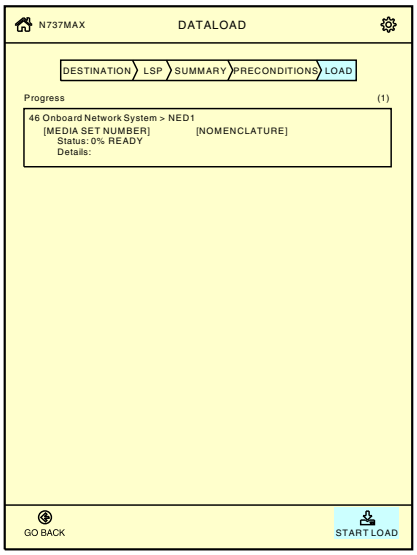
ECCN 9E991 BOEING PROPRIETARY - See title page for details

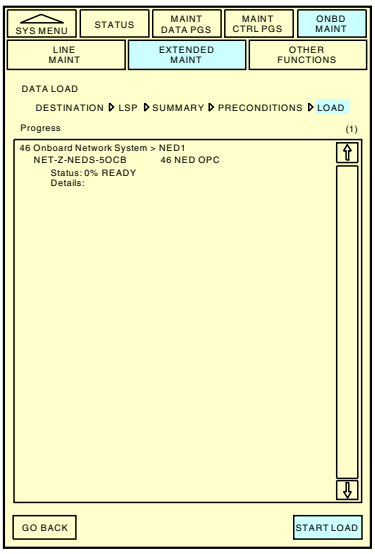
MAINTENANCE LAPTOP



INBOARD FORWARD DISPLAY







NOTE:

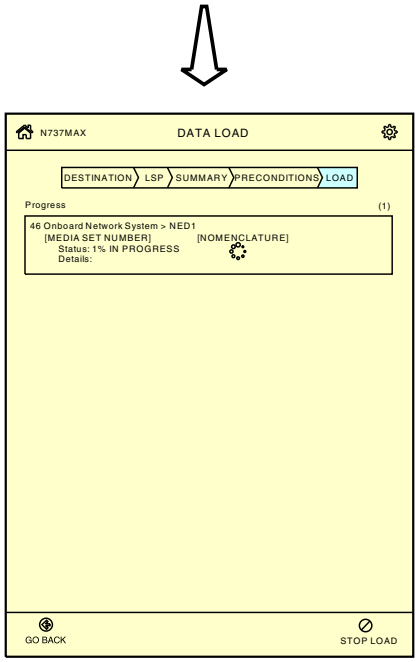
THE DATA ON THIS DISPLAY IS ONLY AN EXAMPLE.

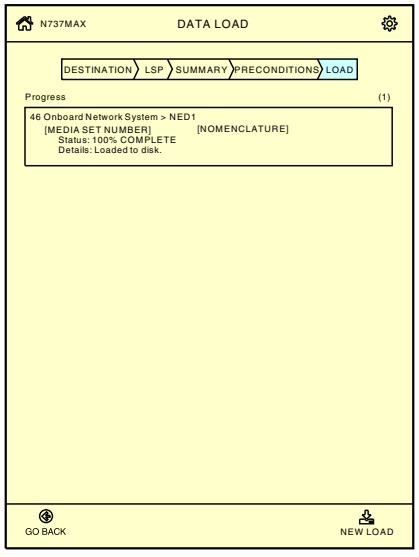
2476937 S0000581252_V1

LRU Software Installation
Figure 207/46-13-00-990-807 (Sheet 3 of 4)

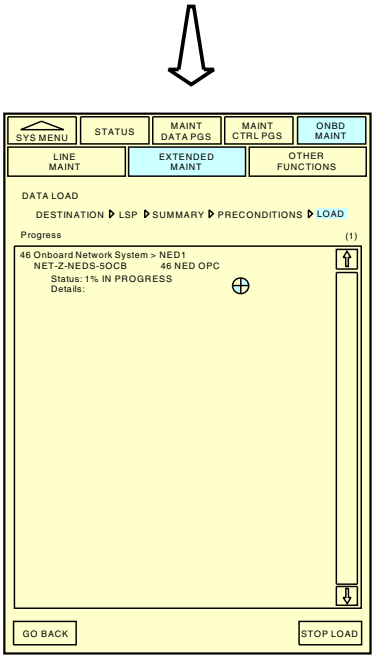
ECCN 9E991 BOEING PROPRIETARY - See title page for details

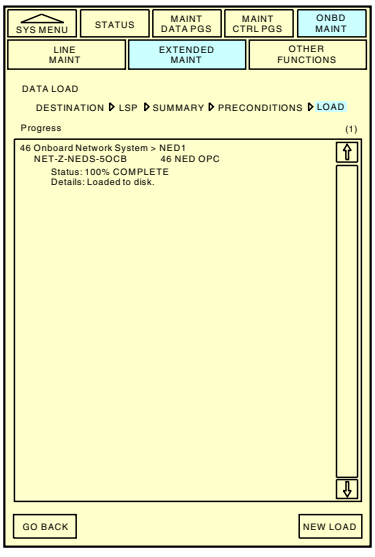
MAINTENANCE LAPTOP





INBOARD FORWARD DISPLAY





NOTE:

THE DATA ON THIS DISPLAY IS ONLY AN EXAMPLE.

2476941 S0000581253_V1

LRU Software Installation
Figure 207/46-13-00-990-807 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-00-860-804

12. Mass Storage Device Software Parts Check

(Figure 208)

A. General

(1) This task gives the steps to do a check of software parts on the Network File Server (NFS) Mass Storage Device (MSD).

(2) The aircraft must be on the ground to do this task.

(3) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Check

SUBTASK 46-13-00-860-022

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

F. Mass Storage Device Software Parts Check

SUBTASK 46-13-00-860-023

(1) For the Mass Storage Device Software Parts Check, do these steps:

(a) Select EXTENDED MAINTENANCE > Mass Storage Device.

(b) Select ONS MSD.

1) Examine the data that shows below the words: Select the Load Part Number.

2) Use the scroll bar to see the complete list of software parts.

3) Make sure that the software parts that show agree with the requirement specified by your airline or operator.

SUBTASK 46-13-00-070-002

(2) If it is necessary to remove software parts from the MSD, do this task: Mass Storage Device Software Parts Removal, TASK 46-13-00-070-801.

SUBTASK 46-13-00-470-013

(3) If it is necessary to stage software parts to the MSD, do this task: Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD), TASK 46-13-00-470-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 46-13-00-470-045

(4) If it is necessary to save a report, do these steps:

(a) Select REPORT.

(b) Select the radio button for Maintenance Device.

(c) Select SEND.

(d) Select the save location.

(e) Save the report.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-008

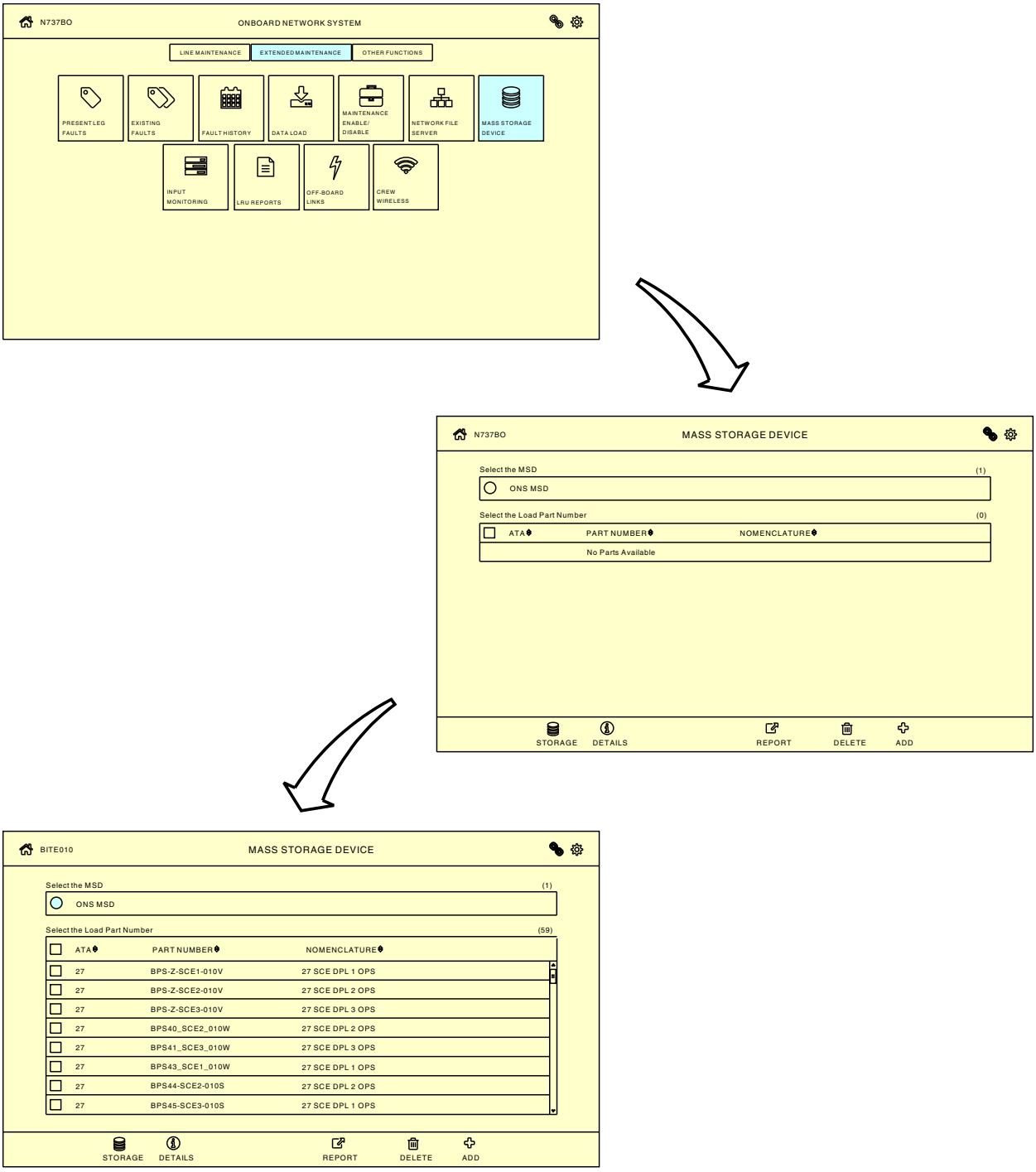
(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

2472980 S0000578915_V3

Mass Storage Device (MSD) Software Parts Check
Figure 208/46-13-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-00-070-801

13. Mass Storage Device Software Parts Removal

(Figure 209)

A. General

(1) This task gives the steps to remove one or more software parts from the Network File Server (NFS) Mass Storage Device (MSD).

(a) Removal prevents that software part from future installation to the applicable Line Replaceable Unit (LRU).

(b) Removal does not uninstall the software from the applicable LRU.

(2) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Software Part Removal

SUBTASK 46-13-00-860-024

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

F. MSD Software Part Removal

SUBTASK 46-13-00-860-025

(1) For the MSD Software Part Removal, do these steps:

(a) Select EXTENDED MAINTENANCE > Mass Storage Device.

(b) Select ONS MSD.

1) Examine the data that shows below Select the Load Part Number.

2) If necessary, use the scroll bar to see the complete record of software parts.

SUBTASK 46-13-00-070-003

(2) To select and to remove one or more parts, do these steps:

(a) Click on the part to put a check mark in the box adjacent to your selection.

1) To deselect, click again on the part.

2) The page will let you select one, more than one, or all of the parts.

(b) Click DELETE.

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(c) When the confirmation box shows, click CONTINUE.

(d) Make sure that the deleted software does not show in the list of software parts.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-009

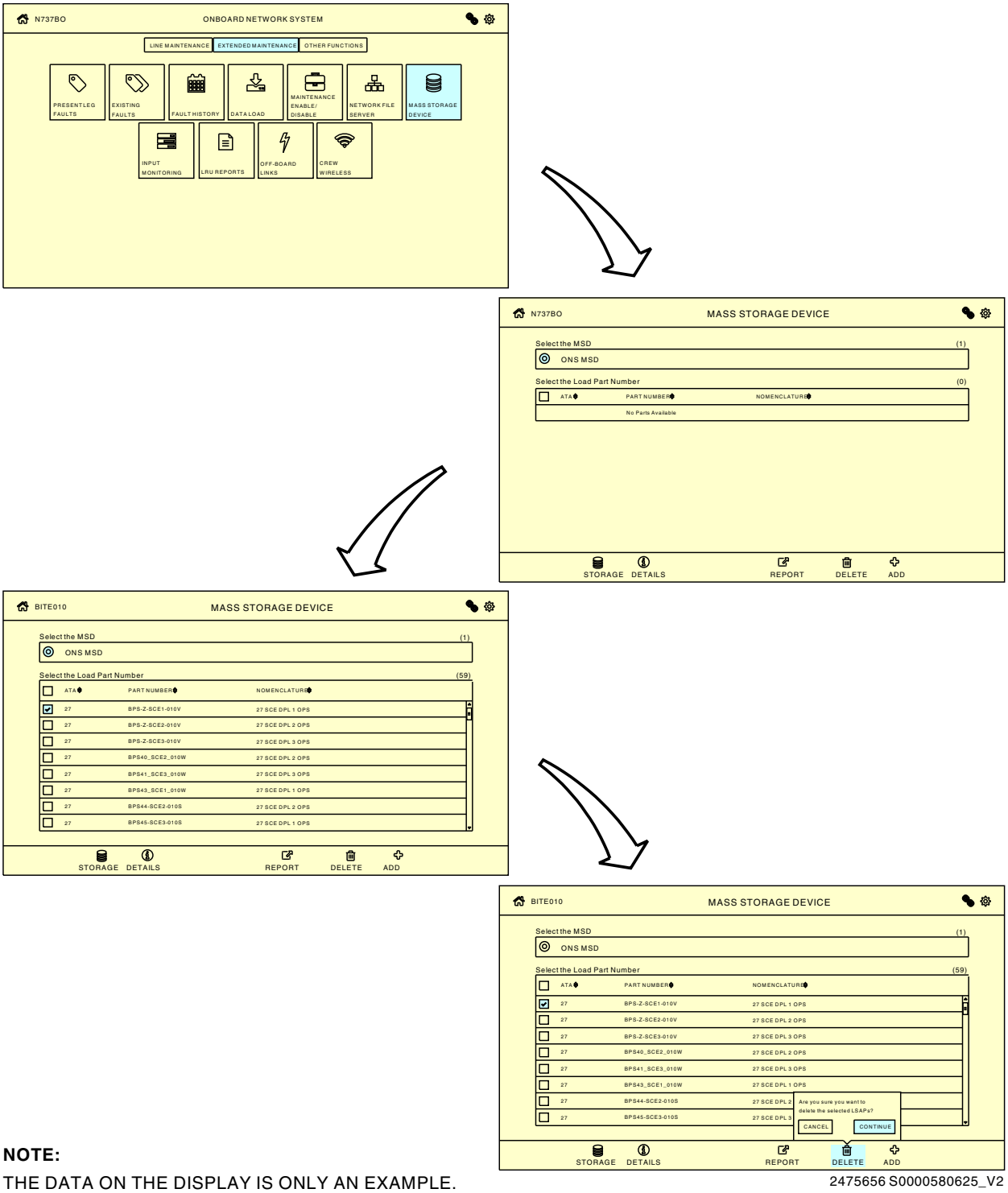
(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

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NOTE:

THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

2475656 S0000580625_V2

Mass Storage Device (MSD) Software Parts Removal
Figure 209/46-13-00-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-00-470-802

14. Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD)

(Figure 210)

A. General

(1) This task gives the steps to stage software parts from the Universal Maintenance Device (UMD) to the Network File Server (NFS) Mass Storage Device (MSD). These parts are for future installation into the applicable Line Replaceable Unit (LRU).

(a) Gather all the software parts you need on a removable media or UMD before going to the airplane.

(2) Typically, the software parts are compressed (or; crated, zipped) into the .zip format.

(3) To do this task the aircraft must be on the ground, with its weight compressing the landing gear.

(4) This task is available when the Onboard Network System (ONS) is accessed with the UMD.

NOTE: This task is not available when you get access to ONS maintenance with the MAX Display System (MDS) two forward Multi Function Display (MFD).

B. References

Reference	Title
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Software Installation

SUBTASK 46-13-00-860-026

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

(a) Use the universal maintenance device, SPL-11321, to get access to the ONS Maintenance browser main menu.

SUBTASK 46-13-00-470-014

(2) To confirm the software is not already installed on the MSD, refer to this task: Mass Storage Device Software Parts Check, TASK 46-13-00-860-804.

F. Mass Storage Device Software Part Installation

SUBTASK 46-13-00-470-015

(1) Put the source media into the UMD, if the software is not already on the UMD.

(a) Your source data can be a compact disk (CD), or a USB memory stick, or other method specified by your airline or operator.

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SUBTASK 46-13-00-470-016

(2) From the ONS Maintenance browser main menu, make your selection:

(a) EXTENDED MAINTENANCE > Mass Storage Device.

(b) From the Mass Storage Device menu, make the selection: ONS MSD.

1) Select the ADD (+ sign icon) button.

2) Make sure that the “Select a file with software parts to load” page shows.

SUBTASK 46-13-00-470-017

(3) Find, select, and upload a software part as follows:

(a) Select the BROWSE button.

(b) When the File Upload window shows, go to the location of the file you want to add to the MSD.

(c) Look for the applicable software File Name and Part Number in the upload window.

(d) Click once on that software part to select it.

1) Then, select OPEN in the file upload window.

NOTE: The software part should start uploading from the UMD.

NOTE: If a window appears that says “Not a valid OBEDS crate”, then refer to the IFIM and do the applicable procedure(s).

(e) When the software transfer operation is complete, make sure these indications show on the current MSD page:

1) The software part that you selected shows in the FILE NAME column.

2) The correct software part number shows in the PART NUMBER column.

3) The STATUS column shows the word SUCCESS.

NOTE: The File transfer status may progress from TRANSFERRED to DELIVERED before showing SUCCESS. Large files may take several minutes to complete before showing SUCCESS.

NOTE: If the operation is unsuccessful, the STATUS column shows the word Error. Refer to the IFIM and do the applicable procedure(s).

(f) To add another software part to the MSD, select the BROWSE icon and do the steps again.

G. Mass Storage Device Software Part Installation Test

SUBTASK 46-13-00-470-018

(1) Select the “Back to ONS MSD LSAPs” button in the top left of the screen and make sure the part(s) are listed.

NOTE: When a part is saved to the MSD, it is available for installation into a Data Loadable LRU. If you must now install this software into the target LRU, refer to the software installation task in the AMM for that specific LRU. This is the task: LRU Software Parts Installation, TASK 46-13-00-470-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-010

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

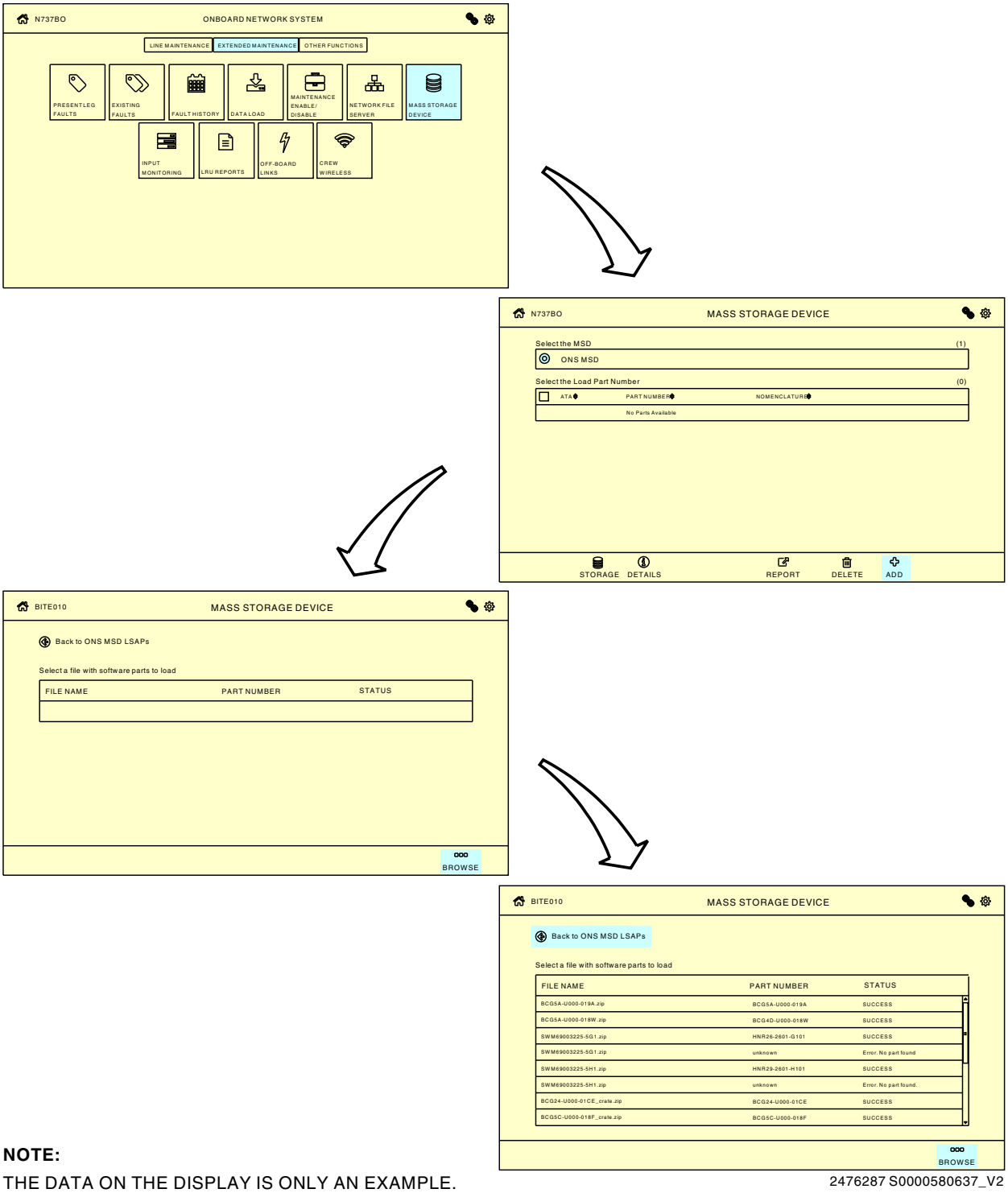
(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

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(2) If the UMD is no longer necessary, stop the ONS maintenance browser session, and disconnect maintenance device.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

2476287 S0000580637_V2

Mass Storage Device (MSD) Software Parts Installation
Figure 210/46-13-00-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-00-470-803

15. Download Files using the Software Management Tool (SMT)

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

B. General

SUBTASK 46-13-00-800-001

(1) This task instructs how to download a file that is stored in the NFS Mass Storage Device (MSD) to the Universal Maintenance Device (UMD).

(2) Do this task when you get access to the Onboard Network System (ONS) maintenance browser with the UMD.

NOTE: This task is not available when you get access to ONS maintenance with the MAX Display System (MDS) two forward Multi Function Display (MFD).

C. Prepare for the Procedure

SUBTASK 46-13-00-480-005

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

(a) Make sure that the universal maintenance device, SPL-11321 is connected to the airplane by a Wired or Wireless connection.

SUBTASK 46-13-00-860-032

(2) Start the Software Management Tool (SMT) on the universal maintenance device, SPL-11321.

(a) Make sure that the desktop of the universal maintenance device, SPL-11321 is shown.

(b) If the “LSAPL-SMT” icon shows on the desktop of the universal maintenance device, SPL-11321, double-click on the icon.

1) Make sure that the Software Management Tool window shows.

(c) If the “LSAPL-SMT” icon does not show, do these steps:

1) Click of the Start button on the desktop menu bar.

NOTE: The Start button is usually located at the bottom left of the desktop window.

2) Click on All Programs.

NOTE: The subsequent menu shows.

3) Click on LSAPL-SMT.

NOTE: The subsequent menu shows.

4) Click on LSAPL-SMT on the menu.

a) Make sure that the Software Management Tool window shows.

(d) Select “Get Data from Airplane” button on the SMT screen.

(e) If the “Unable to connect to the airplane...” message shows in the Status Messages section, do these steps:

1) Wait a minimum of two (2) minutes.

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2) Select the “Get Data from Airplane” button.

3) If the error message still shows, then refer to the IFIM and do the applicable procedure(s).

(f) Make sure the “Successfully connected to airplane...” message with the current date and time shows in the Status Messages section.

(g) Make sure the Airplane Tail ID is correct.

NOTE: The Airplane Tail ID is shown at the upper right corner of the screen as “Airplane: xxxxxx”.

(h) Select the “This SMT” button in the “Redirect Downlinks to” box.

D. Check the Files Downloaded to the Maintenance Laptop

SUBTASK 46-13-00-860-035

(1) Do these steps on the Software Management Tool (SMT) window:

(a) Verify the files in the “List of Airplane Data for XXXXXX” frame are correct.

NOTE: XXXXXX is the airplane Tail ID.

1) Make sure that the File Name, Airplane and Received information of the files are correct.

NOTE: The Location will show the location where the file is stored on the UMD.

2) Make sure that the download is completed.

NOTE: The Status will show SUCCESS when the download is completed.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-011

(1) Close the Onboard Storage Management (OSM) window.

(2) Close the Software Management Tool (SMT) window.

(3) If necessary, disconnect the universal maintenance device, SPL-11321 from the airplane.

(a) To disconnect the Wired Connection, do the following steps:

1) Disconnect the Ethernet cable from the airplane.

2) Disconnect the Ethernet cable from the universal maintenance device, SPL-11321.

(b) To disconnect the Wireless Connection, do the following steps:

1) Left-click on the Wireless Utility icon in the taskbar.

2) Left-click on the current connection.

3) Click the DISCONNECT button.

——— END OF TASK ———

TASK 46-13-00-470-804

16. Stage Software Parts on a Mass Storage Device via Ground Network or Software Management Tool

A. General

(1) Make sure to have all software parts on the Universal Maintenance Device (UMD) before going to the airplane.

(2) This task includes the steps to stage parts to the airplane's Mass Storage Device (MSD) from:

(a) Ground Network:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Loadable Software Airplane Part (LSAP) Librarian
2) LSAP Librarian Proxy Server

(b) Software Maintenance Tool (SMT)

(c) Other

(3) These are the steps to send part(s) to the airplane:

(a) Get the part(s) from the Ground Network, SMT, or other location and store them on the UMD.

(b) Send the part(s) from the UMD to the MSD on the airplane.

(c) Send the data report back to the Ground Network.

(4) A universal maintenance device, SPL-11321 is necessary for this task.

(5) The 46 NFS OAS AIRLINE CERT UMS is required to support staging software to the MSD.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for Loading

SUBTASK 46-13-00-470-020

(1) Start the Software Maintenance Tool (SMT) application on the universal maintenance device, SPL-11321.

(a) To start the SMT, do this task: Download Files using the Software Management Tool (SMT), TASK 46-13-00-470-803.

SUBTASK 46-13-00-470-021

(2) Make sure that the airplane time is correct.

NOTE: Time mismatch between the airplane and the universal maintenance device, SPL-11321 can prevent the staging of the LSAP.

E. Get Scheduled Parts onto the Maintenance Laptop

SUBTASK 46-13-00-480-006

(1) Make sure that the universal maintenance device, SPL-11321 is connected to the Ground Network through a ground-wired Ethernet connection.

SUBTASK 46-13-00-470-022

(2) To get scheduled parts, do these steps:

(a) Select the “Get Parts Scheduled” button on the SMT screen (Figure 211 (Sheet 1)).

(b) Make sure that the status shows “SUCCESS” for all parts.

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(c) Select the Finish button.

NOTE: After you select the Finish button, the “View Parts on SMT” module will show.

NOTE: If you selected the Ground Network location, SMT disconnects from the Ground Network.

F. Get Unscheduled Parts onto the Maintenance Laptop

SUBTASK 46-13-00-470-023

(1) To get parts, do these steps:

(a) If you want to search for parts from the Ground Network, make sure that the universal maintenance device, SPL-11321 is connected to the Ground Network through a ground-wired Ethernet connection.

(b) Select the “Get Parts Unscheduled” button on the SMT screen (Figure 211 (Sheet 2) and Figure 211 (Sheet 3)).

NOTE: The Get Parts Unscheduled screen will show Page 1 of 4.

(c) Select a location to search for part(s) in the lower half of the screen.

1) Select SMT or Ground Network or Other.

NOTE: The SMT option lets you search for parts on SMT residing on the UMD.

The Ground Network option lets you search for parts in LSAP Librarian or LSAP Librarian Proxy Server associated with the current SMT Location.

The Other option lets you to search for parts from other media.

NOTE: You can enter the full or partial of a part name in the “Enter part name” field to filter your search.

(d) Select the Next button.

NOTE: The subsequent screen will show Page 2 of 4.

1) If you select Other, find the part(s) on the directory.

(e) Make sure that the search results are shown.

(f) Select the correct parts in the Part Results table.

1) Select the box adjacent to the parts or click on the parts.

NOTE: You can select multiple parts.

(g) Make sure that a check mark shows in the box adjacent to the selected part(s).

(h) Select the Next button.

NOTE: The subsequent screen will show Page 3 of 4.

SUBTASK 46-13-00-470-024

(2) Select the Tail # and Destination to create uplink tasks for the part(s), do these steps:

(a) Make sure that the selected parts shown in the “Select Parts to Transfer” table.

(b) Select the airplane for the parts.

1) Click on the cell under the “Airplane” column for the part.

NOTE: Tail # should automatically fill in.

a) If the Tail # does not automatically fill in, do these steps:

<1> If a drop-down box shows, select the correct Airplane #.

<2> If a drop-down box does not show, enter the correct Airplane #.

2) Select the Tail # for all parts.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Select a destination system for each part.

NOTE: The destination system for the part should automatically fill in after the Airplane # is selected.

1) If the destination system for a part does not show, do these steps:

a) Click on the cell in the Destination System column for the part.

b) Select a destination system from the drop-down box.

c) Make sure that the destination system is correct.

(d) Select the Next button.

NOTE: The subsequent screen will show Page 4 of 4.

(e) Make sure that the uplink tasks in the Status Messages window are correct.

(f) Select the Finish button.

NOTE: After you select the Finish button, the “View Parts on SMT” module will show.

NOTE: If you selected the Ground Network location, SMT disconnects from the Ground Network.

G. Send Parts to the Airplane

SUBTASK 46-13-00-480-007

(1) Connect the universal maintenance device, SPL-11321 to the airplane. Do these steps:

(a) If the universal maintenance device, SPL-11321 is ground-wired connected to the Ground Network, disconnect the universal maintenance device, SPL-11321 from the Ground Network.

(b) To connect the UMD to the airplane, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

1) Make sure that the UMD is connected to the airplane by a Wired, Wireless Limited or Wireless Full connection.

NOTE: The Connection feature on the Maintenance Control Display Function (MCDF) banner shows how the UMD is connected to the airplane.

SUBTASK 46-13-00-470-025

(2) To send the parts to the airplane, do these steps:

(a) Select the “Send Parts to Airplane” button on the SMT screen (Figure 211 (Sheet 4)).

(b) If the “Unable to connect to the airplane...” message shows in the Status Messages section, do these steps:

1) Wait a minimum of two minutes.

2) Select the “Send Parts to Airplane” button.

(c) Make sure that the “Successfully connected to airplane...” message with the current date and time shows in the Status Messages section.

(d) Make sure that the airplane Tail ID is correct.

NOTE: The airplane Tail ID is shown at the upper right corner of the screen as “Tail #: xxxxx”.

(e) Select the part(s) to send to the airplane from the “Select Part(s) for Transfer to the Airplane” section:

1) Click on the box adjacent to a part or click on a row to select.

NOTE: You can select multiple parts.

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To select all parts, click on the box adjacent to Select All.

(f) Select the Next button.

NOTE: The subsequent screen will show Page 2 of 2.

NOTE: Parts will start uplinking. The Status column shows the progress of the uplink.

(g) Make sure that all uplinks are success.

NOTE: The word “SUCCESS” will show in the Status for each successfully uplinked part.

NOTE: If software parts are not transferred after 60 minutes, the software parts will be deleted and the status will show a timed out error code.

(h) For CSS, to verify parts were delivered, do these steps:

NOTE: For certain CSS fault conditions, the status will not show. Do these steps to make sure that software parts are transferred to the CSS MSD even if the SMT status does not show.

1) Go to a Cabin Attendant Panel (CAP).
2) Select the CABIN MAINTENANCE button.
3) Select the CABIN DATA LOAD button.
4) Select the REGISTRY SET-UP button.
5) Make sure that the CSS parts show.

NOTE: The amount of time it takes for the parts to show in the Registry Set-up screen depends on how many CSS parts are being transferred and how many non-CSS parts are being transferred to other systems. If only CSS parts are being transferred, it can take up to 20 minutes for a full set of CSS parts.

(i) Select the Finish button.

NOTE: After you select the Finish button, the “View Parts on SMT” module will show.

SUBTASK 46-13-00-080-012

(3) Disconnect the universal maintenance device, SPL-11321 from the airplane.

H. Send Data to Ground Network

SUBTASK 46-13-00-480-008

(1) Connect the universal maintenance device, SPL-11321 to the ground network through a ground-wired Ethernet connection.

SUBTASK 46-13-00-470-026

(2) Select the “Send Data to Ground Network” button on the SMT window (Figure 211 (Sheet 5)).

NOTE: When “Send Data to Ground Network” button is selected, SMT automatically connects to the LSAP Librarian or LSAP Librarian Proxy Server. When connected, SMT automatically returns successfully uplinked tasks, failed uplink tasks, downlinked files, and event logs.

NOTE: The Status Messages section will show the status.

(3) Select the Finish button.

NOTE: After you select the Finish button, the “View Parts on SMT” module will show.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-00-080-013

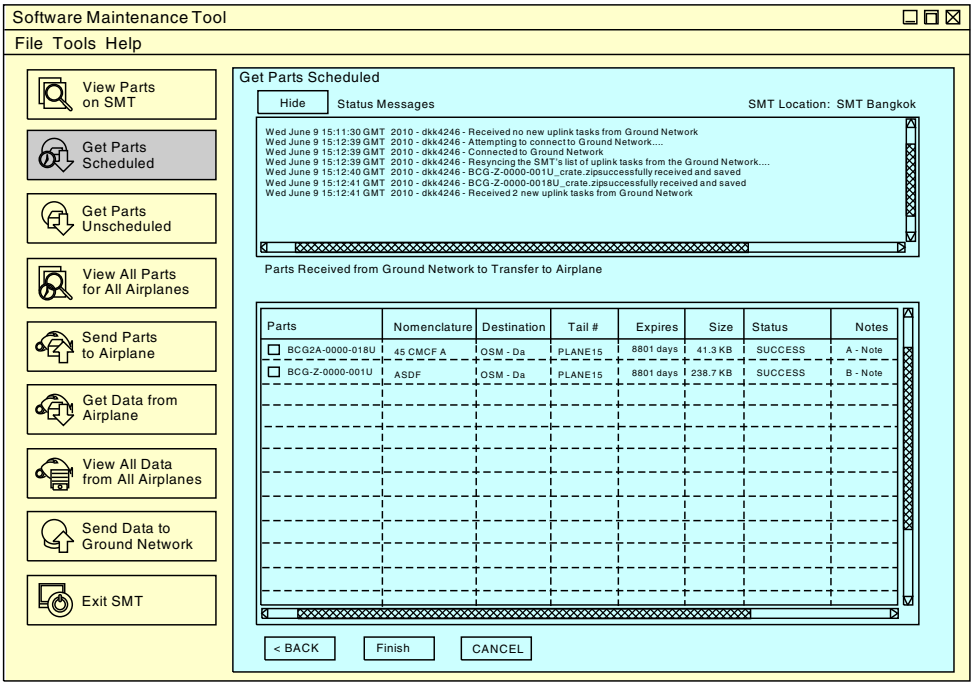
(1) If necessary, close the Software Maintenance Tool.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Select Exit SMT on the SMT screen.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



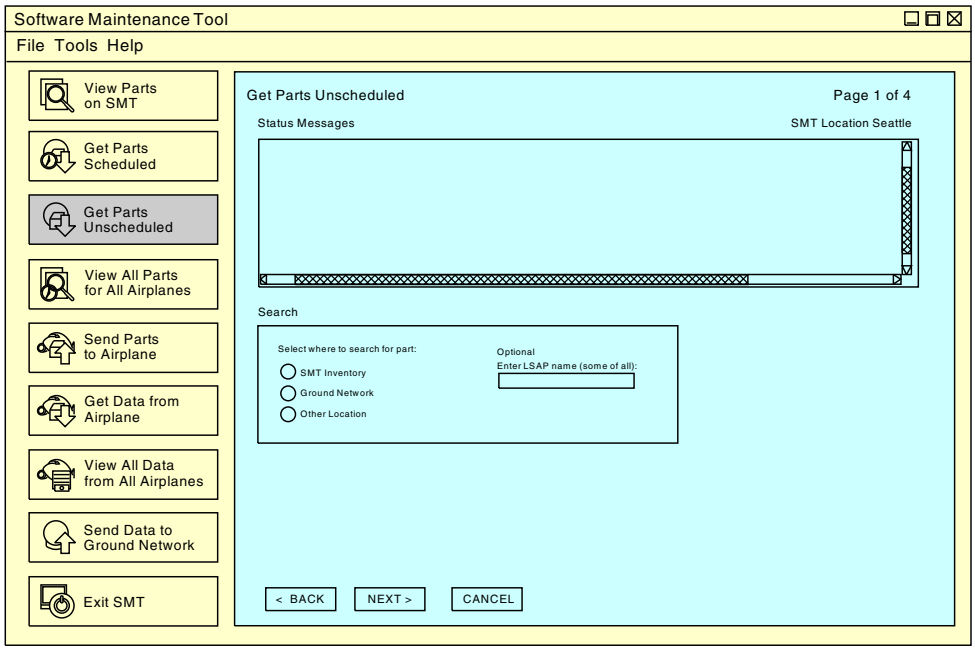
NOTE:

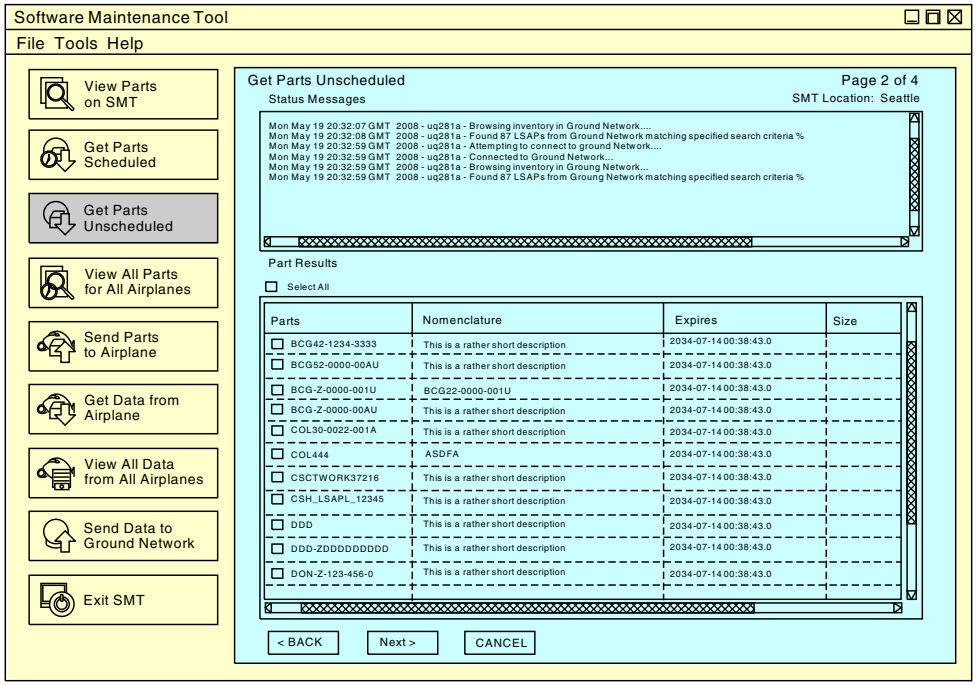
THE DATA SHOWN ON THE DISPLAYS
IS ONLY AN EXAMPLE.

2608068 S0000625870_V2

Software Maintenance Tool
Figure 211/46-13-00-990-811 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





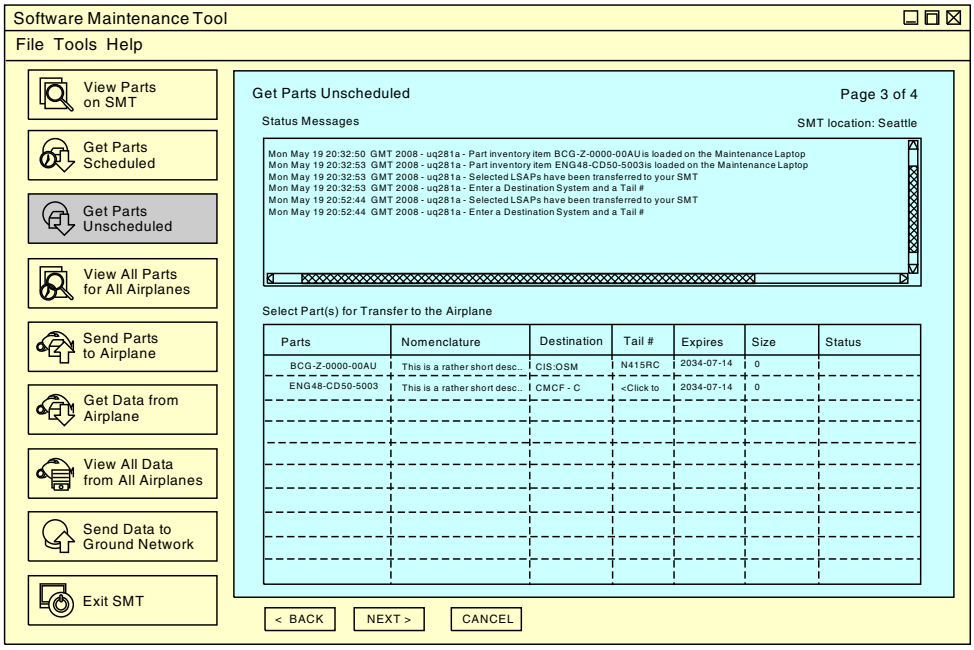
NOTE:

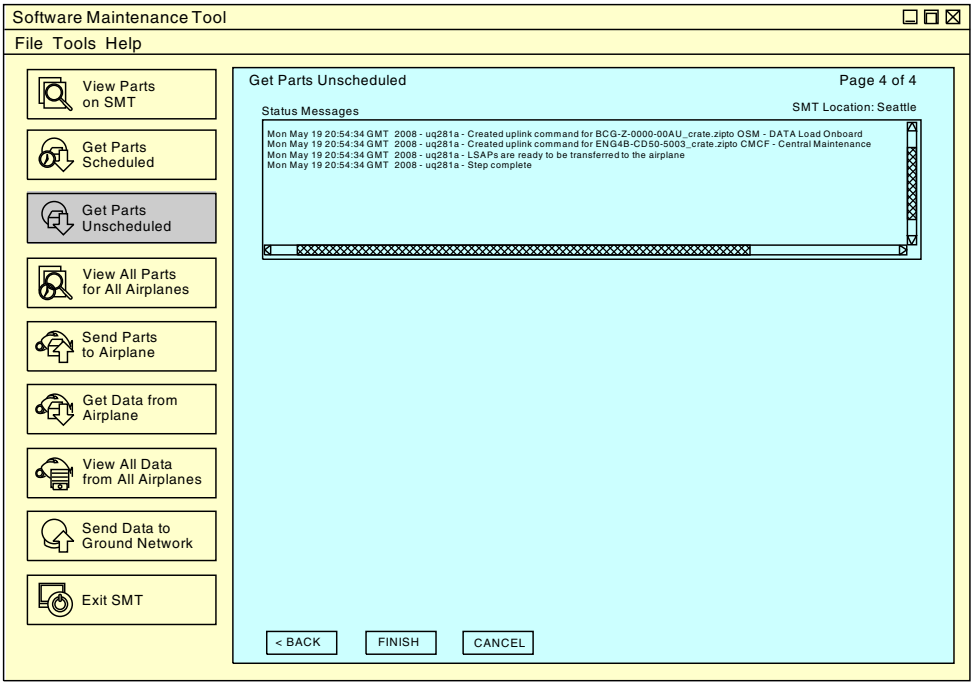
THE DATA SHOWN ON THE DISPLAYS
IS ONLY AN EXAMPLE.

2608069 S0000625871_V1

Software Maintenance Tool
Figure 211/46-13-00-990-811 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





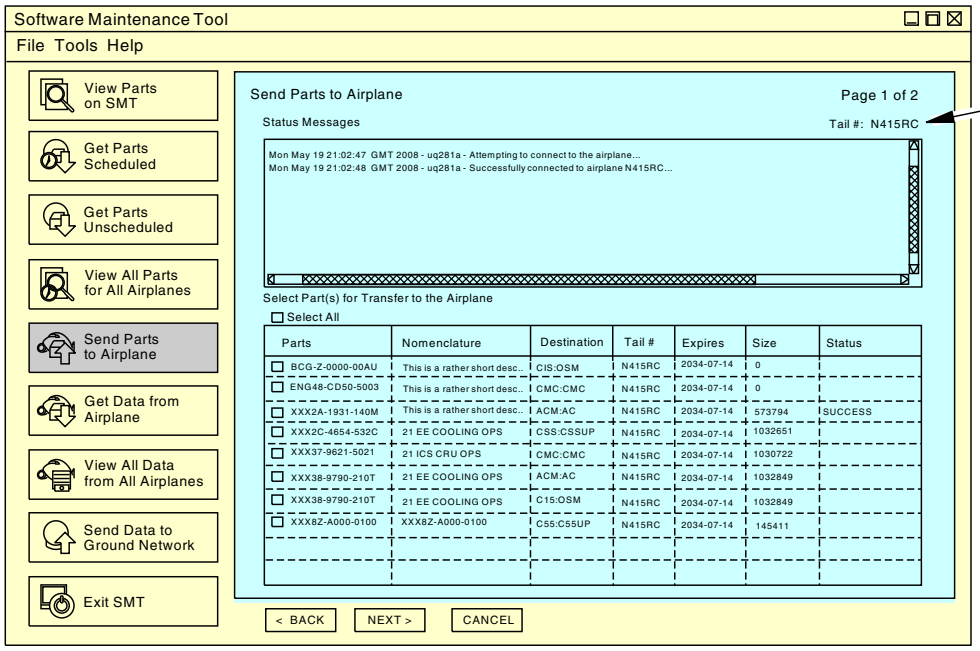
NOTE:

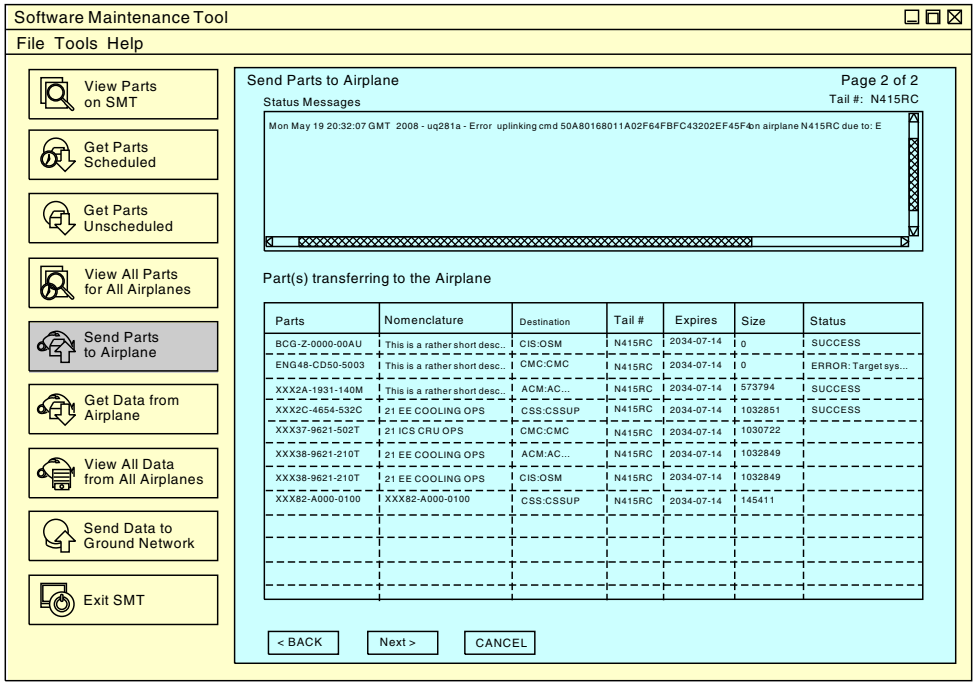
THE DATA SHOWN ON THE DISPLAYS
IS ONLY AN EXAMPLE.

2608070 S0000625872_V1

Software Maintenance Tool
Figure 211/46-13-00-990-811 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





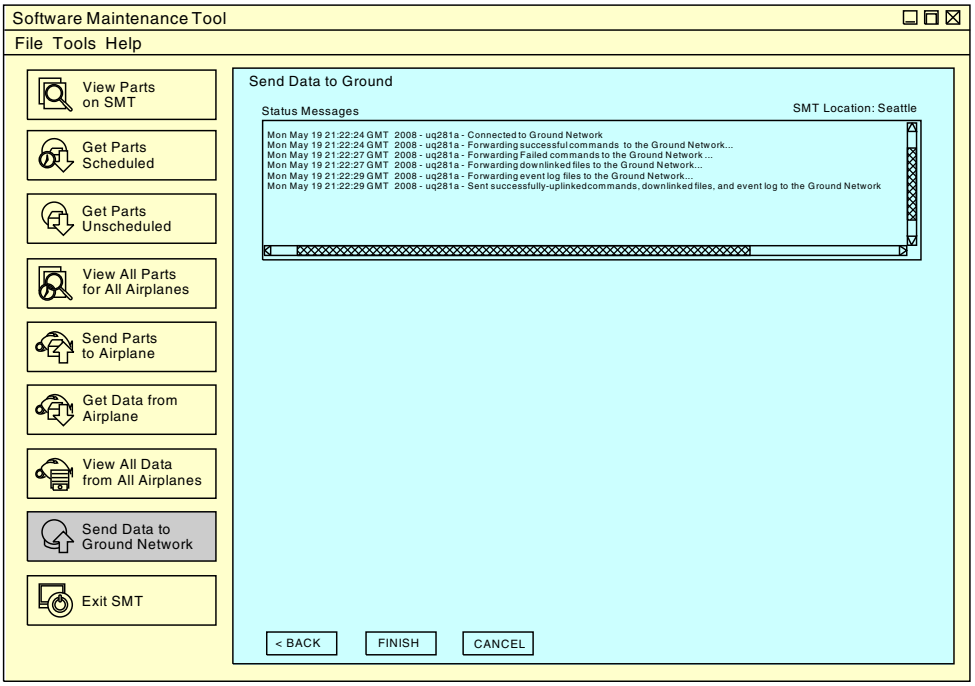
NOTE:

THE DATA SHOWN ON THE DISPLAYS
IS ONLY AN EXAMPLE.

2608072 S0000625873_V1

Software Maintenance Tool
Figure 211/46-13-00-990-811 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

THE DATA SHOWN ON THE DISPLAYS
IS ONLY AN EXAMPLE.

2608073 S0000625874_V1

Software Maintenance Tool
Figure 211/46-13-00-990-811 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NETWORK FILE SERVER - MAINTENANCE PRACTICES

1. General

A. This procedure gives you the tasks to do maintenance to the Network File Server (NFS), M02898. These tasks are as follows:

(1) A software configuration check, to examine the operational software parts installed in the NFS.

(2) A software installation task, to install software that operates the NFS.

(3) A task that deletes software parts from the NFS.

(4) A task that reimages the NFS.

(5) A task to install the minimum operating system software parts to the NFS.

(6) A task to reboot the NFS.

(7) A task to reset NFS data.

(8) A task to remove the NFS SD media card.

(9) A task to install the NFS SD media card.

(10) A task to manually initialize the NFS SD media card.

(11) A task to check the status of the NFS SD media card.

(12) A task to manually backup the NFS to the NFS SD media card.

(13) A task to use automatic NFS Stage 1 restore function.

(14) A task to use manual NFS Stage 1 restore function.

(15) A task to use NFS Stage 2 restore function.

(16) A task to upgrade NFS software OS 7.0 to OS 7.1.

(17) A task to upgrade NFS software OS 7.0 to OS 9.1.

(18) A task to upgrade NFS software OS 7.1 to OS 9.1.

(19) A task to downgrade NFS software OS 9.1 to OS 7.1.

(20) A task to downgrade NFS software OS 9.1 to OS 7.0.

(21) A task to downgrade NFS software OS 7.1 to OS 7.0.

(22) A task to check for the NFS issues with NFS Issue Troubleshooter and Recommended Operations (NITRO) application.

(23) A task to check date and time for the NFS.

(24) A task to cycle power the NFS.

TASK 46-13-01-860-801

2. Network File Server Software Check

A. General

(1) This task gives the steps to examine the software installed in the Network File Server (NFS).

(2) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

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B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Procedure

SUBTASK 46-13-01-970-001

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

SUBTASK 46-13-01-970-009

(2) Make sure that the correct software part numbers for the NFS are known.

F. Network File Server Software Check

SUBTASK 46-13-01-860-001

(1) Do a check of the NFS operational software (TASK 46-13-00-710-802).

NOTE: The NFS operational software is listed under ATA 46.

(a) Make sure that all software parts installed on the NFS agree with the operator requirement.

1) If the NFS NSP OPC and/or NSP OPS software part(s) show “Not Available”, then refer to the IFIM and do the applicable procedure(s).

(b) Save a report if necessary using the REPORT button located in the bottom right-hand corner of the page that shows the system configuration.

G. Put the Airplane Back To Its Usual Condition

SUBTASK 46-13-01-080-001

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

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(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-470-801

3. Network File Server Software Installation

A. General

(1) This task gives the steps to install software into the Network File Server (NFS) from the Mass Storage Device (MSD).

(a) If the software is not already staged to the MSD, gather all the software parts you need on removable media or Universal Maintenance Device (UMD) before leaving for the airplane.

(b) If the software is not already staged to the MSD, a UMD is required to stage the software before installation.

(2) The aircraft must be on the ground to do this task.

(3) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved UMD.

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-00-860-802	Server Credentials Check (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

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E. Prepare for NFS Software Installation

SUBTASK 46-13-01-470-035

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

F. NFS Software Installation - Procedure

SUBTASK 46-13-01-470-004

(1) Install the new NFS software part, or parts (TASK 46-13-00-470-801).

NOTE: During software installation to the NFS, the NFS will install the parts, shut down, and reboot.

(a) Wait approximately ten minutes for the NFS to reboot after the software installation.

G. Software Installation Check

SUBTASK 46-13-01-860-003

(1) Do a check of the NFS software (TASK 46-13-00-710-802).

SUBTASK 46-13-01-710-002

(2) Do this task: Server Credentials Check, TASK 46-13-00-860-802.

NOTE: Do not generate server credentials if a valid server credentials already exists.

SUBTASK 46-13-01-710-001

(3) Do this task: Client Credentials Check, TASK 46-13-00-860-803.

NOTE: Do not generate any client credentials if all LRUs already have valid client credentials.

SUBTASK 46-13-01-810-001

(4) Do this task for ATA 46: Existing Faults Check, TASK 46-13-02-700-805.

(a) If a fault is shown, then do the required maintenance action.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-002

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-070-801

4. Network File Server Software Removal

A. General

(1) This task gives the steps to uninstall loadable software airplane parts (LSAPs) from the Network File Server (NFS).

NOTE: This task does not remove the LSAPs from the NFS Mass Storage Device (MSD).

(2) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

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B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for NFS Software Removal

SUBTASK 46-13-01-970-002

(1) Make sure that a list of NFS software parts to be removed is available from the engineering department.

SUBTASK 46-13-01-070-001

(2) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

F. NFS Software Removal Procedure

SUBTASK 46-13-01-860-004

(1) To do the NFS Software Removal Procedure, do these steps:

(a) Select EXTENDED MAINTENANCE > Network File Server.

(b) Select the UNINSTALL LSAPS tab on the Network File Server menu.

(c) In the column labeled “Select the Load Part Number”, do this step:

1) Select the software parts to remove and make sure that a green check mark shows in the box adjacent to the LSAP.

(d) Select the DELETE icon.

(e) Select the CONTINUE button.

NOTE: The UNINSTALLING LSAPS progress indicator may show and turn until the operation is complete.

NOTE: When the removal operation is complete, the NFS reboot operation will automatically start.

NOTE: The broken chain link icon will show in the upper right corner during the reboot operation. After the NFS reboots, the unbroken chain link icon will show when ONS communication is serviceable.

(f) Wait approximately ten minutes for the NFS to reboot.

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(g) After the NFS reboot is complete, make sure that the removed parts do not show on the UNINSTALL LSAPs tab.

SUBTASK 46-13-01-860-005

(2) View the ATA 46 system configuration to see the NFS operational software (TASK 46-13-00-710-802).

(a) Make sure that none of the removed software parts are installed on the NFS.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-860-006

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-860-802

5. Network File Server Reimage Procedure

A. General

(1) This task gives the instructions to reimage the Network File Server (NFS). The reimage operation erases all software parts and data, except the Boot OS. It is then necessary to install the minimum operating system software parts.

NOTE: Do not stage software parts on the Mass Storage Device (MSD) before the reimage. The reimage operation deletes all files on the MSD. If necessary, make a record of the software parts on the MSD before the NFS reimage.

(2) Do this task only when instructed by your engineering department.

(3) To do this task, the aircraft must be on the ground with its weight compressing the landing gear.

(4) If a manual Initial Data Load (IDL) is necessary after the reimage, remove the SD card from the NFS before the reimage.

(a) To remove the SD card, do this task: Network File Server SD Media Card Removal, TASK 46-13-01-020-801.

(5) If an automatic stage 1 restore is necessary after the reimage, make sure that an SD card is installed to the NFS with a valid backup before the reimage.

(a) To install an SD card, do this task: Network File Server SD Media Card Installation, TASK 46-13-01-420-801.

(b) To backup the SD card, do this task: Network File Server Backup to SD Media Card, TASK 46-13-01-470-803.

(6) This task is available only when the Onboard Network System (ONS) maintenance browser is accessed with the Universal Maintenance Device (UMD).

(7) Gather all the software parts you need on a removable media or UMD before leaving for the airplane.

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B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-02-970-802	Download Manager (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for NFS Reimage

SUBTASK 46-13-01-860-007

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: Do not attempt to connect wirelessly.

(a) Use a universal maintenance device, SPL-11321 to get access to the ONS Maintenance browser.

SUBTASK 46-13-01-970-003

(2) Make sure that any necessary software parts are available on the UMD.

SUBTASK 46-13-01-970-004

(3) If the NFS is serviceable, do a check with your engineering department to download any necessary files that show in the Download Manager (Download Manager, TASK 46-13-02-970-802).

NOTE: Follow your airline's requirement for record keeping of NFS files. The steps that follow will erase all data from the NFS.

F. NFS Reimage Procedure

SUBTASK 46-13-01-800-001

(1) Do the reimage procedure as follows:

(a) Select EXTENDED MAINTENANCE > Network File Server.

(b) From the NETWORK FILE SERVER maintenance menu, select the REIMAGE tab.

(c) Make sure that the REIMAGE Function and Description shows.

(d) Read and understand the REIMAGE Function block and the Description block and then, follow all instructions.

(e) Click on the REIMAGE icon.

(f) Click on the CONTINUE icon when the confirmation pop up window shows.

(g) Wait for the reimage procedure to complete.

NOTE: The reimage procedure takes approximately five minutes. The NFS will reboot to the Initial Data Load (IDL) screen when the reimage procedure is complete.

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(h) Click on the OK icon when the REIMAGE NFS pop up window shows.

(i) If an SD card was not installed, do this task: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

(j) If an SD card was installed, do this task: Network File Server Automatic Stage 1 Restore Function, TASK 46-13-01-470-804.

——— END OF TASK ———

TASK 46-13-01-470-807

6. Network File Server Initial Data Load (IDL) Software Installation

A. General

(1) This task gives the instructions to do the Initial Data Load (IDL) software installation for the Network File Server (NFS). The IDL installs the minimum operating system and configuration software parts.

(a) Gather all the software parts you need on a removable media or Universal Maintenance Device (UMD) before leaving for the airplane.

(b) The software parts must be compressed in .zip format.

(2) Do this task:

(a) On a new or replacement NFS Line Replaceable Unit (LRU) if only a 46 NFS BOOT OS is installed.

(b) Immediately after the NFS Reimage procedure.

(3) Use the IDL page to import the specified parts, one at a time, and then install the parts as a group.

(4) These main software parts will be installed to get the NFS operational:

- 46 NFS SERVER OS
- 46 NFS OPC
- 46 NFS MEF OPS
- 46 NFS NETMANAGER APP
- 46 NSF OAS AIRLINE CERTS UMS
- 46 NFS OAS AIRLINE CRL UMS.

NOTE: These main software parts must be uploaded in the order listed on the IDL page to activate the DATA LOAD button at the bottom right of the IDL page.

(5) To do this task, the aircraft must be on the ground with its weight compressing the landing gear.

(6) This task is available only when the NFS INITIAL DATA LOAD page is accessed with the UMD.

(7) An SD card must not be installed to complete this task. If an SD card with a valid backup is installed at any point while on the NFS INITIAL DATA LOAD screen, do this task: Network File Server Automatic Stage 1 Restore Function, TASK 46-13-01-470-804.

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)
46-13-00-470-802	Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD) (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)

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(Continued)

Reference	Title
46-13-00-860-801	Airplane Credentials Check (P/B 201)
46-13-00-860-802	Server Credentials Check (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)
46-13-02-970-802	Download Manager (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for NFS Initial Data Load

SUBTASK 46-13-01-860-025

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

(a) Use the universal maintenance device, SPL-11321, to get access to the Onboard Network System (ONS) Maintenance browser.

F. NFS Initial Data Load

SUBTASK 46-13-01-470-024

(1) Do the IDL as follows:

(a) Make sure that the NFS has been powered up for at least 3 minutes.

(b) Double-click on the ONS maintenance browser desktop icon.

NOTE: If the ONS maintenance application fails to connect, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(c) Make sure that the IDL screen appears.

1) If the REIMAGE NFS dialogue window shows, select OK to close it.

(d) Make sure that the IDL screen shows the list of software parts to be imported in an amber color under the heading “Select a file with parts to import”.

(e) Make sure that the Boot OS part number shown in the Configuration section of the IDL page matches that given by your engineering department.

- Boot OS 7.0: BCG57-U000-02E3
- Boot OS 7.1: BCG2A-U000-0458
- Boot OS 9.1: BCG27-U000-0895 or BCG24-U000-0CL8

(f) On the IDL page, click the BROWSE button.

(g) Select the directory that contains the software parts.

(h) Select the applicable software part and then click the OPEN button to import it.

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1) Make sure that the software parts are imported in the order listed on the IDL page.

(i) After the part is imported, make sure that the part changes from amber to white.

NOTE: The START LOAD button is disabled (cyan in color) until all of the IDL parts are imported.

(j) Click on the BROWSE button and do the steps above again to select a different software part to import.

(k) Continue to import all applicable software parts.

(l) Make sure that all parts change from amber to white and the START LOAD button is enabled after all necessary parts are imported.

NOTE: The START LOAD button changes from cyan to white in color to show the function is enabled.

SUBTASK 46-13-01-470-025

(2) Do these steps to complete the software installation of the part numbers imported:

(a) Make sure that the applicable target and software part numbers are specified in white under the heading “Select a file with parts to import”.

(b) At the bottom right of the INITIAL DATA LOAD page, select START LOAD.

1) The NFS will do the installation and reboot automatically.

2) Wait ten minutes for the reboot to complete.

NOTE: The NFS is not fully functional until ten minutes have passed.

3) If AIRPLANE IDENTIFICATION MISSING dialog box is shown after IDL reboot, select IGNORE.

(c) Verify at least one System Log file is present under Download Manager. This is the task: Download Manager, TASK 46-13-02-970-802.

NOTE: On the System Log page, select GET CURRENT before completing the verification.

NOTE: It is not necessary to download or delete any system logs.

(d) If a system log is not present after selecting GET CURRENT, the NFS must be reimaged and the IDL must be restarted. Do this task: Network File Server Reimage Procedure, TASK 46-13-01-860-802.

SUBTASK 46-13-01-470-031

(3) Stage all missing Mass Storage Device (MSD) software parts to put the MSD back to normal configuration (TASK 46-13-00-470-802).

SUBTASK 46-13-01-470-057

(4) Dataload all missing NFS software parts. Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

NOTE: A 46 NFS OAS AIRPLANE KEYS UMS that was created before the IDL cannot be dataloaded after the IDL. A new 46 NFS OAS AIRPLANE KEYS UMS must be created using a new Certificate Signing Request (CSR). Refer to this task: Airplane Credentials Check, TASK 46-13-00-860-801.

SUBTASK 46-13-01-470-059

(5) Do this task: Server Credentials Check, TASK 46-13-00-860-802.

(a) Do not make a server credential if one is already made.

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SUBTASK 46-13-01-470-060

(6) Do this task: Client Credentials Check, TASK 46-13-00-860-803.

(a) Do not make new client credentials for any LRU's that already have client credentials.

SUBTASK 46-13-01-470-061

(7) If applicable, do this task: Network File Server SD Media Card Installation, TASK 46-13-01-420-801.

SUBTASK 46-13-01-470-062

(8) If applicable, do this task: Network File Server Backup to SD Media Card, TASK 46-13-01-470-803.

——— END OF TASK ———

TASK 46-13-01-800-801

7. Network File Server Reboot Procedure

A. General

(1) This task gives the steps to reboot the Network File Server (NFS).

NOTE: It is necessary to wait approximately ten minutes for the NFS to reboot.

(2) The NFS should be rebooted at least once every six weeks.

(3) The aircraft must be on the ground to do this task.

(4) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for NFS Reboot

SUBTASK 46-13-01-860-009

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

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F. NFS Reboot Procedure

SUBTASK 46-13-01-800-002

(1) Do the reboot procedure as follows:

(a) Select EXTENDED MAINTENANCE > Network File Server.

(b) Select the REBOOT tab.

(c) Make sure that the REBOOT screen shows the Function and Description and the REBOOT icon in the page footer.

(d) Select the REBOOT icon at the bottom of the page.

(e) Click on the CONTINUE button.

(f) Make sure that the REBOOTING NFS message shows.

(g) Wait approximately ten minutes for the NFS to reboot.

NOTE: The broken chain link icon shows in amber at the top right corner adjacent to the settings icon during the reboot operation. After the NFS reboots, the unbroken chain link shows in white when the ONS is serviceable.

(h) After the reboot is complete, make sure that the screen goes to the ONS maintenance browser home page.

1) If it does not go to the home page after an additional three minutes, refresh the screen.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-860-010

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-800-802

8. Network File Server Reset Data Procedure

A. General

(1) This task gives the steps to reset the Network File Server (NFS) data.

(2) The data reset operation cannot be undone. It restores applications to their default state.

(3) The data reset operation removes application data and system logs. The installed software, Mass Storage Device (MSD) data, security credentials, and security logs are not removed.

(4) The aircraft must be on the ground to do this task.

(5) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

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C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for NFS Data Reset

SUBTASK 46-13-01-860-011

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

F. NFS Data Reset Procedure

SUBTASK 46-13-01-800-003

(1) Do the data reset procedure as follows:

(a) Select EXTENDED MAINTENANCE > Network File Server.

(b) On the Network File Server menu, select the RESET DATA tab.

(c) Make sure that the RESET DATA screen shows the Function and Description and the RESET icon in the page footer.

(d) Select the RESET icon at the bottom of the page.

(e) Make sure that the reset confirmation screen shows.

(f) Click on the CONTINUE button.

(g) Make sure that the RESETTING NFS message is shown.

NOTE: It may take up to 10 minutes for the NFS to reset.

(h) After the reset is complete, make sure that the screen goes to the ONS maintenance browser home page.

1) If it does not go to the home page after three minutes, refresh the screen.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-860-012

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-01-020-801

9. Network File Server SD Media Card Removal

Figure 201

A. General

(1) This task tells you how to remove the Network File Server (NFS) Secure Digital (SD) media card from the NFS.

(2) Wait 3 minutes following a successful NFS Manual Backup to SD Media Card, (Network File Server Backup to SD Media Card, TASK 46-13-01-470-803) before removing the SD Media Card from the NFS.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Removal of SD Media Card

SUBTASK 46-13-01-040-003

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER

SUBTASK 46-13-01-010-008

(2) Get access to the main equipment center.

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 46-13-01-010-009

(3) Loosen thumbscrew on the front panel of the NFS and swing NFS cover downward.

SUBTASK 46-13-01-000-001

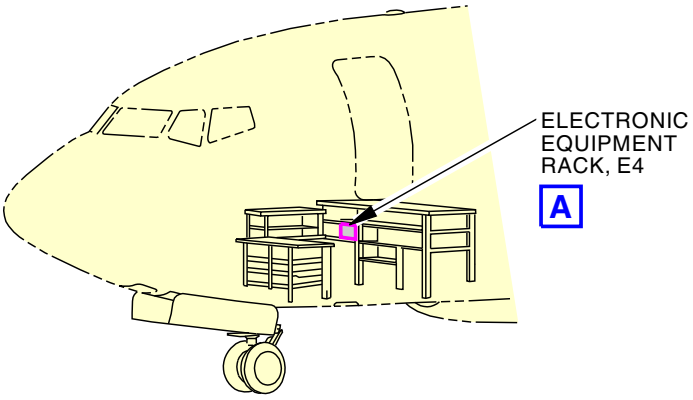
(4) Push SD media card as far as it will go and release pressure on the SD card.

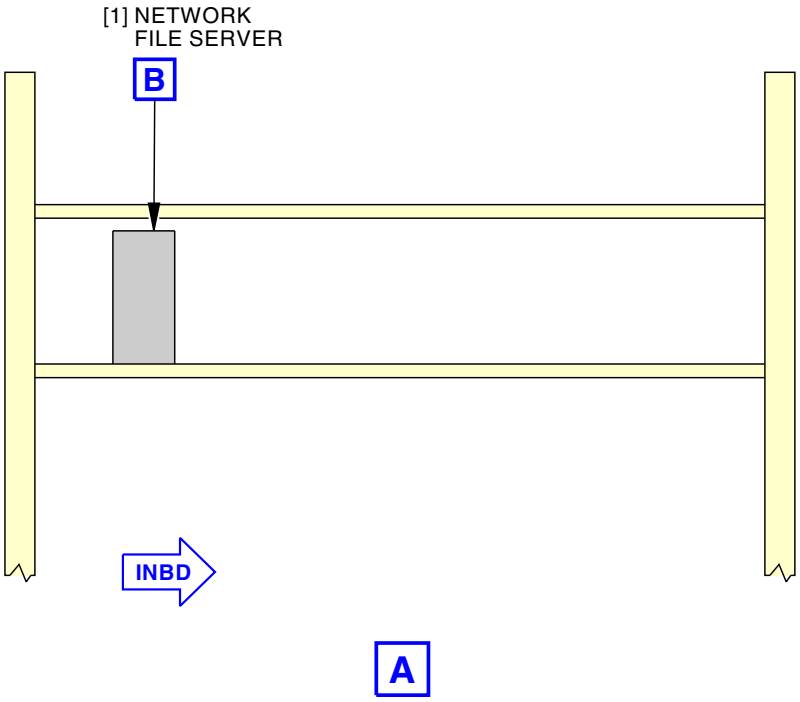
NOTE: Releasing pressure will unlatch the SD media card causing it to protrude from the SD media card slot to enable removal.

(5) If necessary, retain SD media card for installation to the new NFS.

——— END OF TASK ———

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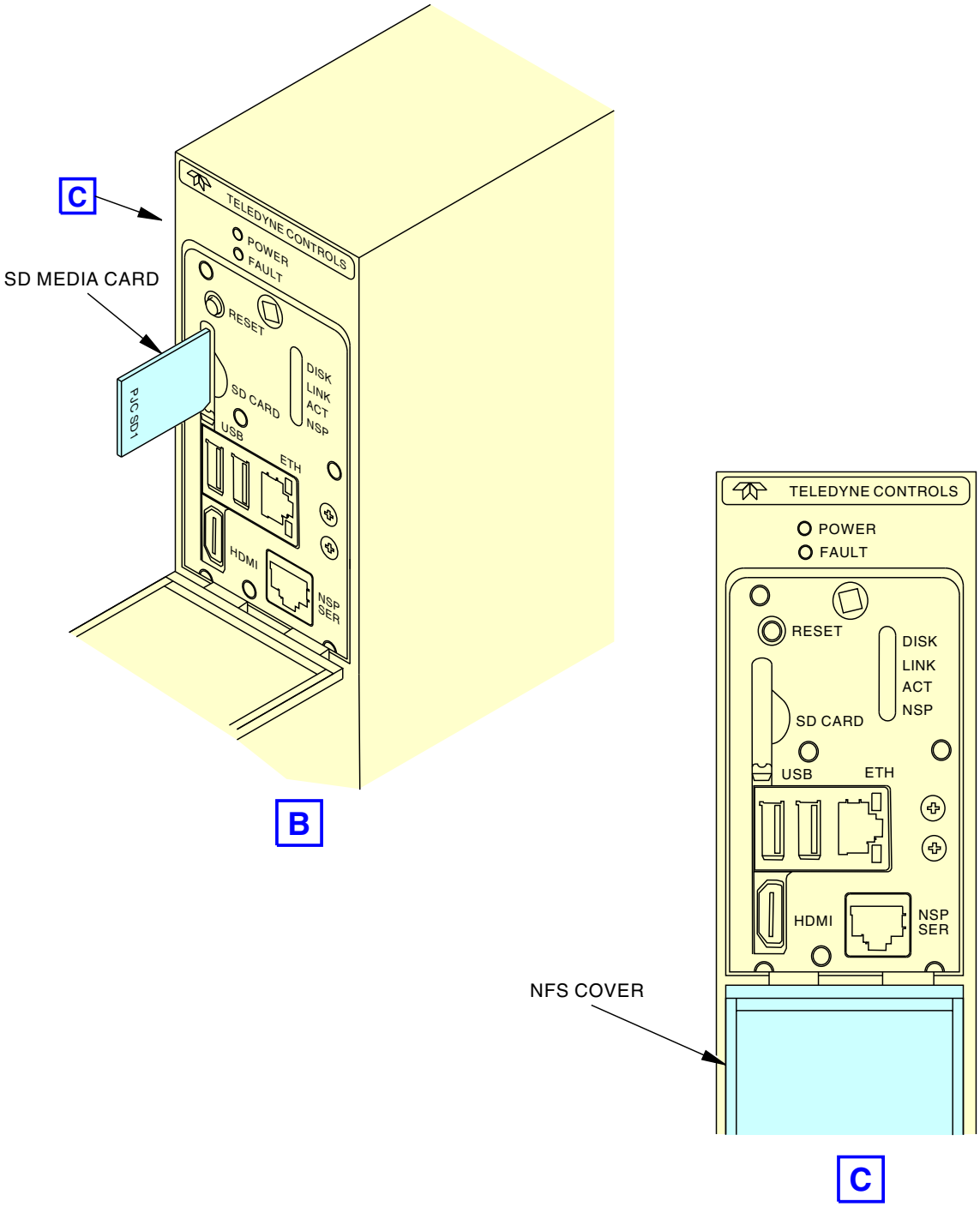




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NFS SD Media Card Installation
Figure 201/46-13-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NFS SD Media Card Installation
Figure 201/46-13-01-990-802 (Sheet 2 of 2)

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TASK 46-13-01-420-801

10. Network File Server SD Media Card Installation

Figure 201

A. General

(1) This task gives the steps to install the Network File Server (NFS) Secure Digital (SD) media card into the NFS.

(a) Do not use a Micro SD Media Card with an SD adapter.

(b) Follow Boeing guidance for SD card sizing and brand.

(2) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-00-750-801	Airplane Identification Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Installation of SD Media Card

SUBTASK 46-13-01-040-002

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER

SUBTASK 46-13-01-010-006

(2) Get access to the main equipment center.

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(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 46-13-01-010-007

(3) Loosen thumbscrew on the front panel of the NFS and swing the NFS cover downward.

SUBTASK 46-13-01-420-002

(4) Put the SD media card into the SD card slot and push the card until the SD card latches in place.

(a) Make sure that the SD media card contacts face left and towards the NFS when inserted into the SD card slot.

SUBTASK 46-13-01-010-012

(5) Close the NFS cover and tighten thumbscrew.

SUBTASK 46-13-01-440-001

(6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER

G. Automatic SD Media Card Initialization

NOTE: If a SD media card has not been initialized by the NFS Backup And Restore Service (BARS), it is automatically initialized when the NFS is booted-up.

NOTE: To complete SD media card initialization, the MAX Display System (MDS) must be powered ON for the NFS to receive the Airplane ID parameter.

NOTE: If the MDS cannot be powered, the Airplane ID parameter can be manually entered to the NFS. Refer to Airplane Identification Check, TASK 46-13-00-750-801.

SUBTASK 46-13-01-480-004

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

SUBTASK 46-13-01-470-021

(2) Do these steps to access BARS to view the automated initialization:

(a) For the MFD, select EXTENDED MAINT > BACKUP AND RESTORE.

(b) For the UMD, select EXTENDED MAINTENANCE > Network File Server > BACKUP & RESTORE.

(c) While the initialization is in progress, the Media Status shows the current status of the operation.

NOTE: The SD media card initialization may take up to three minutes.

(d) When the initialization is complete, the Media Card Status shows “Initialized”.

(e) Do a manual backup to the SD Media card (TASK 46-13-01-470-803).

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-010

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-470-802

11. Network File Server SD Media Card Manual Initialization

A. General

(1) This procedure gives the steps to manually initialize the Network File Server (NFS) Secure Digital (SD) media card.

(2) If the DPC is inactive, the Airplane ID parameter can be manually entered to the NFS. Refer to Airplane Identification Check, TASK 46-13-00-750-801.

(3) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
31-65-00-970-804	Airplane Config Data (ACD) Page Function (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-00-750-801	Airplane Identification Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for Initialization of SD Media Card

SUBTASK 46-13-01-970-012

(1) To verify the Airplane ID on the MFD, do these steps:

(a) Do this task: Multi Function Display Connection, TASK 46-13-00-480-803.

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(b) On the MFD main function menu:

1) Select MAINT.

2) Select MAINT CTRL PGS.

3) Select AIRPLANE CONFIG DATA (ACD).

(c) Set the DISPLAY SOURCE switch on the P5-28 overhead panel to the ALL ON 1 position and record the data.

NOTE: When the DISPLAY SOURCE switch is set to ALL ON 1, Display Processing Computer (DPC) 1 will be the source of the ACD data. When the DISPLAY SOURCE switch is set to ALL ON 2, DPC 2 will be the source of the ACD data.

(d) Set the DISPLAY SOURCE switch on the P5-28 overhead panel to the ALL ON 2 position and record the data.

1) Make sure that the ACD data is correct for the airplane.

a) If the ACD data is not correct, do this task: Airplane Config Data (ACD) Page Function, TASK 31-65-00-970-804.

SUBTASK 46-13-01-970-011

(2) To verify the Airplane ID on the UMD, do these steps:

(a) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

(b) Select LINE MAINTENANCE > Airplane ID.

(c) Enter the correct Airline ID, Aircraft Type, and Airplane ID and select SAVE. If the correct Airline ID, Aircraft Type, and Airplane ID are already present, no action is required.

NOTE: If the DPCs are powered, the Airline ID and Airplane ID will be grayed out and they must be changed via the ACD page on the MFD.

F. Manual Initialization of SD Media Card

SUBTASK 46-13-01-470-038

(1) To do the manual initialization of the SD media card, do these steps:

(a) For the UMD, select EXTENDED MAINTENANCE > Network File Server > BACKUP & RESTORE.

(b) For the MFD, select EXTENDED MAINT > BACKUP AND RESTORE.

(c) Select MEDIA CARD DETAILS in the lower left corner of the page.

(d) Select INITIALIZE MEDIA CARD on Media Card Details window.

NOTE: This process will remove all data on the SD card.

(e) Select CONFIRM.

NOTE: The SD Media Card initialization may take up to three minutes.

1) Do not remove the SD Media Card from the Network File Server while a manual initialization is in progress.

(f) While the initialization is in progress, the INITIALIZE MEDIA CARD window will show the current status of the operation.

(g) For the UMD, “INITIALIZE MEDIA CARD COMPLETE” is shown when the initialization is complete.

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(h) For the MFD, “The Media Card Initialization was completed successfully” is shown when the initialization is complete.

(i) Select OK to close the INITIALIZE MEDIA CARD window.

(j) Do this task: Network File Server Backup to SD Media Card, TASK 46-13-01-470-803.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-011

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-700-801

12. Network File Server SD Media Card Status Check

A. General

(1) This procedure gives the steps to check the status of the Secure Digital (SD) media card that is installed in the Network File Server (NFS).

(2) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

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E. Prepare for the Status Check

SUBTASK 46-13-01-470-039

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

F. SD Media Card Status Check

SUBTASK 46-13-01-700-001

(1) To check the SD Media Card Status, do these steps:

(a) For the UMD, select EXTENDED MAINTENANCE > Network File Server > BACKUP & RESTORE.

(b) For the MFD, select EXTENDED MAINT > BACKUP AND RESTORE.

(c) Select the MEDIA CARD DETAILS button in the lower left corner of the Backup And Restore Service (BARS) page.

1) Examine the summary of the SD media card content.

(d) Select CANCEL to close the MEDIA CARD DETAILS window.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-012

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-470-803

13. Network File Server Backup to SD Media Card

A. General

(1) Encrypted Backup:

(a) For airplanes with Wireless WAN Unit (WWU) and/or Crew Wireless LAN Unit (CWLU), the Network File Server (NFS) will have the 46 NFS OAS AIRPLANE KEYS UMS software part installed. The Backup And Restore Service (BARS) requires this software part to be back-up encrypted. To enable encrypted backup, the 46 NFS MEF UMS and 46 BARS SECURITY UMS software parts must be installed to the NFS.

(b) If the Onboard Network System (ONS) is not configured to support encrypted backup, the 46 NFS OAS AIRPLANE KEYS UMS software part will not be backed-up by BARS.

(2) Automatic Backup:

(a) The ONS is configured to automatically perform a backup of the NFS to the Secure Digital (SD) media card whenever the airplane transitions from Ground to Air.

(3) Manual Backup:

(a) This procedure gives the steps to manually backup the NFS to the Secure Digital (SD) media card.

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(b) The SD media card must be installed and initialized before doing a backup.

(c) This task is available when the ONS is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)
46-13-00-470-802	Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD) (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. NFS Encrypted Backup to SD Media Card

SUBTASK 46-13-01-470-011

(1) To enable encrypted back-up capability, do these steps:

(a) Make sure that 46 NFS MEF UMS and 46 BARS SECURITY UMS software parts are installed to the NFS by checking the NFS system configuration. Do this task: System Configuration Check, TASK 46-13-00-710-802.

1) If the software parts are not installed, stage the 46 NFS MEF UMS and 46 BARS SECURITY UMS to the NFS. Do this task: Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD), TASK 46-13-00-470-802.

NOTE: This step cannot be done on the MFD.

2) Dataload the 46 NFS MEF UMS and 46 BARS SECURITY UMS software parts to the NFS. Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

F. NFS Manual Backup to SD Media Card

SUBTASK 46-13-01-480-005

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

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SUBTASK 46-13-01-470-042

(2) To do an NFS manual backup to a SD Media Card, do these steps:

(a) For the UMD, select EXTENDED MAINTENANCE > Network File Server > BACKUP & RESTORE.

(b) For the MFD, select EXTENDED MAINT > BACKUP AND RESTORE.

(c) Verify that the Status in the Median Card box shows “Initialized.

1) If the SD card status shows “Failed”, do the following tasks:

a) Remove the SD card (TASK 46-13-01-020-801).

b) Install a new SD card (TASK 46-13-01-420-801).

c) Manually initialize the new SD card (TASK 46-13-01-470-802).

2) If the SD card status shows “Not Installed”, do the following tasks:

a) Install a new SD card (TASK 46-13-01-420-801).

b) Manually initialize the SD card (TASK 46-13-01-470-802).

3) If the Status shows “Pending” or “INITIALIZATION IN PROGRESS”, wait at least 5 minutes for the status to change to “Initialized'.

(d) Select the BACKUP TO MEDIA CARD button in the lower portion of BARS page.

1) Do not remove the SD Media Card from the Network File Server while a manual backup is in progress.

NOTE: The backup status is shown on the BACKUP TO MEDIA CARD window.

(e) Make sure that a backup to media card message is shown when the backup is complete.

(f) Select OK to close the BACKUP TO MEDIA CARD window.

(g) Select MEDIA CARD DETAILS in the lower corner of the page to view a summary of the SD media card content.

(h) Make sure that the size of the last backup is not zero.

1) If the size is 0, attempt the backup again.

2) If the size is still 0 after the second attempt, retrieve and install a new SD card. Do this task: Network File Server SD Media Card Installation, TASK 46-13-01-420-801.

(i) Select CANCEL to close the MEDIA CARD DETAILS window.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-013

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-01-470-804

14. Network File Server Automatic Stage 1 Restore Function

A. General

(1) The automatic Stage 1 restore process for the Network File Server (NFS) automatically starts when the NFS turns ON when:

(a) The NFS is installed with BOOT OS 7.1 (BCG2A-U000-0458) or BOOT OS 9.1 (BCG27-U000-0895 or BCG24-U000-0CL8).

(b) The BOOT OS version for the media card backup is the same version of the BOOT OS installed in the NFS.

(c) The NFS does not have the Initial Data Load (IDL) software parts installed.

(d) A Secure Digital (SD) media card with the Backup and Restore Service (BARS) backup image is installed in the NFS.

(e) The Airplane Identification on the SD media card matches the Airplane Identification on the Display Processing Computer (DPC).

NOTE: The BARS SD card contains the Tail ID from the airplane where the backup images were created, and the restore process will not be allowed if the Airplane Identification on the SD card does not agree with the Airplane Identification on the DPC.

(2) The automatic Stage 1 restores all of the Onboard Network System (ONS) software parts to the NFS that do not need encryption. The automatic Stage 1 restore also restores all software parts to the Mass Storage Device (MSD).

(3) The automatic Stage 1 restore does not need any user interaction to initiate or complete.

(a) During the restore process, ONS functions will not be available.

(b) The amber fault light on the front panel of the NFS flashes in a pattern of three seconds on and one-half second off.

NOTE: The restore process may take over 30 minutes to complete and may take longer depending on the MSD contents. If the amber fault light is steady ON (not flashing) for more than 5 minutes, the restore process has failed.

NOTE: The restore process is complete when any of the following are shown:

- The Universal Maintenance Device (UMD) shows an active home page,
- The Multi Function Display (MFD) is able to access ONS functions.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Automatic Stage 1 Restore Procedure

SUBTASK 46-13-01-010-010

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 46-13-01-470-014

(2) To do the automatic stage 1 restore, do these steps:

(a) Make sure that the SD media card with the BARS backup image is installed in the NFS.

1) If the SD media card is not installed, do this task: Network File Server SD Media Card Installation, TASK 46-13-01-420-801.

(b) Reboot the NFS:

1) Open and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER

a) Make sure that the circuit breaker is open for at least five seconds before closing it.

(c) Make sure that the amber fault light on the front panel of the NFS begins flashing in a pattern of three seconds ON and one-half second OFF.

(d) For the MFD, make sure that the amber ONBD MAINT IS NOT AVAILABLE window is shown for approximately 30 minutes during the automated restore and reboot.

(e) For the UMD, wait for approximately 30 minutes since the ONS will not be accessible during the automated restore and reboot.

NOTE: When the stage 1 restore is complete, the MFD and UMD can be used to access ONS functions.

(f) After 30 minutes, access the ONS.

1) To access the ONS via the forward display, do this task: Multi Function Display Connection, TASK 46-13-00-480-803.

a) If a “NFS RESTORE COMPLETE” modal shows, select OK.

b) If directed by your engineering department to conduct a stage 2 restore, do this task: Network File Server Stage 2 Restore, TASK 46-13-01-470-806.

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2) To access the ONS via UMD, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

NOTE: Do not attempt to connect wirelessly.

a) If a “NFS RESTORE COMPLETE” modal shows, select OK.

b) If a “NFS RESTORE PENDING FOR CONNECTIVITY shows, do this task: Network File Server Stage 2 Restore, TASK 46-13-01-470-806.

(g) If ONS is not accessible after 30 minutes, the Stage 1 restore has failed and a manual Initial Data Load must be completed. Do the following tasks:

1) Remove the SD media card (TASK 46-13-01-020-801).

2) Conduct a manual Initial Data Load (TASK 46-13-01-470-807).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-014

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-470-805

15. Network File Server Manual Stage 1 Restore

A. General

(1) This procedure gives the steps to manually restore the Network File Server (NFS) with a Secure Digital (SD) media card with a Backup and Restore Service (BARS) backup image.

NOTE: The NFS must be configured with Onboard Network System (ONS) OS7.1 BOOT OS (BCG2A-U000-0458) or OS9.1 BOOT OS (BCG27-U000-0895 or BCG24-U000-0CL8) to enable the BARS function.

NOTE: The BOOT OS version used when creating the Media Card Backup must be the same version of BOOT OS installed in the NFS.

(2) This task is available when the ONS is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Manual Stage 1 Restore

SUBTASK 46-13-01-470-048

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321, is required for access.

NOTE: Do not attempt to connect wirelessly.

F. Manual Stage 1 Restore

SUBTASK 46-13-01-470-051

(1) For the Manual Stage 1 Restore do these steps:

(a) For the UMD, do these steps:

1) Select EXTENDED MAINTENANCE > Network File Server > BACKUP & RESTORE.

2) Verify the Status in the Media Card box shows “Initialized”.

a) If the SD card status shows “Failed”, do the following tasks:

<1> Remove the SD card (TASK 46-13-01-020-801).

<2> Install a new SD card (TASK 46-13-01-420-801).

<3> Manually initialize the new SD card (TASK 46-13-01-470-802).

<4> Manually backup the new SD card (TASK 46-13-01-470-803).

b) If the SD card status shows “Not Installed”, do the following tasks:

<1> Install a new SD card (TASK 46-13-01-420-801).

<2> Manually initialize the new SD card (TASK 46-13-01-470-802).

<3> Manually backup the new SD card (TASK 46-13-01-470-803).

c) If the Status shows “Pending” or “INITIALIZATION IN PROGRESS”, wait at least 5 minutes for the status to change to “Initialized'.

3) Verify the “Last Restorable Backup” in the Media Card box shows a valid date and time. If it shows an invalid date and time, or there is no backup, do the following tasks:

a) Manually initialize the new SD card (TASK 46-13-01-470-802).

b) Manually backup the new SD card (TASK 46-13-01-470-803).

4) Select RESTORE TO NFS in the lower right portion of the page.

5) Select CONFIRM.

6) Make sure that “Restoring from the [AIRPLANE ID] Media Card” window is shown.

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(b) For the MFD, do these steps:

1) Select EXTENDED MAINT > BACKUP AND RESTORE.

2) Select RESTORE TO NFS in the lower right portion of the page.

3) Select CONFIRM.

4) Make sure that an amber ONBD MAINT IS NOT AVAILABLE window is shown while the stage 1 restore is in process.

NOTE: On the UMD, the connection status icon in the top right corner will change from a white connected chain to an amber broken chain-link during the restore process.

NOTE: The ONS functions that are normally accessible from the UMD and MFD will not be available while the phase 1 restore is in process. The restore process may take over 30 minutes to complete and may take longer depending on the Mass Storage Device (MSD) contents.

SUBTASK 46-13-01-470-052

(2) After 30 minutes, verify that the stage 1 restore is complete:

(a) For the UMD, do these steps:

1) If a “NFS RESTORE COMPLETE” modal shows, select OK.

2) If a “NFS RESTORE PENDING FOR CONNECTIVITY” modal shows, do this task: Network File Server Stage 2 Restore, TASK 46-13-01-470-806.

(b) For the MFD, do these steps:

1) If a “NFS RESTORE COMPLETE” modal shows, select OK.

2) If it is necessary to conduct a stage 2 restore, do this task: Network File Server Stage 2 Restore, TASK 46-13-01-470-806.

(3) If ONS is not accessible after 30 minutes, the Stage 1 restore has failed and a manual Initial Data Load must be completed. Do the following tasks:

(a) Remove the SD media card (TASK 46-13-01-020-801).

(b) Conduct a manual Initial Data Load (TASK 46-13-01-470-807).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-015

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-470-806

16. Network File Server Stage 2 Restore

A. General

(1) This procedure tells you how to do the Stage 2 restoration of the Network File Server (NFS) with a Secure Digital (SD) media card containing the Backup And Restore Service (BARS) backup image. Stage 2 restore can only be done using a Universal Maintenance Device (UMD).

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(a) Gather all the software parts you need on a removable media or UMD before leaving the airplane.

NOTE: While a Stage 2 Restore is in progress, do not use the MAX forward displays (MFDs) to access or use ONS functionality.

(2) The Stage 2 restore is required to support off-board communications.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-00-750-801	Airplane Identification Check (P/B 201)
46-13-00-860-802	Server Credentials Check (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Stage 2 Restore of NFS Using a Universal Maintenance Device (UMD)

SUBTASK 46-13-01-470-018

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

NOTE: Do not attempt to connect wirelessly.

(2) If a “NFS RESTORE COMPLETE” modal shows, select OK.

(3) An “NFS RESTORE PENDING FOR CONNECTIVITY” modal will show with the following options:

NOTE: If the modal does not show, then refer to the IFIM and do the applicable procedure(s).

(a) DISABLE

NOTE: DISABLE is used to cancel the stage 2 restore. Once the stage 2 restore is cancelled, Crew Wireless or Off-Board Communication functionality will not be available until a new CSR is generated and a new 46 NFS OAS AIRPLANE KEYS UMS software part is created.

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(b) NOT NOW

NOTE: NOT NOW is used to pause the Stage 2 restore process, which allows navigation to other Onboard Network System (ONS) functions. To resume the stage 2 restore, return to the BACKUP AND RESTORE menu page to continue with the installation of the 46 NFS BARS SECURITY UMS part. This allows Crew Wireless or Off-Board Communication functionality.

(c) CONTINUE

(4) Select CONTINUE to proceed with the stage 2 restore.

(5) Select BROWSE and select the 46 NFS BARS SECURITY UMS software part.

(6) Select CONTINUE to initiate the stage 2 restore.

NOTE: The RESTORE TO NFS window appears while the stage 2 restore is in progress.

NOTE: The RESTORE TO NFS window displays “The Restore from the [Airplane ID] Media Card completed Successfully.” when stage 2 restore is complete.

(7) Select OK to close the RESTORE TO NFS window.

(a) If the BARS stage 2 restore fails, then refer to the IFIM and do the applicable procedure(s).

(8) Verify that there are valid server credentials. Do this task: Server Credentials Check, TASK 46-13-00-860-802.

NOTE: Do not generate new server credentials if there are valid server credentials present.

(9) Verify that there are valid client credentials for all LRUs. Do this task: Client Credentials Check, TASK 46-13-00-860-803.

NOTE: Do not generate new client credentials if there are valid client credentials present for all LRUs.

(10) Verify all expected software is dataloaded to the NFS. Do this task: System Configuration Check, TASK 46-13-00-710-802.

(11) Verify the Aircraft Type is set. Do this task: Airplane Identification Check, TASK 46-13-00-750-801.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-016

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the UMD is no longer necessary, stop the Onboard Network System (ONS) maintenance browser session and disconnect the maintenance device.

——— END OF TASK ———

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TASK 46-13-01-470-808

17. Network File Server OS Upgrade OS7.0 to OS7.1

A. General

(1) This task gives the steps to upgrade the Network File Server (NFS) OS7.0 to OS7.1.

(a) Gather all the software parts you need on removable media or Universal Maintenance Device (UMD) before leaving for the airplane.

NOTE: The OS7.0 46 NFS Boot OS is: BCG57-U000-02E3.

NOTE: The OS7.1 46 NFS Boot OS is: BCG2A-U000-0458.

(2) To do this task, the weight-on-wheels discrete must be in the GROUND condition.

(3) Do this task when you get access to the Onboard Network System (ONS) maintenance browser with the UMD.

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)
46-13-00-470-802	Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD) (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Procedure

SUBTASK 46-13-01-470-030

(1) Use a universal maintenance device, SPL-11321, to do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

SUBTASK 46-13-01-470-068

(2) If the NFS is in an Initial Data Load state with only the OS7.0 Boot OS loaded, do this task to load the core OS7.0 software parts: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

NOTE: After the core OS7.0 software is loaded, do not continue the IDL Software Installation task and instead return here to continue the upgrade.

NOTE: If the NFS already has a full suite of the OS7.0 parts loaded, skip this step.

NOTE: Only an NFS with all the core OS7.0 software parts can be upgraded.

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(a) Stage the OS7.1 46 NFS BOOT OS to the Mass Storage Device (MSD). Do this task: Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD), TASK 46-13-00-470-802.

(b) Install the OS7.1 46 NFS BOOT OS to the NFS. Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

NOTE: The Dataload status of the OS7.1 46 NFS BOOT OS will say “Installing…” for approximately ten minutes, and then that status will update to “Not Installed.” This status may be incorrect and the part may be installed.

(c) Run a system configuration of the NFS to ensure the OS7.1 46 NFS BOOT OS software part has successfully installed. Do this task: System Configuration Check, TASK 46-13-00-710-802.

(d) Remove all OS7.0 software parts from the NFS. Do this task: Network File Server Reimage Procedure, TASK 46-13-01-860-802.

(e) Install the OS7.1 software parts and return the NFS to a fully operational state. Do this task: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

(f) Run a system configuration of the NFS to ensure all OS7.0 software parts have been replaced with OS7.1 software parts. Do this task: System Configuration Check, TASK 46-13-00-710-802.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-019

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the UMD is no longer necessary, stop the ONS maintenance browser session and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-470-809

18. Network File Server OS Upgrade OS7.0 to OS9.1

A. General

(1) This task gives the steps to upgrade the Network File Server (NFS) OS7.0 to OS9.1.

(a) Gather all the software parts you need on removable media or Universal Maintenance Device (UMD) before leaving for the airplane.

NOTE: The OS7.0 46 NFS Boot OS is: BCG57-U000-02E3.

NOTE: The OS9.1 46 NFS Boot OS is: BCG27-U000-0895 or BCG24-U000-0CL8.

(2) To do this task, the weight-on-wheels discrete must be in the GROUND condition.

(3) Do this task when you get access to the Onboard Network System (ONS) maintenance browser with the UMD.

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)

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(Continued)

Reference	Title
46-13-00-470-802	Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD) (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Procedure

SUBTASK 46-13-01-470-054

(1) Use a universal maintenance device, SPL-11321, to do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

SUBTASK 46-13-01-470-053

(2) If the NFS is in an Initial Data Load state with only the OS7.0 Boot OS loaded, do this task to load the core OS7.0 software parts: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

NOTE: After the core OS7.0 software is loaded, do not continue the Initial Data Load (IDL) Software Installation task and instead return here to continue the upgrade.

NOTE: If the NFS already has a full suite of the OS7.0 parts loaded, skip this step.

NOTE: Only an NFS with all the core OS7.0 software parts can be upgraded.

(a) Stage the OS9.1 46 NFS BOOT OS to the Mass Storage Device (MSD). Do this task: Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD), TASK 46-13-00-470-802.

(b) Install the OS9.1 46 NFS BOOT OS to the NFS. Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

NOTE: The Dataload status of the OS9.1 46 NFS BOOT OS will say “Installing…” for approximately ten minutes, and then that status will update to “Not Installed.” This status may be incorrect and the part may be installed.

(c) Run a system configuration of the NFS to ensure the OS9.1 46 NFS BOOT OS software part has successfully installed. Do this task: System Configuration Check, TASK 46-13-00-710-802.

(d) Remove all OS7.0 software parts from the NFS. Do this task: Network File Server Reimage Procedure, TASK 46-13-01-860-802.

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(e) Install the OS9.1 software parts and return the NFS to a fully operational state. Do this task: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

(f) Run a system configuration of the NFS to ensure all OS7.0 software parts have been replaced with OS9.1 software parts. Do this task: System Configuration Check, TASK 46-13-00-710-802.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-017

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(2) If the UMD is no longer necessary, stop the ONS maintenance browser session and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-470-811

19. Network File Server OS Upgrade OS7.1 to OS9.1

A. General

(1) This task gives the steps to upgrade the Network File Server (NFS) OS7.1 to OS9.1.

(a) Gather all the software parts you need on removable media or Universal Maintenance Device (UMD) before leaving for the airplane.

NOTE: The OS7.1 46 NFS Boot OS is: BCG2A-U000-0458.

NOTE: The OS9.1 46 NFS Boot OS is: BCG27-U000-0895 or BCG24-U000-0CL8.

(2) To do this task, the weight-on-wheels discrete must be in the GROUND condition.

(3) Do this task when you get access to the Onboard Network System (ONS) maintenance browser with the UMD.

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)
46-13-00-470-802	Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD) (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

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D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Procedure

SUBTASK 46-13-01-470-066

(1) Use a universal maintenance device, SPL-11321, to do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

SUBTASK 46-13-01-470-067

(2) If the NFS is in an Initial Data Load (IDL) state with only the OS7.1 Boot OS loaded, do this task to load the core OS7.1 software parts: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

NOTE: After the core OS7.1 software is loaded, do not continue the IDL Software Installation task and instead return here to continue the upgrade.

NOTE: If the NFS already has a full suite of the OS7.1 parts loaded, skip this step.

NOTE: Only an NFS with all the core OS7.1 software parts can be upgraded.

(a) Stage the OS9.1 46 NFS BOOT OS to the Mass Storage Device (MSD). Do this task: Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD), TASK 46-13-00-470-802.

(b) Install the OS9.1 46 NFS BOOT OS to the NFS. Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

NOTE: The Dataload status of the OS9.1 46 NFS BOOT OS will say “Installing…” for approximately ten minutes, and then that status will update to “Not Installed.” This status may be incorrect and the part may be installed.

(c) Run a system configuration of the NFS to ensure the OS9.1 46 NFS BOOT OS software part has successfully installed. Do this task: System Configuration Check, TASK 46-13-00-710-802.

(d) Remove all OS7.1 software parts from the NFS. Do this task: Network File Server Reimage Procedure, TASK 46-13-01-860-802.

(e) Install the OS9.1 software parts and return the NFS to a fully operational state. Do this task: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

(f) Run a system configuration of the NFS to ensure all OS7.1 software parts have been replaced with OS9.1 software parts. Do this task: System Configuration Check, TASK 46-13-00-710-802.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-021

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(2) If the UMD is no longer necessary, stop the ONS maintenance browser session and disconnect the maintenance device.

END OF TASK

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TASK 46-13-01-470-812

20. Network File Server OS Downgrade OS9.1 to OS7.1

A. General

(1) This task gives the steps to downgrade the Network File Server (NFS) OS9.1 to OS7.1.

(a) Gather all the software parts you need on removable media or Universal Maintenance Device (UMD) before leaving for the airplane.

NOTE: The OS7.1 46 NFS Boot OS is: BCG2A-U000-0458.

NOTE: The OS9.1 46 NFS Boot OS is: BCG27-U000-0895 or BCG24-U000-0CL8.

(2) The aircraft must be on the ground to do this task.

(3) This task is available when the Onboard Network System (ONS) is accessed with the UMD.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Procedure

SUBTASK 46-13-01-470-055

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

NOTE: Do not attempt to connect wirelessly.

SUBTASK 46-13-01-470-056

(2) If the NFS is in a fully operational OS9.1 state, reimage the NFS. Do this task: Network File Server Reimage Procedure, TASK 46-13-01-860-802.

NOTE: If only the OS9.1 Boot OS is present on the NFS (the NFS is in an Initial Data Load (IDL) state), skip this step.

(3) Install the core OS7.1 software parts. Do this task: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

NOTE: The IDL Software Installation task can be used to install the remaining OS7.1 software parts and return the NFS to a fully operational state.

NOTE: When staging and dataloading the rest of the OS7.1 software parts after the IDL, make sure to stage and install the OS7.1 Boot OS as well.

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(4) Run a system configuration of the NFS to ensure all OS9.1 software parts have been replaced with OS7.1 software parts. Do this task: System Configuration Check, TASK 46-13-00-710-802.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-018

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-470-813

21. Network File Server OS Downgrade OS9.1 to OS7.0

A. General

(1) This task gives the steps to downgrade the Network File Server (NFS) OS9.1 to OS7.0.

(a) Gather all the software parts you need on removable media or Universal Maintenance Device (UMD) before leaving for the airplane.

NOTE: The OS7.0 46 NFS Boot OS is: BCG57-U000-02E3.

NOTE: The OS9.1 46 NFS Boot OS is: BCG27-U000-0895 or BCG24-U000-0CL8.

(2) The aircraft must be on the ground to do this task.

(3) This task is available when the Onboard Network System (ONS) is accessed with the UMD.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

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E. Procedure

SUBTASK 46-13-01-470-069

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

NOTE: Do not attempt to connect wirelessly.

SUBTASK 46-13-01-470-070

(2) If the NFS is in a fully operational OS9.1 state, reimage the NFS. Do this task: Network File Server Reimage Procedure, TASK 46-13-01-860-802.

NOTE: If only the OS9.1 Boot OS is present on the NFS (the NFS is in an Initial Data Load (IDL) state), skip this step.

(3) Install the core OS7.0 software parts. Do this task: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

NOTE: The IDL Software Installation task can be used to install the remaining OS7.0 software parts and return the NFS to a fully operational state.

NOTE: When staging and dataloading the rest of the OS7.0 software parts after the IDL, make sure to stage and install the OS7.0 Boot OS as well.

(4) Run a system configuration of the NFS to ensure all OS9.1 software parts have been replaced with OS7.0 software parts. Do this task: System Configuration Check, TASK 46-13-00-710-802.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-020

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-01-470-814

22. Network File Server OS Downgrade OS7.1 to OS7.0

A. General

(1) This task gives the steps to downgrade the Network File Server (NFS) OS7.1 to OS7.0.

(a) Gather all the software parts you need on removable media or Universal Maintenance Device (UMD) before leaving for the airplane.

NOTE: The OS7.0 46 NFS Boot OS is: BCG57-U000-02E3.

NOTE: The OS7.1 46 NFS Boot OS is: BCG2A-U000-0458.

(2) The aircraft must be on the ground to do this task.

(3) This task is available when the Onboard Network System (ONS) is accessed with the UMD.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)

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(Continued)

Reference	Title
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Procedure

SUBTASK 46-13-01-470-071

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

NOTE: Do not attempt to connect wirelessly.

SUBTASK 46-13-01-470-072

(2) If the NFS is in a fully operational OS7.1 state, reimage the NFS. Do this task: Network File Server Reimage Procedure, TASK 46-13-01-860-802.

NOTE: If only the OS7.1 Boot OS is present on the NFS (the NFS is in an Initial Data Load state), skip this step.

(3) Install the core OS7.0 software parts. Do this task: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

NOTE: The IDL Software Installation task can be used to install the remaining OS7.0 software parts and return the NFS to a fully operational state.

NOTE: When staging and dataloading the rest of the OS7.0 software parts after the IDL, make sure to stage and install the OS7.0 Boot OS as well.

(4) Run a system configuration of the NFS to ensure all OS7.1 software parts have been replaced with OS7.0 software parts. Do this task: System Configuration Check, TASK 46-13-00-710-802.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-022

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-01-700-802

23. NFS Issue Check (with NITRO Application)

A. General

(1) This task includes the steps to use the NFS Issue Troubleshooter and Recommended Operations (NITRO) application for the troubleshooting check of known Onboard Network System (ONS) issues.

(a) NITRO is not a replacement for acceptance testing. Maintenance tasks must be signed off in accordance with airline policy.

(2) For the NITRO to function correctly:

(a) Use NITRO only with the 737MAX.

(b) Use NITRO only with Network File Server (NFS) OS9.1.

(c) Download and install the NITRO to the universal maintenance device, SPL-11321.

1) Contact your Engineering department to install NITRO.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Prepare for the Check

SUBTASK 46-13-01-420-004

(1) Connect the universal maintenance device, SPL-11321, through the Ethernet cable as follows (TASK 46-13-00-480-801):

NOTE: Wireless connection is not an option for NITRO at this time.

(a) Connect one end of the RJ-45 Ethernet cable, to the RJ45 port of the universal maintenance device, SPL-11321.

(b) Connect the opposite end to one of these data ports:

1) Flight Compartment - On the P61 Panel, find the RJ45 ONS port on the Maintenance Built-In-Test Equipment (BITE) Panel.

2) E/E Bay - On the E4-2 rack, find the RJ45 port behind the door on the NFS front panel.

(c) Supply the electrical power to the universal maintenance device, SPL-11321.

(d) Log-in as follows:

1) Push the CTRL, ALT and DELETE keys at the same time.

a) Make sure that it shows: Log On to Windows.

2) Enter the User Name and Password and then select OK.

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E. NFS Issue Check

SUBTASK 46-13-01-470-075

(1) Double-click on the “737 MAX NITRO” desktop icon to open the NITRO application.

SUBTASK 46-13-01-470-076

(2) Select “Run Troubleshooting Check”.

NOTE: Warning window will appear to make sure that the NITRO is used on a 737MAX with ONS OS9.1 or newer.

(a) Press “Continue” if the conditions are met.

(b) If the conditions are not met, press “Cancel” and exit NITRO.

SUBTASK 46-13-01-860-031

(3) In the next window, if necessary, select options “ACARS” or “Cell or Wi-Fi Link”.

(a) Select “Continue” to run the troubleshooting check.

NOTE: If there is a problem with the NFS connection to the Universal Maintenance Device (UMD), a failure window will show.

(b) Wait about 5 seconds for each check.

SUBTASK 46-13-01-970-014

(4) Follow the troubleshooting recommendations in the next window.

(a) If the troubleshooting check status shows a green check mark, “V”, no troubleshooting is recommended for the associated troubleshooting check.

(b) If the troubleshooting check status shows a red “X”, troubleshooting action is recommended with the associated IFIM TASK.

(c) If the troubleshooting check status shows a blue “?”, do these steps:

1) Make sure that the NFS has been powered for 10 minutes.

2) Make sure that Server Credentials were not recently generated. If they were, wait 10 minutes.

3) Make sure that the amber NFS Fault light is OFF.

4) Make sure that the NFS connection to the UMD is through Ethernet cable.

5) Rerun NITRO.

(d) Save the report as necessary.

——— END OF TASK ———

TASK 46-13-01-700-803

24. Network File Server Date and Time Check

A. General

(1) This task gives the steps to determine if DPC reads correct time from MMR.

B. References

Reference	Title
31-65-00-970-804	Airplane Config Data (ACD) Page Function (P/B 201)
34-58-00-710-801	Global Positioning System - Operational Test (P/B 501)

C. Location Zones

Zone	Area
211	Flight Compartment - Left

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(Continued)

Zone	Area
212	Flight Compartment - Right

D. Network File Server Date and Time Check

SUBTASK 46-13-01-860-032

(1) Do these steps:

(a) Push the SYS button, on the MFD panel to access the MAINT page, on the forward displays.

(b) Select MAINT on the forward displays.

(c) Select MAINT CTRL PGS > AIRPLANE CONFIG DATA (ACD).

(d) Under the DATE/TIME header, note the TIME and DATE.

1) Make sure that the date and time is correct.

a) If the date and time is correct, then no action is required.

b) If the date and time is missing or incorrect, then do these steps:

<1> Press “UNLOCK PAGE”.

<2> If the date and time is editable, manually correct the date and time on the DPCs (TASK 31-65-00-970-804).

NOTE: The DPCs will only let the mechanic update the date and time if the MMR is not providing a valid date and time.

c) If the date and time is incorrect, but present and not editable, do the MMR Functional Test: Global Positioning System - Operational Test, TASK 34-58-00-710-801.

<1> If the MMR is operational, do these steps:

<a> Select MAINT CTRL PGS > AIRPLANE CONFIG DATA (ACD).

<b> Make sure that the date and time is correct.

<2> If the MMR is not operational, refer to the IFIM and do the applicable procedure(s).

——— END OF TASK ———

TASK 46-13-01-440-801

25. Network File Server (NFS) Power Cycle

A. General

(1) This task contains the procedure to power cycle the Network File Server (NFS).

B. Location Zones

Zone	Area
212	Flight Compartment - Right

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C. NFS Power Cycle

SUBTASK 46-13-01-440-002

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER

(2) Wait for 60 seconds.

(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER

——— END OF TASK ———

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NETWORK FILE SERVER - REMOVAL/INSTALLATION

1. General

A. This procedure contains two tasks:

(1) Removal of the network file server (NFS), M02898.

(2) Installation of the NFS.

TASK 46-13-01-000-801

2. Network File Server Removal

(Figure 401)

A. General

(1) This task gives the steps to remove the Network File Server (NFS).

(a) The Network File Server, M02898, is installed on shelf E4-2.

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
46-13-00-470-803	Download Files using the Software Management Tool (SMT) (P/B 201)
46-13-01-020-801	Network File Server SD Media Card Removal (P/B 201)
46-13-01-860-802	Network File Server Reimage Procedure (P/B 201)
46-13-02-970-802	Download Manager (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Removal

SUBTASK 46-13-01-970-005

(1) Before you remove the NFS, The Boeing Company recommends that you make a back-up copy of the applicable files to include Log files (Security, Syslog, and Performance) from the NFS to the universal maintenance device, SPL-11321, do one of these tasks:

- Download Files using the Software Management Tool (SMT), TASK 46-13-00-470-803

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- Download Manager, TASK 46-13-02-970-802.

NOTE: Know and follow your airline or operator's procedure for record retention.

NOTE: If the NFS has a solid amber fault light, it will not be possible to download files. The NFS must be removed without downloading files.

SUBTASK 46-13-01-860-013

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER
C	2	C04067	ONS DATALOAD RLY

SUBTASK 46-13-01-010-001

(3) Get access to the main equipment center.

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

G. NFS Removal

SUBTASK 46-13-01-020-003

(1) Do this task: Network File Server SD Media Card Removal, TASK 46-13-01-020-801.

SUBTASK 46-13-01-020-001

(2) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 46-13-01-020-002



CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(3) Remove the network file network file server [1] (TASK 20-10-07-000-801).

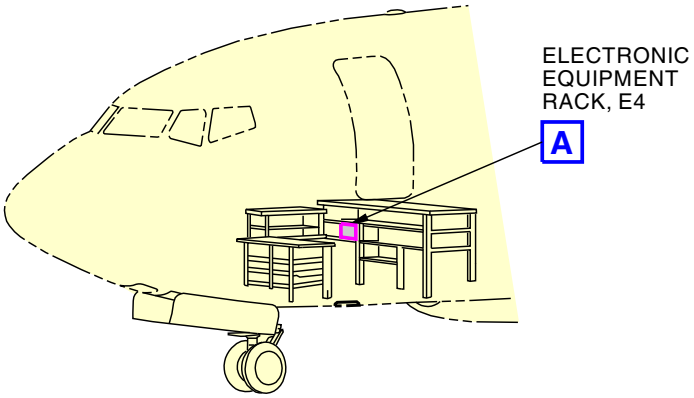
(a) Put protective covers on the electrical connectors.

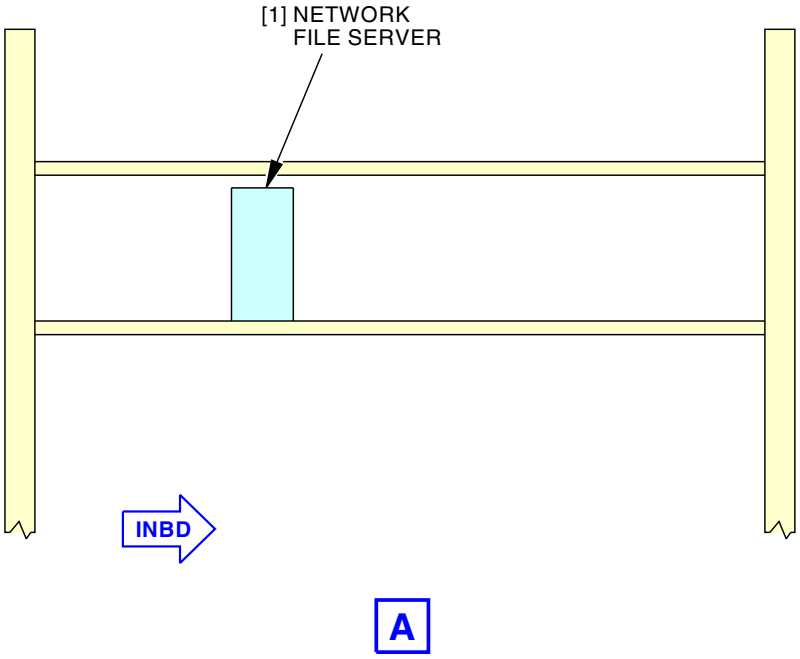
SUBTASK 46-13-01-070-002

(4) If the network file server [1] is sent for a repair or decommissioning, reimage the network file server [1], do this task: Network File Server Reimage Procedure, TASK 46-13-01-860-802.

——— END OF TASK ———

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Network File Server Installation
Figure 401/46-13-01-990-801

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TASK 46-13-01-400-801

3. Network File Server Installation

Figure 401

A. General

(1) This task gives the steps to install the network file server (NFS), M02898.

(2) The NFS is installed on shelf E4-2 in the E/E bay.

(3) To complete this task, a universal maintenance device, SPL-11321 is required.

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
46-13-00-860-801	Airplane Credentials Check (P/B 201)
46-13-01-020-801	Network File Server SD Media Card Removal (P/B 201)
46-13-01-420-801	Network File Server SD Media Card Installation (P/B 201)
46-13-01-470-804	Network File Server Automatic Stage 1 Restore Function (P/B 201)
46-13-01-470-806	Network File Server Stage 2 Restore (P/B 201)
46-13-01-470-807	Network File Server Initial Data Load (IDL) Software Installation (P/B 201)
46-13-01-710-801	Network File Server Operational Test (P/B 501)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Network file server	46-13-01-01-010	PTF ALL

E. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

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G. Prepare for the Installation

SUBTASK 46-13-01-860-014

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER
C	2	C04067	ONS DATALOAD RLY

SUBTASK 46-13-01-010-002

(2) Make sure that this access panel is open:

Number	Name/Location
117A	Electronic Equipment Access Door

H. NFS Installation Procedure

SUBTASK 46-13-01-860-015

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(1) Do this task: (ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802).

SUBTASK 46-13-01-010-003

(2) Remove the protective covers from the connectors and examine the electrical connectors and wires for damage, and dirt.

(a) Repair the problems that you find.

SUBTASK 46-13-01-420-001

(3) Install the network file server [1]. This is the task: (E/E Box Installation, TASK 20-10-07-400-801).

SUBTASK 46-13-01-470-020

(4) If applicable, do this task: Network File Server SD Media Card Installation, TASK 46-13-01-420-801.

SUBTASK 46-13-01-860-016

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER
C	2	C04067	ONS DATALOAD RLY

(a) If the installed NFS was removed in a serviceable state to conduct maintenance unrelated to the NFS, do these steps:

1) Do this task: Network File Server Operational Test, TASK 46-13-01-710-801.

2) Skip to step “Put the Airplane Back to Its Usual Condition”.

NOTE: If the NFS contains an SD Card with a valid backup, and has ONS 7.1 or later, the NFS will automatically initiate a stage 1 restore upon power up, as described in Network File Server Automatic Stage 1 Restore Function, TASK 46-13-01-470-804. The NFS will automatically install software parts, shutdown, and reboot. The process will take approximately 30 minutes.

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(6) If the NFS does not automatically restore data from the SD Media Card, or an SD Media Card is not present, do this task: Network File Server Initial Data Load (IDL) Software Installation, TASK 46-13-01-470-807.

SUBTASK 46-13-01-860-028

(7) If the automatic Stage 1 Restore fails, remove the SD Media Card (TASK 46-13-01-020-801) and conduct a manual Initial Data Load (TASK 46-13-01-470-807).

NOTE: The automatic Stage 1 Restore has failed if the NFS does not become operational on the UMD or the MFD 30 minutes after closing the NFS circuit breaker.

SUBTASK 46-13-01-470-029

(8) If the automatic Stage 1 Restore completes successfully, and the airplane requires the 46 NFS OAS AIRPLANE KEYS UMS, do this task: Network File Server Stage 2 Restore, TASK 46-13-01-470-806.

(9) If the automatic Stage 1 Restore failed, and the Network File Server (NFS) was manually loaded, a new CSR must be generated and a new 46 NFS OAS AIRPLANE KEYS UMS must be created. This is the task: Airplane Credentials Check, TASK 46-13-00-860-801.

(10) Do this task: Network File Server Operational Test, TASK 46-13-01-710-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-860-019

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) Stop the ONS maintenance browser, and disconnect the laptop.

SUBTASK 46-13-01-410-001

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

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NETWORK FILE SERVER - ADJUSTMENT/TEST

1. General

A. This procedure has two tasks:

(1) An operational test to make sure that the network file server (NFS), M02898, is serviceable.

(2) A functional test to make sure that the NFS, M02898, is serviceable.

TASK 46-13-01-710-801

2. Network File Server Operational Test

A. General

(1) This task makes sure that the Network File Server (NFS) is serviceable.

(2) The airplane must be on the ground to do this task.

(3) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Operational Test

SUBTASK 46-13-01-860-020

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

NOTE: Do not attempt to connect wirelessly.

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SUBTASK 46-13-01-010-004

(2) Get access to the main equipment center.

(a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

G. Network File Server Operational Test

SUBTASK 46-13-01-810-002

(1) For the UMD, make sure that the Onboard Maintenance browser main menu is shown.

(a) Make sure that the unbroken chain link icon shows in the top right corner next to the settings icon.

NOTE: When the chain is unbroken and is white in color, this indicates that the NFS interface is active and the ONS system is serviceable.

NOTE: When the chain is broken and is amber in color, this shows the NFS has lost communication or the NFS is rebooting. Wait approximately ten minutes before you continue.

SUBTASK 46-13-01-810-004

(2) For the MFD, make sure that the window OBND MAINT IS UNAVAILABLE does not show.

SUBTASK 46-13-01-210-003

(3) If the access method is not able to connect, examine the LED on the NFS front panel.

(a) Make sure that the POWER LED (green in color) is ON.

1) If the POWER LED is not ON, close the NFS circuit breaker and wait 10 minutes before checking the POWER LED again.

Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER

(b) Make sure that the FAULT LED (amber in color) is OFF.

1) Refer to the IFIM and do the applicable procedure(s) if any of the following conditions are observed:

- The FAULT LED is ON
- The FAULT LED flashes ON once per second
- The FAULT LED flashes ON every three seconds

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-004

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-01-720-801

3. Network File Server Functional Test

A. General

(1) This task includes the steps to make sure that the Network File Server (NFS) is serviceable.

- The NFS, M02898 is in the E/E bay on shelf E4-2.

(2) The aircraft must be on the ground before you do this task.

(3) To do this task, use the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-00-750-801	Airplane Identification Check (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Functional Test

SUBTASK 46-13-01-480-001

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

(a) Use the universal maintenance device, SPL-11321, to get access to the Onboard Network System (ONS) maintenance browser.

NOTE: Do not attempt to connect wirelessly.

SUBTASK 46-13-01-480-003

(2) If desired at any point below, instead of manually adding an Airplane Data Monitor parameter, the JSON file “Network File Server Functional Test - ADM Parameters.json” file can be imported using the following steps:

(a) Go to OTHER FUNCTIONS > Airplane Data Monitor

(b) Select IMPORT

(c) Select the JSON file

(d) Select OPEN

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(e) All parameters relevant to the bus check steps should populate the ADM page

SUBTASK 46-13-01-860-021

(3) On the UMD, make sure that the Onboard Maintenance browser main menu shows.

(a) Look at the page that shows and make sure that the unbroken chain link icon shows in the top right corner next to the settings icon.

NOTE: When the chain is unbroken and is white in color, this indicates that the NFS interface is active and ONS is serviceable.

NOTE: When the chain is broken and is amber in color, this shows the NFS has lost communication or the NFS is rebooting. Wait approximately ten minutes before you continue.

SUBTASK 46-13-01-010-005

(4) Get access to the main equipment center.

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 46-13-01-720-002

(5) In the equipment bay, look at the NFS front panel and examine the LEDs.

(a) Make sure that the POWER LED (green in color) is on.

1) If the POWER LED is not ON, close the NFS circuit breaker and wait 10 minutes before checking the POWER LED again.

Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER

(b) Make sure that the FAULT LED (amber in color) is OFF.

1) Refer to the IFIM and do the applicable procedure(s) if any of the following conditions are observed:

- The FAULT LED is ON
- The FAULT LED flashes ON once per second
- The FAULT LED flashes ON every three seconds

G. Airplane Identification Check

SUBTASK 46-13-01-720-003

(1) Do this task: Airplane Identification Check, TASK 46-13-00-750-801.

(a) Make sure that the airline identification in the “Airline ID” field is correct and is the same as the airline Identification on the MAX Display System (MDS) Airplane Config Data Screen.

(b) Make sure that the data in the “Airplane ID” field is correct and is the same as data on the MDS Airplane Config Data Screen.

(c) Make sure that the data in the “Aircraft Type” field is correct. If the field is not filled out, use the drop down menu to select the correct Aircraft Type and hit SAVE.

(d) Make sure that the correct tail number shows on the top left of the “ONBOARD MAINTENANCE” screen.

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H. ADIRU ADR-4 and IR-3 Bus Checks

SUBTASK 46-13-01-720-004

(1) Do these steps to make sure that the ADIRU ADR-4 and IR-3 Buses are functional.

(a) Set the Left and Right ADIRUs to the NAV mode.

(b) On the ONS Main menu, select “OTHER FUNCTIONS” and then “Airplane Data Monitor”.

(c) Select “ADD”, “ARINC 429” and enter the following data:

1) Parameter Name: adrequipid

2) Bus: W-ADIRU-L ADR-4

3) Label: 377

4) SDI: 01

5) Data Type: Binary Representation (BIN)

6) MSB-LSB: 22-11

(d) Select SAVE.

1) Make sure that the time that shows with adrequipid is updating.

(e) On the Airplane Data Monitor page, select “ADD”, “ARINC 429” and enter the following data:

1) Parameter Name: irequipid

2) Bus: W-ADIRU-L IR-3

3) Label: 377

4) SDI: 01

5) Data Type: Binary Representation (BIN)

6) MSB-LSB: 22-9

(f) Select SAVE.

1) Make sure that the time displayed with irequipid is updating.

I. DPC GP Bus Check

SUBTASK 46-13-01-720-005

(1) Do these steps to make sure that the DPC GP Bus is functional.

(a) On the Airplane Data Monitor page, select “ADD”, “ARINC 429” and enter the following data:

1) Parameter Name: dsw01a

2) Bus: W-DPC-1 MDS-GP 1

3) Label: 270

4) SDI: 01

5) Data Type: Binary Representation (BIN)

6) MSB-LSB: 31-9

(b) Select SAVE.

1) Make sure that the time that shows with dsw01a is updating.

NOTE: It is not necessary to confirm the value that shows. This check is to make sure that the data is present and you see the time change.

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(c) Open these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	10	C04022	DISPLAY DPC 1 HOLDUP

(d) Make sure that the time that shows with dsw01a is not updating.

(e) Close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	10	C04022	DISPLAY DPC 1 HOLDUP

NOTE: The wait time for DPC 1 power-up is approximately three minutes.

(f) On the Airplane Data Monitor page, select “Add”, “ARINC 429” and enter the following data:

1) Parameter Name: dsw02a
2) Bus: W-DPC-2 MDS-GP 1
3) Label: 270
4) SDI: 01
5) Data Type: Binary Representation (BIN)
6) MSB-LSB: 31-9

(g) Select SAVE.

1) Make sure that the “Value” that shows with dsw02a is updating.

NOTE: It is not necessary to confirm the value that shows. This check is to make sure that the data is present and you see the time change.

(h) Open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	11	C04025	DISPLAY DPC 2 PRI

(i) Make sure that the time that shows with dsw02a is not updating.

(j) Close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	11	C04025	DISPLAY DPC 2 PRI

NOTE: The wait time for DPC 2 power-up is approximately three minutes.

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J. FMC-02 and FMC-09 Bus Check

SUBTASK 46-13-01-860-022

(1) Set the Flight Management Computer (FMC) source select switch, on the P5 forward overhead panel, to BOTH ON L.

SUBTASK 46-13-01-720-006

(2) Do these steps to make sure that the FMC-02 and FMC-09 Busses are functional.

(a) On the Airplane Data Monitor page, select “ADD”, “ARINC 429” and enter the following data:

1) Parameter Name: fmcdisc1
2) Bus: W-FMC-1 Relay FMC-02
3) Label: 270
4) SDI: XX
5) Data Type: Binary Representation (BIN)
6) MSB-LSB: 31-9

(b) Select SAVE.

(c) Select “ADD”, “ARINC 429” and enter the following data:

1) Parameter Name: fmcdisc2a
2) Bus: W-FMC-1 Relay FMC-09
3) Label: 271
4) SDI: XX
5) Data Type: Binary Representation (BIN)
6) MSB-LSB: 31-9

(d) Select SAVE.

1) Make sure that the time that shows with fmcdisc1 and fmcdisc2a is updating.

NOTE: It is not necessary to confirm the value that shows. This check is to make sure that the data is present and you see the time change.

(e) Open this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	6	C01017	FMCS CMPTR 1

(f) Make sure that the time that shows with fmcdisc1 and fmcdisc2a is not updating.

(g) Close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	6	C01017	FMCS CMPTR 1

K. ATC Bus Check

SUBTASK 46-13-01-720-007

(1) Do these steps to make sure that the Air Traffic Control (ATC) Bus is functional.

(a) On the Airplane Data Monitor page, select “ADD”, “ARINC 429” and enter the following data:

1) Parameter Name: atcdata

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2) Bus: W-ATC-1 General Output 1
3) Label: 275
4) SDI: XX
5) Data Type: Binary Representation (BIN)
6) MSB-LSB: 31-9

(b) Select SAVE.

1) Make sure that the time that shows with atcdata is updating.

NOTE: It is not necessary to confirm the value that shows. This check is to make sure that the data is present and you see the time change.

(c) Open this circuit breaker:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
B	5	C00186	ATC 1

(d) Make sure that the time that shows with atcdata is not updating.

(e) Close this circuit breaker:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
B	5	C00186	ATC 1

L. MMR Bus Check

SUBTASK 46-13-01-720-008

(1) Do these steps to make sure that the Multi-Mode Receiver (MMR) Bus is functional.

(a) On the Airplane Data Monitor page, select “ADD”, “ARINC 429” and enter the following data:

1) Parameter Name: mmrequipid
2) Bus: W-MMR-1 GNSS Data 3
3) Label: 377
4) SDI: XX
5) Data Type: Binary Representation (BIN)
6) MSB-LSB: 31-9

(b) Select SAVE.

1) Make sure that the time that shows with mmrequipid is updating.

NOTE: It is not necessary to confirm the value that shows. This check is to make sure that data is present and you see the time change.

(c) Open this circuit breaker:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
A	2	C01479	RADIO NAVIGATION MMR 1

(d) Make sure that the time that shows with mmreqipid is not updating.

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(e) Close this circuit breaker:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
A	2	C01479	RADIO NAVIGATION MMR 1

M. CMU Bus Check

SUBTASK 46-13-01-720-009

(1) Do these steps to make sure that the Communications Management Unit (CMU) Bus is functional.

NOTE: The CMU is an optional Line Replaceable Unit (LRU). This test is not required if CMU is not installed.

(a) On the Airplane Data Monitor page, select “ADD”, “ARINC 429” and enter the following data:

1) Parameter Name: OutStatusWord1
2) Bus: W-CMU Output Bus 3
3) Label: 270
4) SDI: 00
5) Data Type: Binary Representation (BIN)
6) MSB-LSB: 31-9

(b) Select SAVE.

1) Make sure that the time that shows with OutStatusWord1 is updating.

NOTE: It is not necessary to confirm the value that shows. This check is to make sure that the data is present and you see the time change.

(c) Open this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

(d) Make sure that the time that shows with OutStatusWord1 is not updating.

(e) Close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

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N. Weight on Wheels Discrete Check

SUBTASK 46-13-01-720-014

(1) Make sure that the Weight-On-Wheels (WOW) Discrete pin is functional.

(a) On the Airplane Data Monitor page, select "ADD".

(b) Select "PREDEFINED".

(c) Select “wow”.

(d) Select SAVE.

(e) Make sure that the wow value is true.

(f) Make sure that the time that shows with wow is updating.

(g) To use the MDS two inboard displays (MFDs), do this task: Multi Function Display Connection, TASK 46-13-00-480-803 .

(h) Push the “SYS” button on the P9-82 Multi Function Display (MFD) panel.

(i) Select "OTHER FUNCTIONS" on the MDS.

(j) Select “SPECIAL FUNCTIONS” .

(k) Select “32 - Proximity Switch Electronic Unit" under Select ATA System.

(l) Select “PSEU - Air/Ground Override” under Select System Function.

(m) Select Continue.

(n) Select Continue under PRECONDITIONS.

(o) Set "GND TEST ENABLE SWITCH" to "ENABLE" on the MAINT/BITE PANEL (P61).

(p) Select "START FUNCTION" under FUNCTION.

(q) Select "Set SYS 1 and 2 IN AIR".

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE CONTROL SURFACES AND LANDING GEAR DOOR AREAS. THE CONTROL SURFACES, THE LANDING GEAR, AND THE LANDING GEAR DOORS CAN MOVE WHEN YOU DO THE AIR MODE SIMULATION. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(r) Select CONTINUE.

(s) Make sure that the following is observed on the MDS: Air/Gnd Override is ACTIVE, System 1: AIR, and System 2: AIR.

(t) Make sure that the WOW value is “false” on the UMD.

(u) Make sure that the time that shows with WOW is updating on the UMD.

(v) Go back to the MDS.

(w) Select "OPERATE TEST AGAIN" .

(x) Select CONTINUE.

(y) Select "Set SYS 1 and 2 ON GND".

(z) Select CONTINUE.

(aa) Make sure that the following is observed on the MDS: Air/Gnd Override is NOT ACTIVE, System 1: GND, and System 2: GND.

(ab) Select STOP FUNCTION.

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O. Ethernet Interfaces Check

SUBTASK 46-13-01-720-015

(1) Make sure that the Ethernet Interfaces are functional.

(a) Use a UMD to do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: Do not attempt to connect wirelessly.

(b) Select “EXTENDED MAINTENANCE” and then “Input Monitoring” (on the maintenance device “ONBOARD MAINTENANCE” screen).

(c) Select “NFS” under "Select an item to monitor".

(d) Select “ETHERNET” on the bottom of the webpage.

(e) Make sure that the “Status” of DPC-1 is “Up”.

(f) Make sure that the “Status” of DPC-2 is “Up”.

(g) Open these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	10	C04022	DISPLAY DPC 1 HOLDUP

1) Make sure that the “Status” of DPC-1 is “Down”.

(h) Close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	10	C04022	DISPLAY DPC 1 HOLDUP

(i) Make sure that the “Status” of DFDAU is “Up”.

(j) Make sure that the “Status” of MMR-L is “Up”.

(k) Make sure that the “Status” of MMR-R is “Up”.

PTF ALL POST SB 737-44-1020

(l) Make sure that the “Status” of WAP-1 is “Up”.

PTF ALL

(m) To use the MDS two inboard displays (MFDs), do this task: Multi Function Display Connection, TASK 46-13-00-480-803.

(n) Select “MAINT CTRL PGS”.

(o) Select “MISC SYSTEM CTRLS (MSC)”.

(p) Select the "TEST" screen element Diamond for ENG 1 (under EEC MAINT POWER box)

(q) Select the "TEST" screen element Diamond for ENG 2 (under EEC MAINT POWER box)
.

(r) Make sure that the “STATUS: PWR” is now displayed below ENG 1 and 2.

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(s) Check the following items in input monitoring on the UMD:

1) Make sure that the “Status” of EEC-1A is “UP”.
2) Make sure that the “Status” of EEC-1B is “UP”.
3) Make sure that the “Status” of EEC-2A is “UP”.
4) Make sure that the “Status” of EEC-2B is “UP”.
5) Make sure that the “Status” of EGPWS is “UP”.

P. DFDAU A717 Receive Busses Check

SUBTASK 46-13-01-720-012

(1) Make sure that the A717 receive buses from the Digital Flight Data Acquisition Unit (DFDAU) to the NFS are functional.

(a) Select the MENU button on Control Display Unit (CDU).
(b) Select ACMS.
(c) Select MAINTENANCE.
(d) Select INTERFACE TESTS.
(e) Select RECORDER.
(f) Select CONFIG DAR OUTPUT.
(g) Select LSK2 LEFT "ON".

NOTE: The preceding instructions enable DAR output from DFDAU, the SUBFRAME counts up on the bottom of the CDU screen.

NOTE: There is a delay before INPUT MONITORING page shows "A717 STATUS ACTIVE".

(h) Use a UMD to do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: Do not attempt to connect wirelessly.

(i) Select "EXTENDED MAINTENANCE" and then "Input Monitoring" (on the maintenance device "ONBOARD MAINTENANCE" screen).
(j) Select NFS.
(k) Select QAR RECEIVE on the bottom of the webpage.

1) Make sure that the status of the DAR and RS422 Buses show ACTIVE.

Q. DFDAU Avionic Data Stream to NFS Check

SUBTASK 46-13-01-720-013

(1) Do these steps to make sure that the DFDAU Avionic Data Stream to NFS is functional.

(a) Select the horizontal striped menu icon on the top left of the screen, then select the Home Button within the ONS window.
(b) Select "LINE MAINTENANCE".
(c) Select "Security".
(d) Select the "DFDAU" checkbox.
(e) Select the “GENERATE" button.
(f) Select “CONTINUE”.
(g) Make sure that the DFDAU shows "Successfully retrieved and installed credentials" under CREDENTIAL GENERATION STATUS.

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R. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-01-080-005

(1) Exit the ONS maintenance browser and disconnect the UMD.

SUBTASK 46-13-01-040-001

(2) If it is necessary, remove the power from the EEC (TASK 73-21-00-800-801-G00).

SUBTASK 46-13-01-410-002

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

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ONBOARD MAINTENANCE FUNCTION - MAINTENANCE PRACTICES

1. General

A. This document has the Onboard Maintenance Function (OMF) tasks that follow:

(1) OMF Ground Test
(2) Fault Guidance Check
(3) Inbound Flight Deck Effects Check
(4) Existing Flight Deck Effects Check
(5) Present Leg Faults Check
(6) Existing Faults Check
(7) Fault History Check
(8) Maintenance Enable/Disable Operation
(9) Special Functions
(10) Download Manager
(11) LRU Reports
(12) Off-board Links
(13) Shop Faults
(14) Maintenance Data Pages
(15) Erasing a Latched Status Message - Verification Check

TASK 46-13-02-710-801

2. Onboard Maintenance Function Ground Test

(Figure 201)

A. General

(1) This task gives the steps to do the Onboard Maintenance Function (OMF) Ground Test.

(2) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

(3) There are three types of ground tests:

- System Test.
- Operational Test.
- Line Replaceable Unit (LRU) Replacement Test.

NOTE: It is possible that there is more than one ATA system test, or operational test, or LRU replacement test. It is necessary to first make a selection of the ATA system. Then, select the applicable test to operate the ground test function.

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(a) All three test types are not available for all ATA systems. The types of ground tests that are available to you automatically show after you make the ATA system selection.

NOTE: Ground tests are interactive or non-interactive. It is possible that the ground test you select is an Interactive Test. Interactive Tests have instructions and steps that you do while the test is in operation. These steps and instructions include opening and closing circuit breakers, moving switch settings, and other types of actions. Non-interactive tests do not require actions while the test is in progress.

NOTE: If the Network File Server circuit breaker is opened at any time, wait at least 5 seconds before closing the circuit breaker.

B. References

Reference	Title
29-11-00 P/B 201	HYDRAULIC SYSTEMS A AND B - MAINTENANCE PRACTICES
31-65-00-970-804	Airplane Config Data (ACD) Page Function (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Test

SUBTASK 46-13-02-840-001

(1) These are the ground tests available:

GROUND TESTS

ATA	TEST TYPE	TEST NAME	INTERACTIVE TEST?
21 - Air Conditioning	System Test	Air Conditioning IBIT - Left	NO
21 - Air Conditioning	System Test	Air Conditioning IBIT - Right	NO
21 - Air Conditioning	System Test	Air Conditioning IBIT with Flow Sensor Test - Left	NO
21 - Air Conditioning	System Test	Air Conditioning IBIT with Flow Sensor Test - Right	NO
27 - Flight Controls - Stall Management Yaw Damper System	System Test	SMYD 1 - Self Test	YES

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GROUND TESTS (Continued)

ATA	TEST TYPE	TEST NAME	INTERACTIVE TEST?
27 - Flight Controls - Stall Management Yaw Damper System	System Test	SMYD 1 - Servo Test	YES
27 - Flight Controls - Stall Management Yaw Damper System	System Test	SMYD 2 - Self Test	YES
27 - Flight Controls - Stall Management Yaw Damper System	System Test	SMYD 2 - Servo Test	YES
27 - Spoiler Control	Operational Test	Control Wheel	YES
27 - Spoiler Control	Operational Test	SCE Test	YES
27 - Spoiler Control	Operational Test	Speedbrake Lever	YES
27 - Spoiler Control	System Test	Flight Spoilers	YES
27 - Spoiler Control	System Test	Ground Spoilers	YES
31 - Indicating System	System Test	DPC-1 CMU Chime Test	YES
31 - Indicating System	System Test	DPC-1 MAINT Light Test	YES
31 - Indicating System	System Test	DPC-2 CMU Chime Test	YES
31 - Indicating System	System Test	DPC-2 MAINT Light Test	YES
31 - Indicating System	LRU Replacement Test	DPC-1 NVM Erase Function	NO
31 - Indicating System	LRU Replacement Test	DPC-1 Replacement Test	NO
31 - Indicating System	LRU Replacement Test	DPC-2 NVM Erase Function	NO
31 - Indicating System	LRU Replacement Test	DPC-2 Replacement Test	NO
32 - Proximity Switch Electronic Unit	Operational Test	PSEU - LGTV Test	NO
32 - Proximity Switch Electronic Unit	Operational Test	PSEU - Self Test	NO
32 - Proximity Switch Electronic Unit	LRU Replacement Test	PSEU - Replacement Test	YES
32 - Antiskid/Autobrake system	Operational Test	AACU Operational Test	YES
36 - Pneumatics	Operational Test	Protection Gate - Left	YES
36 - Pneumatics	Operational Test	Protection Gate - Right	YES
36 - Pneumatics	System Test	Pneumatic Engine ON - Left	YES
36 - Pneumatics	System Test	Pneumatic Engine ON - Right	YES
36 - Pneumatics	LRU Replacement Test	Electrical LRU - Left	YES

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GROUND TESTS (Continued)

ATA	TEST TYPE	TEST NAME	INTERACTIVE TEST?
36 - Pneumatics	LRU Replacement Test	Electrical LRU - Right	YES
36 - Pneumatics	LRU Replacement Test	Pneumatic Engine OFF - Left FAMV	YES
36 - Pneumatics	LRU Replacement Test	Pneumatic Engine OFF- Right FAMV	YES
36 - Pneumatics	LRU Replacement Test	Pneumatic Engine OFF - Left HPSOV	YES
36 - Pneumatics	LRU Replacement Test	Pneumatic Engine OFF - Right HPSOV	YES
36 - Pneumatics	LRU Replacement Test	Pneumatic Engine OFF - Left PRSOV	YES
36 - Pneumatics	LRU Replacement Test	Pneumatic Engine OFF- Right PRSOV	YES
71 - 80 Left Engine	Operational Test	Actuator Test	YES
71 - 80 Left Engine	Operational Test	Thrust Reverser Interlock Test	YES
71 - 80 Left Engine	System Test	Electronic Engine Controller Blower Test	YES
71 - 80 Left Engine	System Test	Electronic Engine Controller Test	NO
71 - 80 Left Engine	System Test	Engine Idle Test	YES
71 - 80 Left Engine	System Test	Harness Test	YES
71 - 80 Left Engine	System Test	Ignition Channel A Test	YES
71 - 80 Left Engine	System Test	Ignition Channel B Test	YES
71 - 80 Left Engine	System Test	Wet Motor Test	YES
71 - 80 Right Engine	Operational Test	Actuator Test	YES
71 - 80 Right Engine	Operational Test	Thrust Reverser Interlock Test	YES
71 - 80 Right Engine	System Test	Electronic Engine Controller Blower Test	YES
71 - 80 Right Engine	System Test	Electronic Engine Controller Test	NO
71 - 80 Right Engine	System Test	Engine Idle Test	YES
71 - 80 Right Engine	System Test	Harness Test	YES
71 - 80 Right Engine	System Test	Ignition Channel A Test	YES
71 - 80 Right Engine	System Test	Ignition Channel B Test	YES
71 - 80 Right Engine	System Test	Wet Motor Test	YES

SUBTASK 46-13-02-480-013

(2) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 can access the ONS.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 46-13-02-860-053

(3) If hydraulic power was removed, do these steps:

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Supply hydraulic power to the applicable system (PAGEBLOCK 29-11-00/201).

(b) Cycle the LRU/flight control before performing the ground test(s).

NOTE: Ground tests may fail if LRU/flight controls are not cycled before running the ground test.

F. Ground Test Procedure

SUBTASK 46-13-02-700-019

(1) Verify the correct time and date is shown on the Airplane Configuration Data (ACD) page. This is the task: Airplane Config Data (ACD) Page Function, TASK 31-65-00-970-804.

SUBTASK 46-13-02-700-001

(2) Do the applicable ground test:

(a) On the P61 maintenance Built-In-Test Equipment (BITE) panel, lift the switchguard and set the GND TEST switch to ENABLE.

(b) For the UMD, select LINE MAINTENANCE > Ground Test.

(c) For the MFD, select LINE MAINT > GROUND TESTS.

(d) On the Ground Test page, select the ATA system.

NOTE: Use the scroll bar, the up or down arrows, or the page numbers (1, 2, 3) to find the ATA if it does not show.

(e) Select the test type (System, Operational, or LRU Replacement).

NOTE: All three test types are not available for all ATA systems. The types of ground tests that are available show after you make the ATA system selection.

(f) Select the applicable test.

(g) Select the CONTINUE button.

1) Make sure that the test PRECONDITIONS screen shows.

(h) Follow all steps and instructions that show on the PRECONDITIONS screen.

(i) After you do a check to make sure that all preconditions that show are set and are correct on this page, select the CONTINUE button.

(j) Make sure the TEST screen shows and continues to show through the test sequence.

(k) Make sure that TEST CONDITION: READY shows.

NOTE: If TEST CONDITION: READY does not show, the preconditions have not been met.

(l) Select the START TEST button.

NOTE: It is possible that a percent -complete progress indicator shows.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that TEST CONDITION: OPERATING shows and that the START TEST button changes to STOP TEST.

NOTE: The STOP TEST button may be grey indicating that the test is stoppable or cyan indicating that the test is not stoppable.

2) Make sure that you do not see TEST CONDITION: INHIBITED, or TEST INTERFERENCE, or CANNOT OPERATE, or STOPPED.

3) If the test is interactive, make sure that you do all the steps that show.

NOTE: If interactive test is not performed in a timely manner, it may result in the test timing out and not completing.

(m) Make sure that when the test is complete, there are no faults and you see the TEST CONDITION: PASS message.

(n) If you see TEST CONDITION: FAILED, then review the fault messages on the current screen. Refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(o) To see the POST CONDITIONS screen, select the POST CONDITIONS icon on the footer at the bottom of the page.

(p) Follow all steps and instructions that show on the POST CONDITIONS screen to put the airplane to its normal condition.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-860-002

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

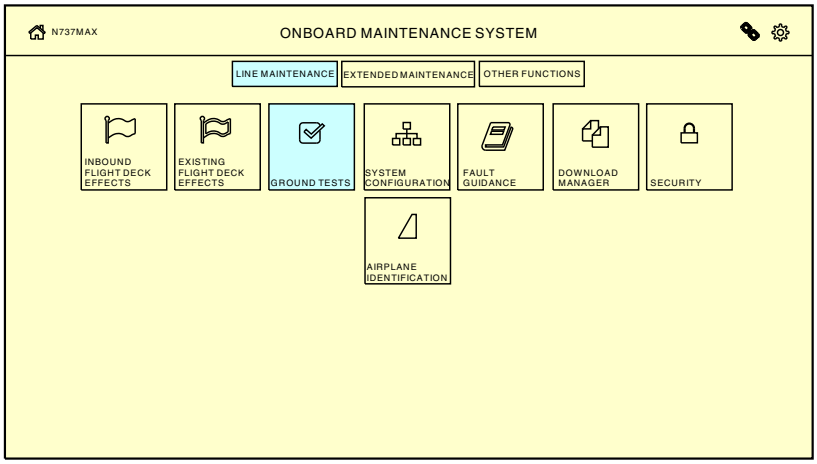
(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

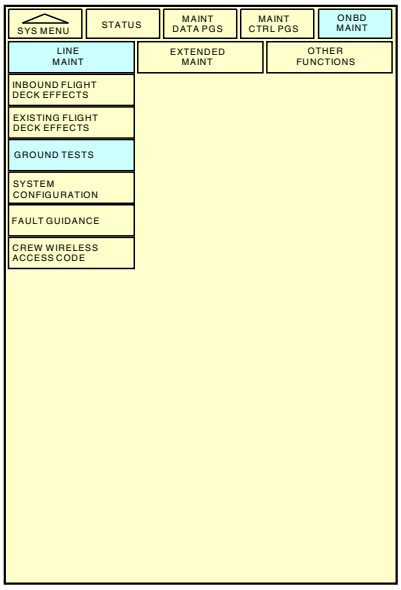
(3) On the P61 Panel, set the GND TEST switch to NORM and close the switchguard.

——— END OF TASK ———

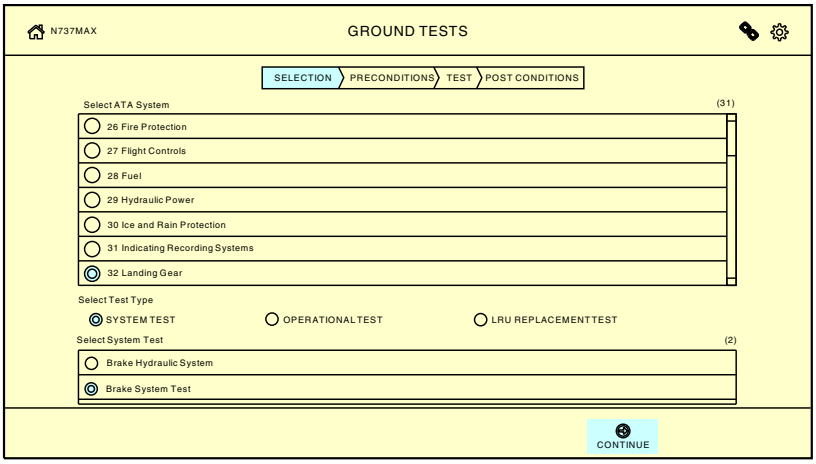
ECCN 9E991 BOEING PROPRIETARY - See title page for details



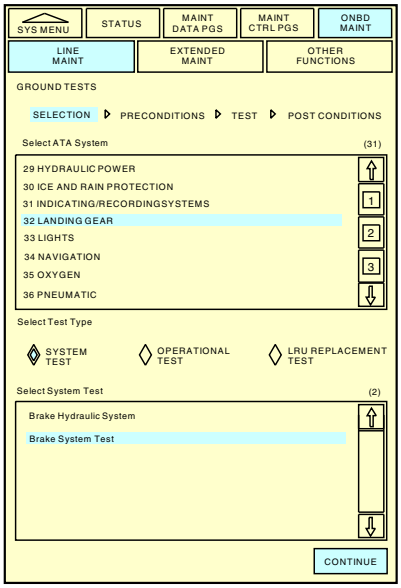
UNIVERSAL MAINTENANCE DEVICE (UMD)



INBOARD FORWARD DISPLAY



UNIVERSAL MAINTENANCE DEVICE (UMD)



INBOARD FORWARD DISPLAY

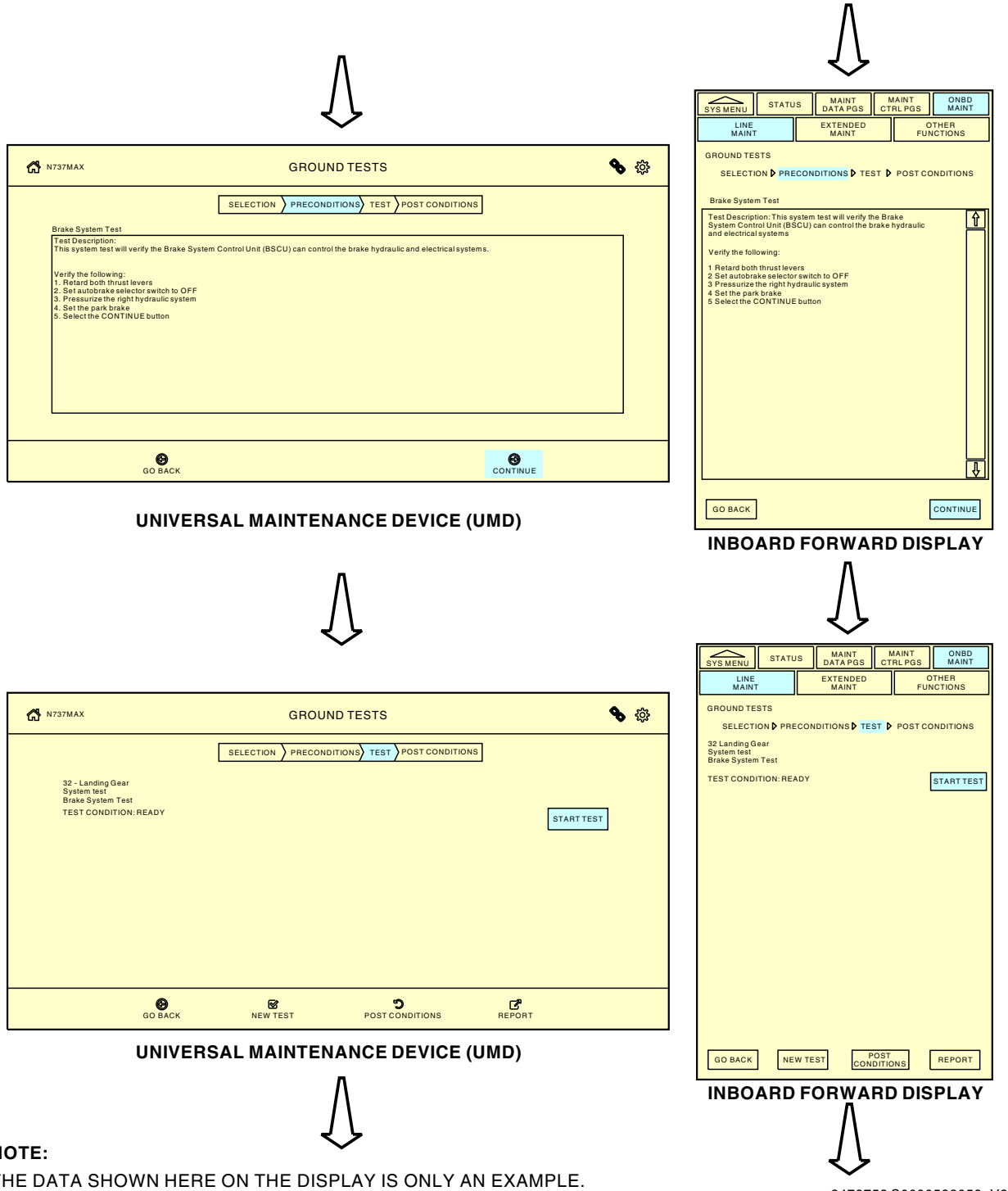
NOTE:

THE DATA SHOWN HERE ON THE DISPLAY IS ONLY AN EXAMPLE.

2478749 S0000582015_V3

Onboard Maintenance Function Ground Test
Figure 201/46-13-02-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



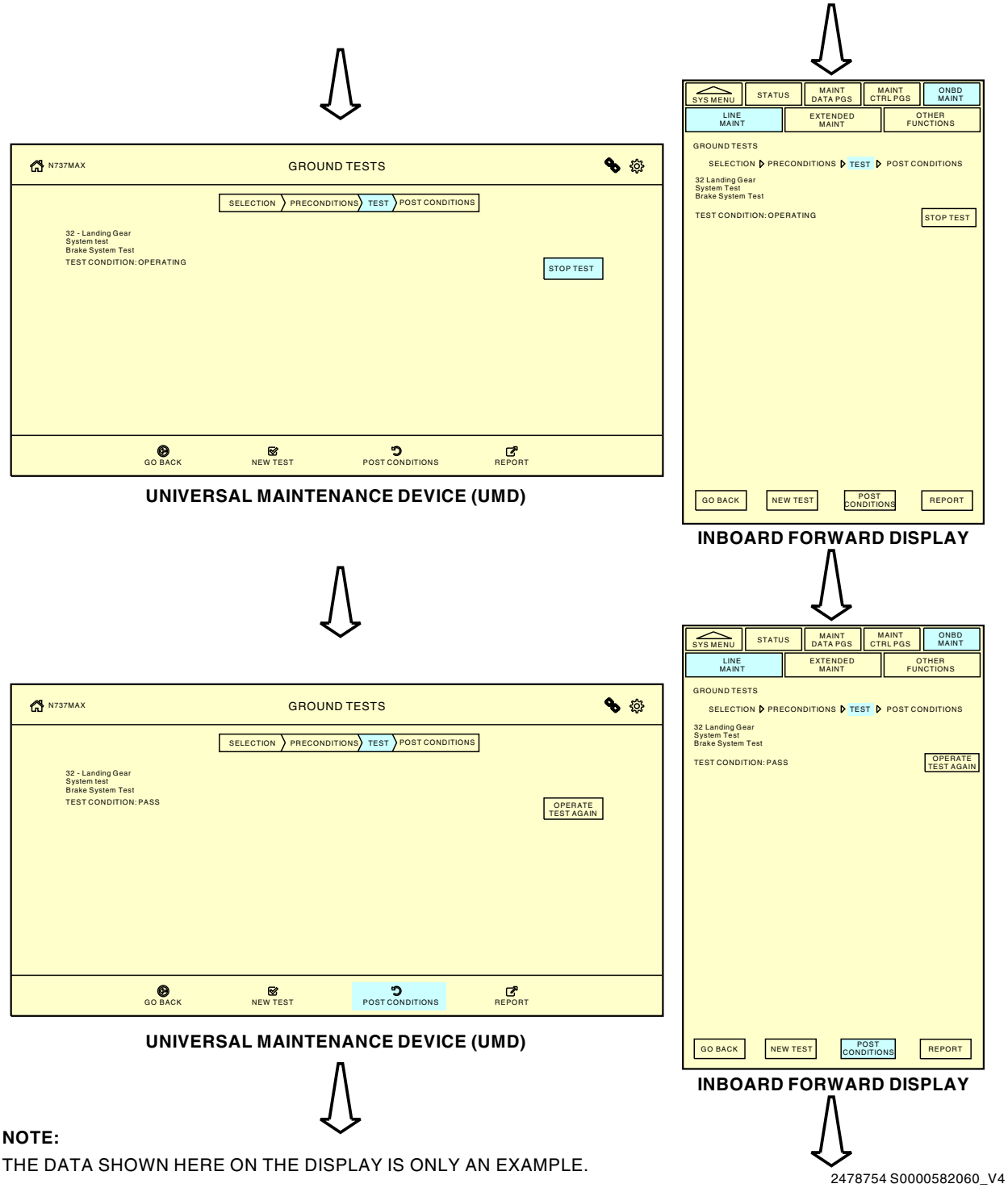
NOTE:

THE DATA SHOWN HERE ON THE DISPLAY IS ONLY AN EXAMPLE.

2478753 S0000582059_V3

Onboard Maintenance Function Ground Test
Figure 201/46-13-02-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



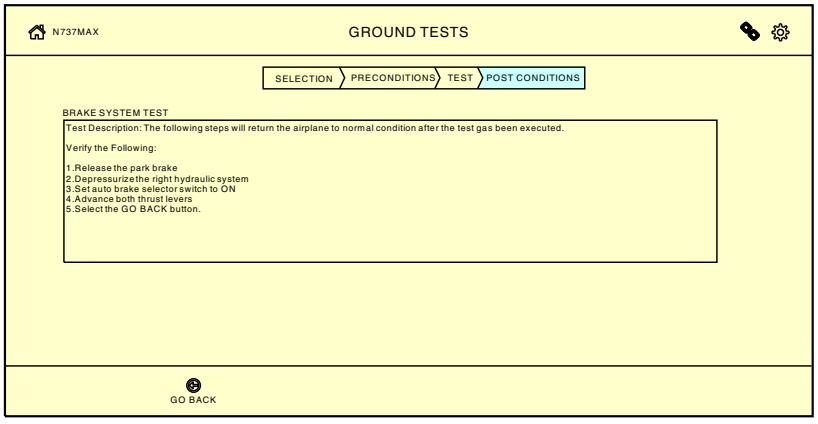
NOTE:

THE DATA SHOWN HERE ON THE DISPLAY IS ONLY AN EXAMPLE.

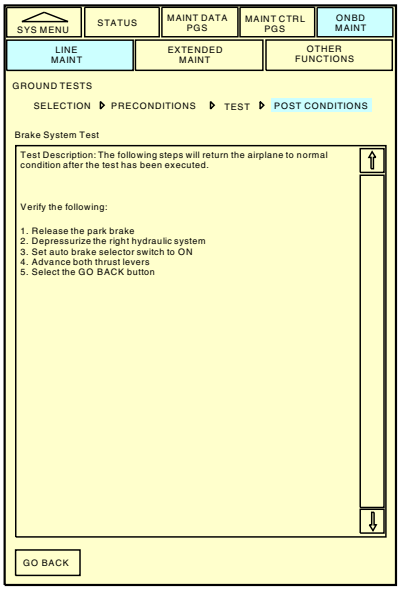
2478754 S0000582060_V4

Onboard Maintenance Function Ground Test
Figure 201/46-13-02-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



UNIVERSAL MAINTENANCE DEVICE (UMD)



INBOARD FORWARD DISPLAY

NOTE:
THE DATA SHOWN HERE ON THE DISPLAY IS ONLY AN EXAMPLE.

2478755 S0000582061_V3

Onboard Maintenance Function Ground Test
Figure 201/46-13-02-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-02-700-801

3. Fault Guidance Check

(Figure 202)

A. General

(1) This procedure gives the instructions to do a fault guidance check.

(2) Fault guidance shows the information that follows:

- ATA Chapter-Section.
- ATA Abbreviated System Name.
- Indication if maintenance information is available at the LRU BITE Panel, CDU and/or via Onboard Maintenance Function (OMF) and where information resides.
- There is a placeholder for IFIM reference, but it is blank for all systems currently and may be used for future functionality.

(3) Use this task when you require general information about faults at the ATA level.

(4) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Procedure

SUBTASK 46-13-02-480-016

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

F. Procedure

SUBTASK 46-13-02-700-002

(1) Do the Fault Guidance Check as follows:

(a) For the UMD, select LINE MAINTENANCE > Fault Guidance.

(b) For the MFD, select LINE MAINT > FAULT GUIDANCE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Review the Fault Guidance page to see troubleshooting information.

(d) Use the scroll bar if necessary to show additional faults.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-080-002

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

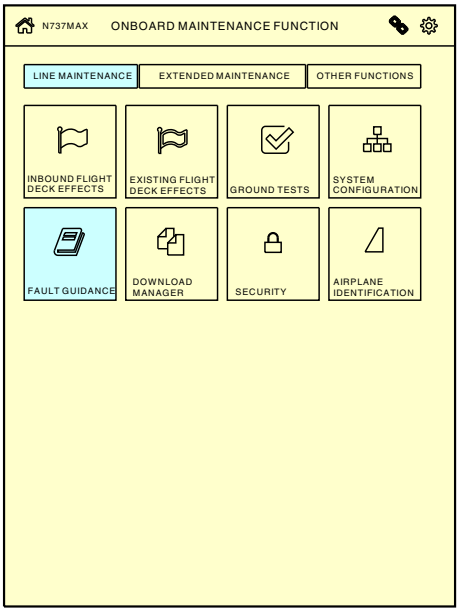
(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

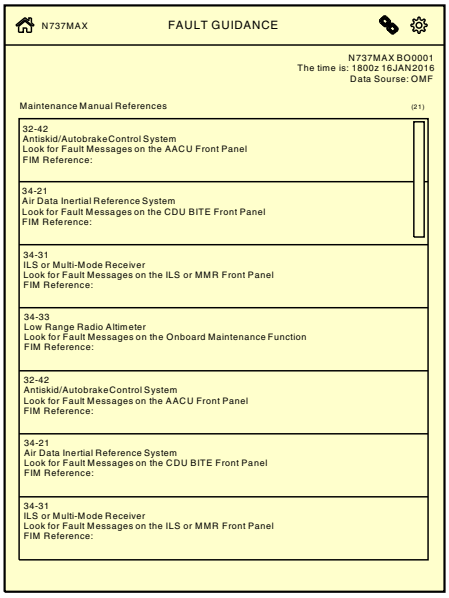
(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

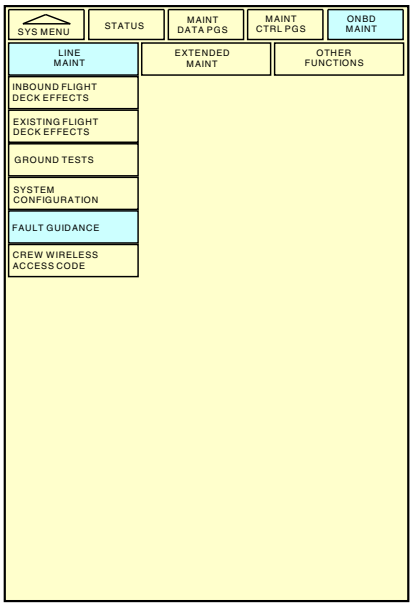
ECCN 9E991 BOEING PROPRIETARY - See title page for details

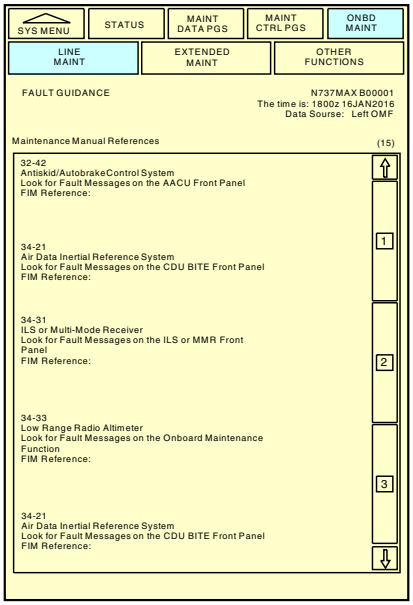
UNIVERSAL MAINTENANCE DEVICE (UMD)





MULTIFUNCTION DISPLAY





NOTE:

THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

2478058 S0000581841_V4

Fault Guidance Check
Figure 202/46-13-02-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-02-700-802

4. Inbound Flight Deck Effects Check

Figure 203

A. General

(1) This procedure gives the instructions to do an inbound flight deck effects check.

NOTE: If performance on the Inbound Flight Deck Effects page is slow, this may be due to a large amount of maintenance message instances. To fix this issue, delete the fault history (TASK 46-13-02-700-806). If no action is taken, all functionality on the Inbound Flight Deck Effects page will still be available but performance may be slower.

(2) Inbound flight deck effects (FDE) show the FDE's recorded during the present leg.

(3) The information that is shown includes:

(a) An initial summary and status of correlated FDE's with date and time, fault description, type, fault code, and maintenance message(s).

(b) A summary and status of all FDE's with date and time, fault description, type, fault code and correlated maintenance message(s), if applicable.

(c) An option to make a selection from the FDE summary of a single maintenance message to see these details:

1) Date, time and flight phase when message occurred.

2) If message is currently active.

3) Number of occurrences during the current leg.

4) If message occurred in previous leg.

NOTE: When selecting a single maintenance message, do not navigate away from the selection until the maintenance message has fully loaded.

(d) Recommended maintenance actions to fix the message.

(e) Correlated FDE's.

(f) Additional selections for Fault History and Flight Legs, if available.

NOTE: When you select the FLIGHT LEGS option, there is a SHOW LEG tab and SHOW DATA tab.

NOTE: The FLIGHT LEGS icon, SHOW LEG tab, and SHOW DATA tab do not show on the Multi Function Display (MFD).

1) A flight leg graph shows when you select the SHOW LEG tab.

2) A flight leg data table shows when you select the SHOW DATA tab.

(4) Use this task when you want to see faults for the inbound leg grouped by FDE and sorted by time with the most recent FDE first.

(5) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- MFD,
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Procedure

SUBTASK 46-13-02-480-037

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

F. Procedure

SUBTASK 46-13-02-860-007

(1) Do the Inbound Flight Deck Effects check as follows:

(a) For the UMD, select LINE MAINTENANCE > Inbound Flight Deck Effects.

(b) For the MFD, select LINE MAINT > INBOUND FLIGHT DECK EFFECTS.

(c) Find the faults and make a record of them, if necessary.

NOTE: It is possible to look at correlated flight deck effects (FDE's). These are FDEs with a corresponding maintenance message. You can also look at all FDEs that are with and without a correlated maintenance message.

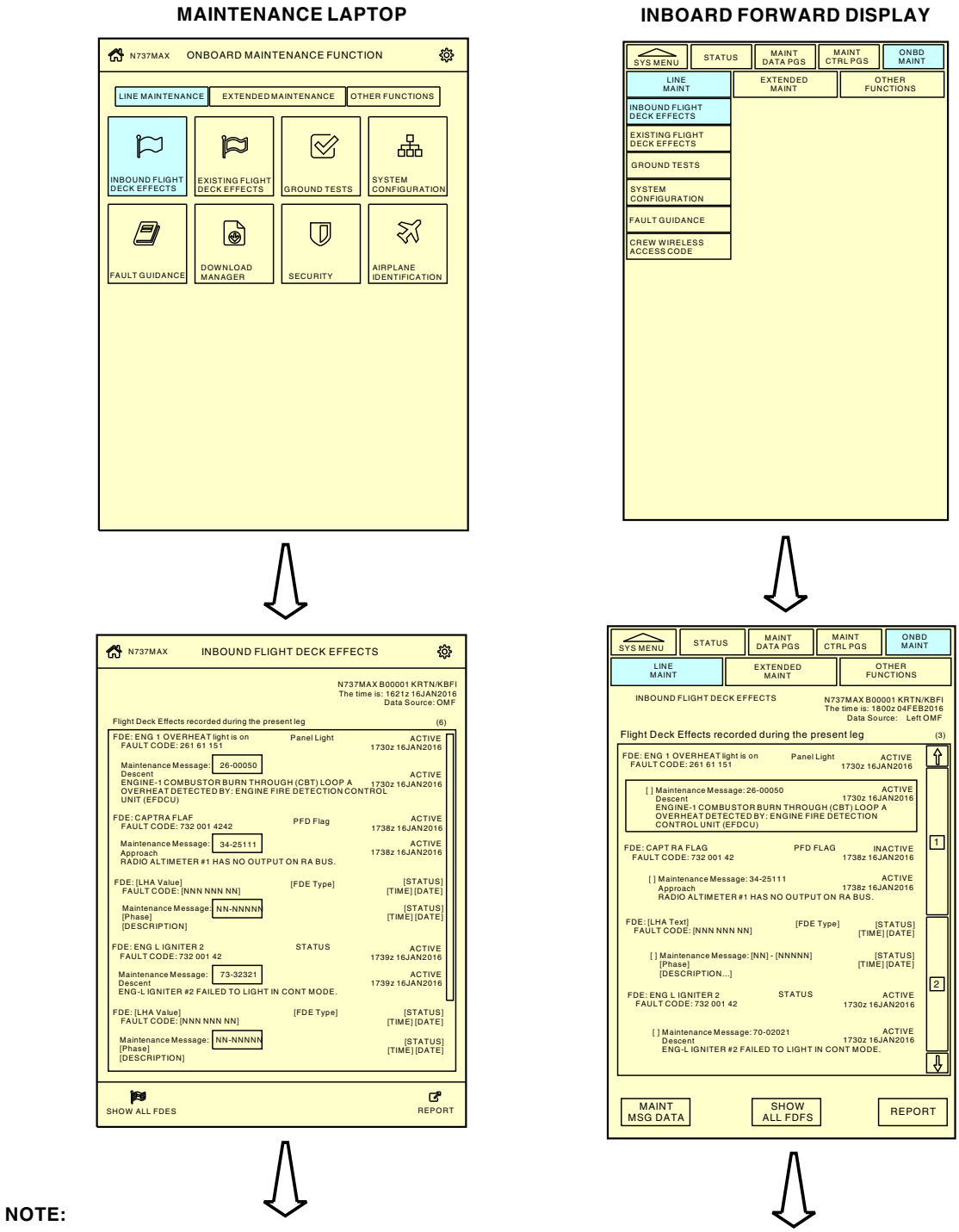
(d) Use the scroll bar if necessary to show additional faults.

(e) If applicable, select the REPORT button to make a record of the inbound FDE's.

(f) If applicable, refer to the IFIM and do the applicable procedure(s) to troubleshoot and correct the faults that show.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



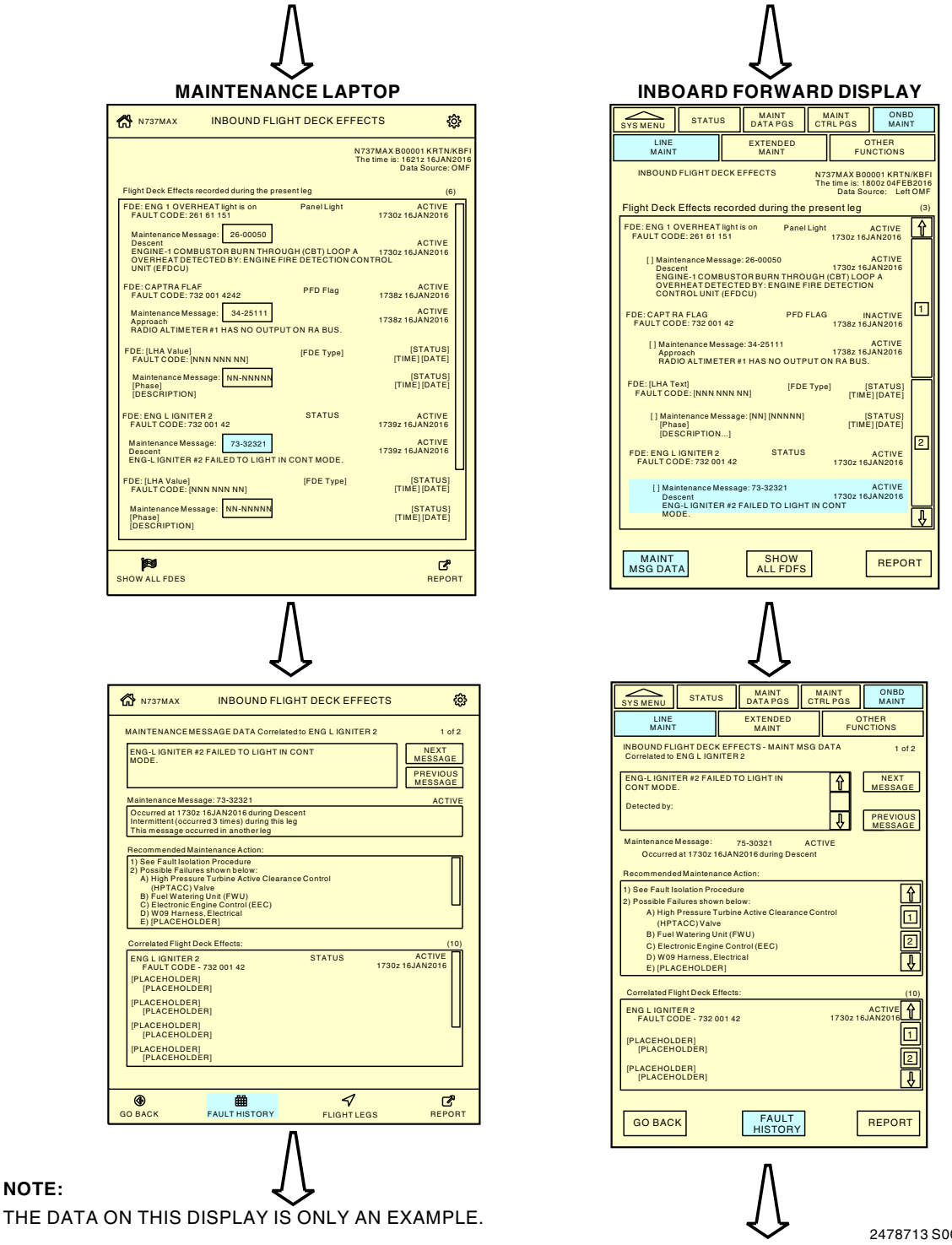
NOTE:

THE DATA ON THIS DISPLAY IS ONLY AN EXAMPLE.

2478705 S0000582072_V1

Inbound Flight Deck Effect Check
Figure 203/46-13-02-990-804 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

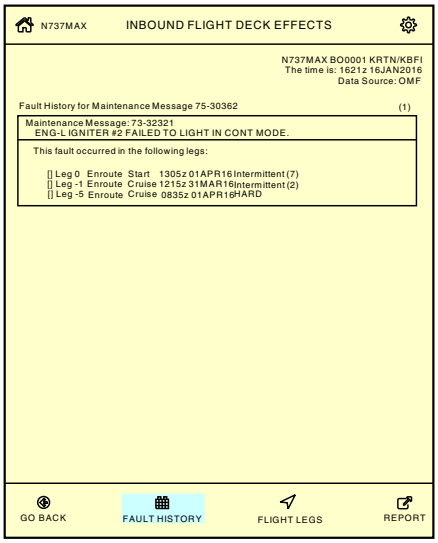
THE DATA ON THIS DISPLAY IS ONLY AN EXAMPLE.

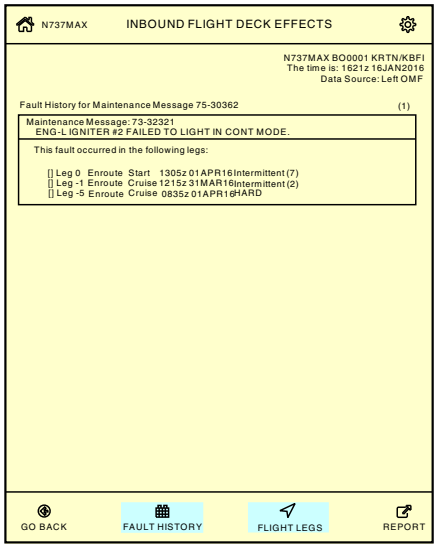
2478713 S0000582121_V1

Inbound Flight Deck Effect Check
Figure 203/46-13-02-990-804 (Sheet 2 of 4)

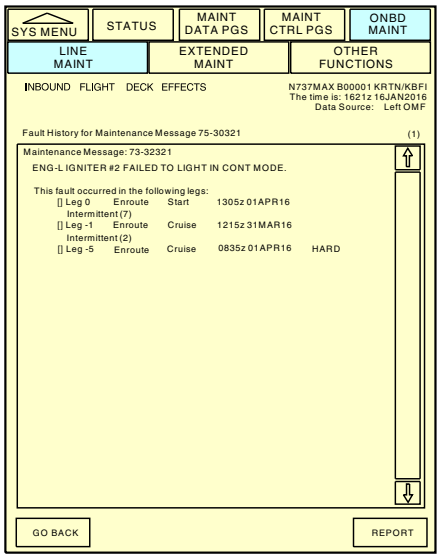
ECCN 9E991 BOEING PROPRIETARY - See title page for details

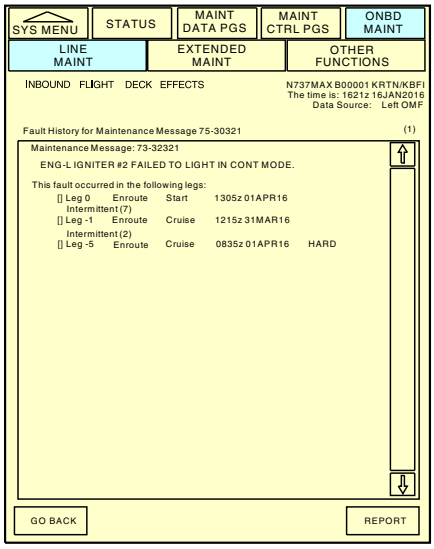
MAINTENANCE LAPTOP





INBOARD FORWARD DISPLAY





STOP! THE SCREENS ON THE PAGE THAT FOLLOWS DO NOT SHOW ON THE MAX DISPLAY SYSTEM (MDS).

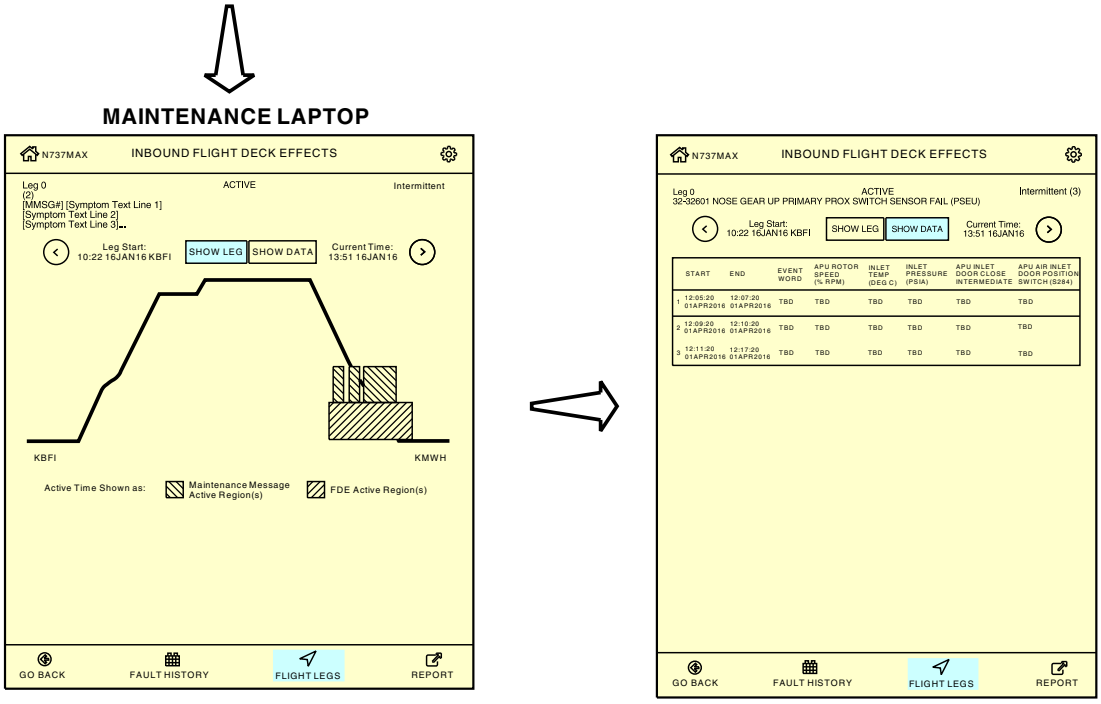
NOTE:

THE DATA ON THIS DISPLAY IS ONLY AN EXAMPLE.

2478715 S0000582122_V1

Inbound Flight Deck Effect Check
Figure 203/46-13-02-990-804 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

THE DATA ON THIS DISPLAY IS ONLY AN EXAMPLE.

2478720 S0000582123_V1

Inbound Flight Deck Effect Check
Figure 203/46-13-02-990-804 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-02-700-803

5. Existing Flight Deck Effects Check

(Figure 204)

A. General

(1) This procedure gives the instructions to do an existing flight deck effects check.

NOTE: If performance on the Existing Flight Deck Effects page is slow, this may be due to a large amount of maintenance message instances. To fix this issue, delete the fault history (TASK 46-13-02-700-806). If no action is taken, all functionality on the Existing Flight Deck Effects page will still be available but performance may be slower.

(2) Existing Flight Deck Effects (EFDE) show active faults that are grouped by Flight Deck Effect (FDE). The EFDE list shows the newest FDE first.

(3) The information that you see includes:

(a) An initial summary and status of correlated FDE's with date and time, fault description, type, fault code, and maintenance message(s).

(b) A summary and status of all FDE's with date and time, fault description, type, fault code and correlated maintenance message(s), if applicable.

(c) An option to make a selection from the FDE summary of a single maintenance message to see these details:

1) Date, time and flight phase when message occurred.

2) If message is currently active.

3) Number of occurrences during the current leg.

4) If message occurred in previous leg.

NOTE: When selecting a single maintenance message, do not navigate away from the selection until the maintenance message has fully loaded.

NOTE: If GPS time is received after the time has been manually set, or if GPS time jumps occur, the change of time may cause inactive faults to be listed when entering this page.

(d) Recommend maintenance actions to fix the message.

(e) Correlated FDE's.

(f) Additional selections for Fault History and Flight Legs, if available.

NOTE: When you select the FLIGHT LEGS option, there is a SHOW LEG tab and SHOW DATA tab.

NOTE: The FLIGHT LEGS icon, SHOW LEG tab, and SHOW DATA tab do not show on the Multi Function Display (MFD).

1) A flight leg graph shows when you select the SHOW LEG tab.

2) A flight leg data table shows when you select the SHOW DATA tab.

(4) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- MFD,
- Other type of approved Universal Maintenance Device (UMD).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Procedure

SUBTASK 46-13-02-480-004

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

F. Procedure

SUBTASK 46-13-02-860-009

(1) Do the EFDE check as follows:

(a) For the UMD, select LINE MAINTENANCE > Existing Flight Deck Effects.

(b) For the MFD, select LINE MAINT > EXISTING FLIGHT DECK EFFECTS.

(c) Find the faults and make a record of them, if necessary.

NOTE: It is possible to look at correlated FDEs. These are FDEs with a corresponding maintenance message. You can also look at all FDEs that have and do not have a correlated maintenance message.

(d) Use the scroll bar if necessary to show additional faults.

(e) If applicable, select the REPORT button or icon to make a record of the inbound FDEs.

(f) If applicable, refer to the IFIM and do the applicable procedure(s) to troubleshoot and correct the faults that show.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-860-048

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

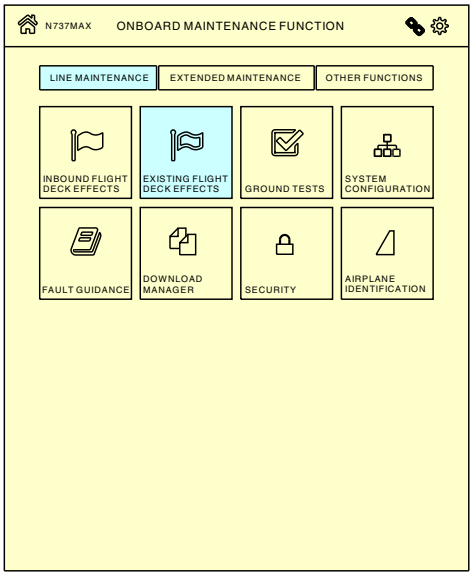
(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

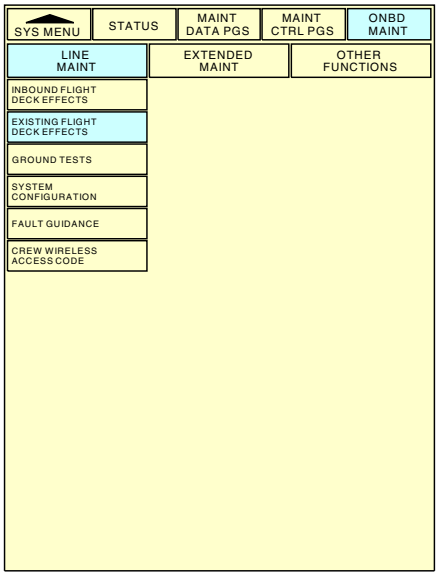
END OF TASK

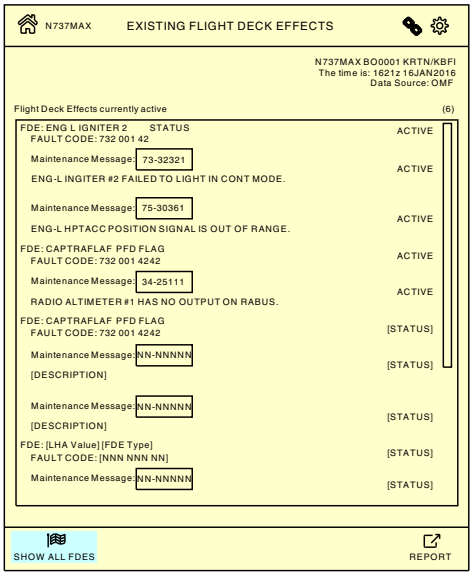
ECCN 9E991 BOEING PROPRIETARY - See title page for details

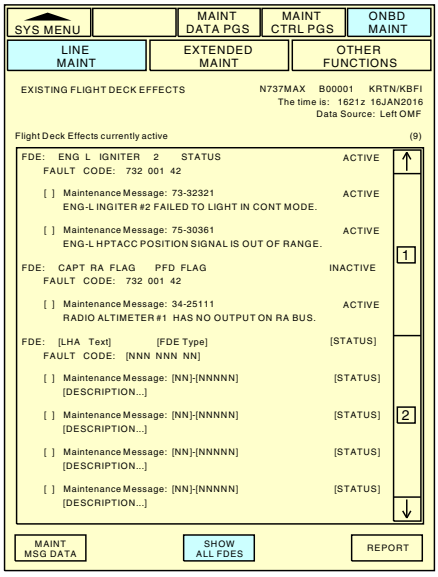
MAINTENANCE LAPTOP



INBOARD FORWARD DISPLAY







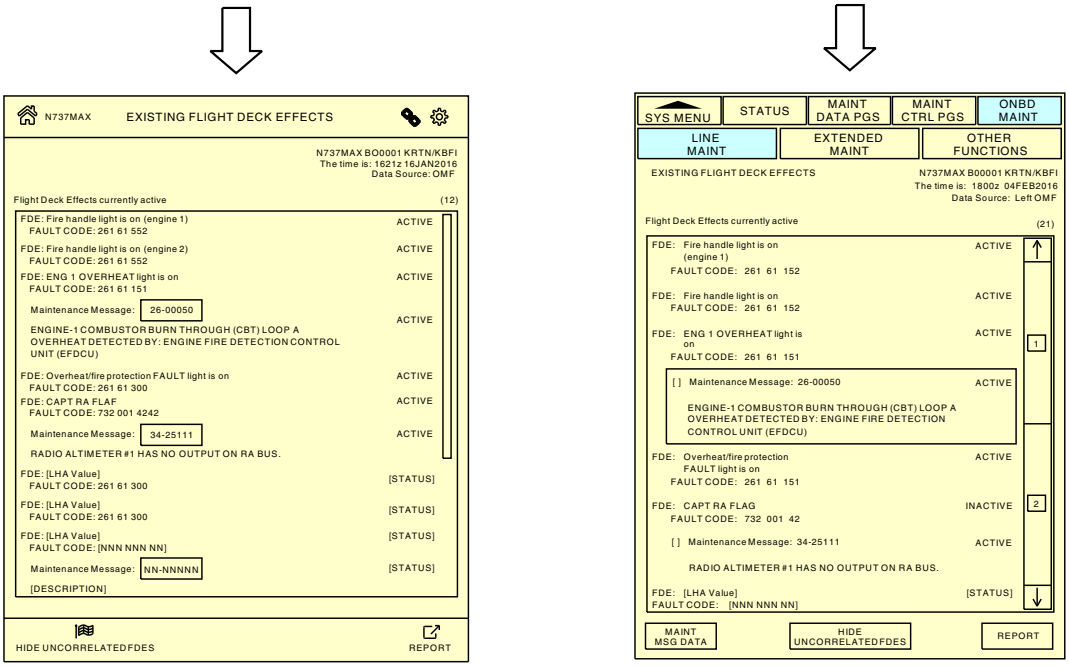
NOTE:

THE DATA ON THE DISPLAY IS AN EXAMPLE.

2480740 S0000582785_V2

Existing Flight Deck Effects Check
Figure 204/46-13-02-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

THE DATA ON THE DISPLAY IS AN EXAMPLE.

2480775 S0000582886_V2

Existing Flight Deck Effects Check
Figure 204/46-13-02-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-02-700-804

6. Present Leg Faults Check

Figure 205

A. General

(1) This procedure gives the instructions to do a check of the present leg faults.

NOTE: If performance on the Present Leg Faults page is slow, this may be due to a large amount of maintenance message instances. To fix this issue, delete the fault history (TASK 46-13-02-700-806). If no action is taken, all functionality on the Present Leg Faults page will still be available but performance may be slower.

(2) The Present Leg Faults page shows faults for the current leg grouped by flight deck effect (FDE). The faults are sorted by time with the newest FDEs showing first.

(a) Faults from the present leg that do not correlate to an FDE are grouped together at the end of the correlated list as non-correlated faults.

(b) Another option is to show faults grouped by ATA and sorted by ATA chapter number.

(c) It is also possible to select a maintenance message from this screen. After you select the maintenance message, you see details that include possible cause, status of the message (active, inactive), date and time of occurrence, etc.

NOTE: When selecting a single maintenance message, do not navigate away from the selection until the maintenance message has fully loaded.

(3) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Procedure

SUBTASK 46-13-02-480-021

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access with a Maintenance Laptop (ML) or Portable Maintenance Device (PMD).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Procedure

SUBTASK 46-13-02-860-011

(1) Do the Present Leg Faults check as follows:

(a) For the UMD, select EXTENDED MAINTENANCE > Present Leg Faults.

(b) For the MFD, select EXTENDED MAINT > PRESENT LEG FAULTS.

(c) Make sure that the display shows a list of data (faults) recorded during the present leg.

NOTE: The list can be sorted by ATA or by flight deck effect (FDE).

(d) Use the scroll bar if necessary to show additional faults.

NOTE: After a maintenance message is selected, details of the fault message are shown. The details include date and time of the message, flight phase when the message was triggered, possible causes, and message status.

(e) If applicable, select the REPORT button to make a record of the Present Leg Faults.

(f) If applicable, refer to the IFIM and do the applicable procedure(s) to troubleshoot and correct the faults that show.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-080-004

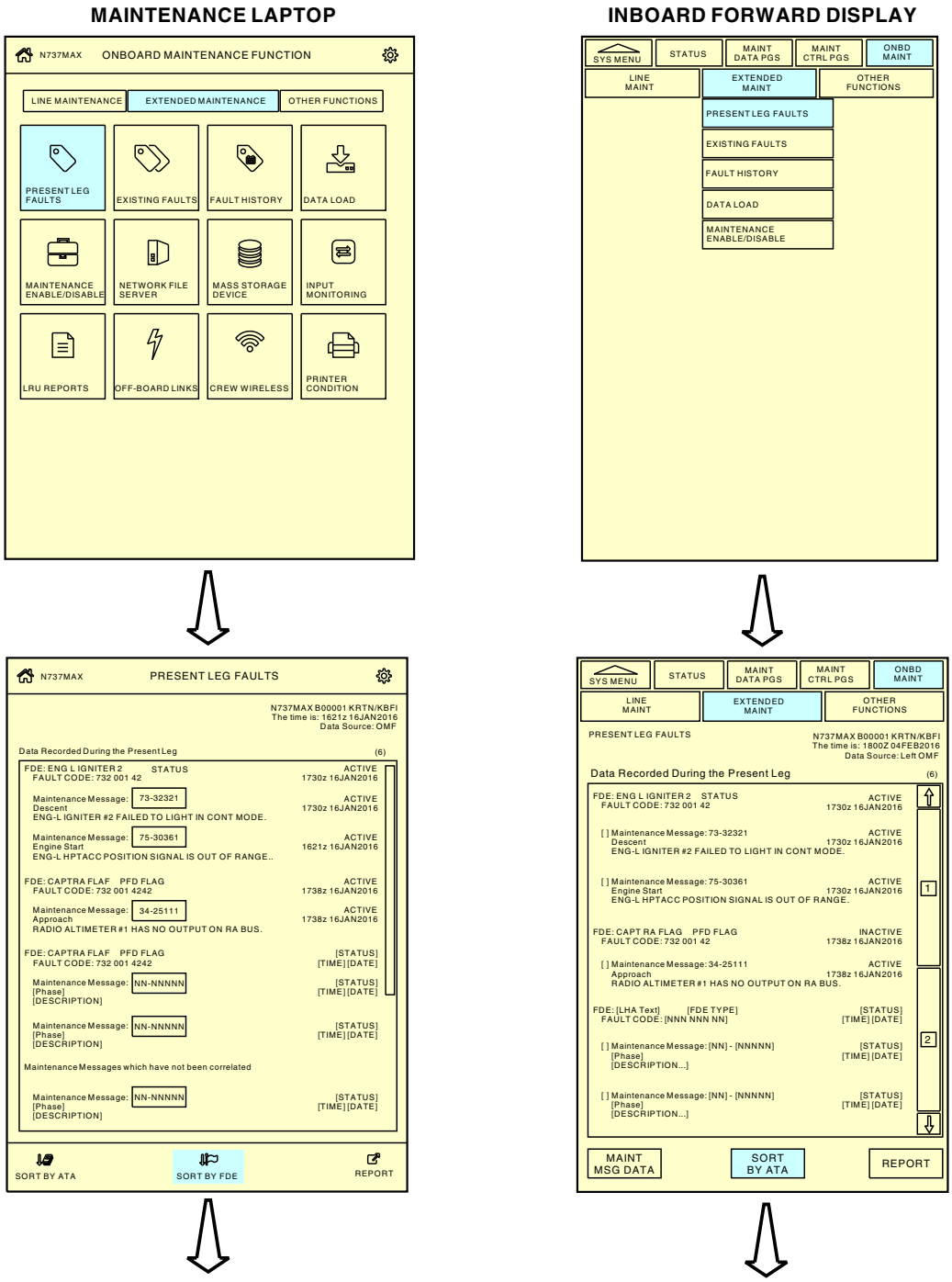
(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

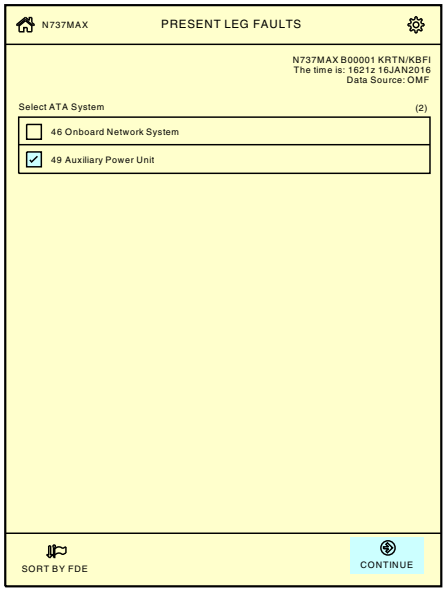
THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

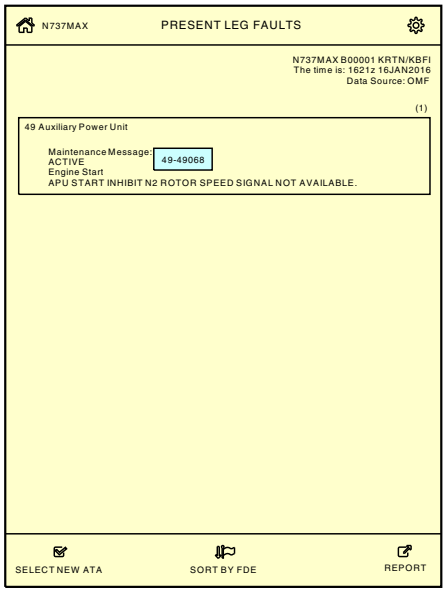
2483872 S0000584146_V1

Present Leg Faults Check
Figure 205/46-13-02-990-807 (Sheet 1 of 2)

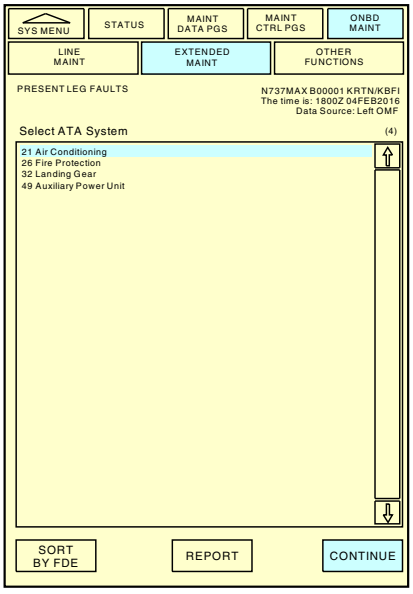
ECCN 9E991 BOEING PROPRIETARY - See title page for details

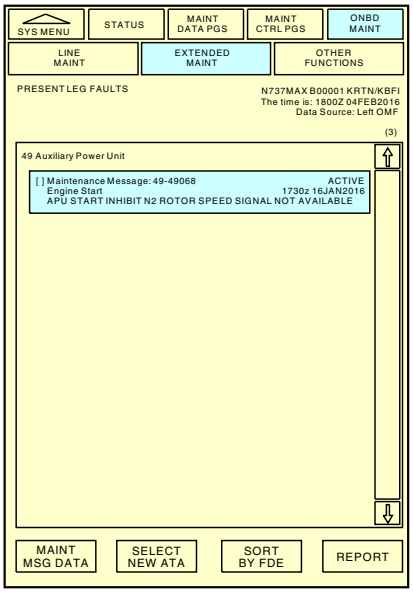
MAINTENANCE LAPTOP





INBOARD FORWARD DISPLAY





NOTE:

THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

2483878 S0000584147_V1

Present Leg Faults Check
Figure 205/46-13-02-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-02-700-805

7. Existing Faults Check

A. General

(1) This procedure gives the instructions to do a check of existing faults.

NOTE: If performance on the Existing Faults page is slow, this may be due to a large amount of maintenance message instances. To fix this issue, delete the fault history (TASK 46-13-02-700-806). If no action is taken, all functionality on the Existing Faults page will still be available but performance may be slower.

(2) Existing faults show the information that follows:

- Currently active maintenance messages on the airplane.
- Currently active FDE's and their correlated active maintenance messages.

NOTE: If GPS time is received after the time has been manually set, or if GPS time jumps occur, the change of time may cause inactive faults to be listed when entering this page.

(3) Use this task when you need active maintenance messages for a selected ATA, or active Flight Deck Effects (FDEs) and their correlated active maintenance messages.

(4) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
05-00-00-750-801	Scheduled Maintenance Task (SMT) - Repair Time Limits (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Procedure

SUBTASK 46-13-02-480-024

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Procedure

SUBTASK 46-13-02-700-003

(1) Do the Existing Faults check as follows:

(a) For the UMD, select EXTENDED MAINTENANCE > Existing Faults.

(b) For the MFD, select EXTENDED MAINT > EXISTING FAULTS.

(c) Make sure that the Select ATA System menu shows.

NOTE: It is possible to sort by flight deck effect.

(d) From the Select ATA System menu, make your ATA selection(s).

NOTE: It is possible to select more than one ATA at the same time.

(e) Select the CONTINUE option.

NOTE: When you select the specific ATA chapter and then the CONTINUE button, this shows the list of existing faults for that ATA system.

(f) Select a maintenance message.

1) Do not navigate away from the selection until the maintenance message has fully loaded.

(g) If applicable, refer to the IFIM and do the applicable procedure(s) to troubleshoot and correct the faults that show.

(h) For Scheduled Maintenance Tasks (SMT) flight deck effects, go to Scheduled Maintenance Task (SMT) - Repair Time Limits, TASK 05-00-00-750-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-860-047

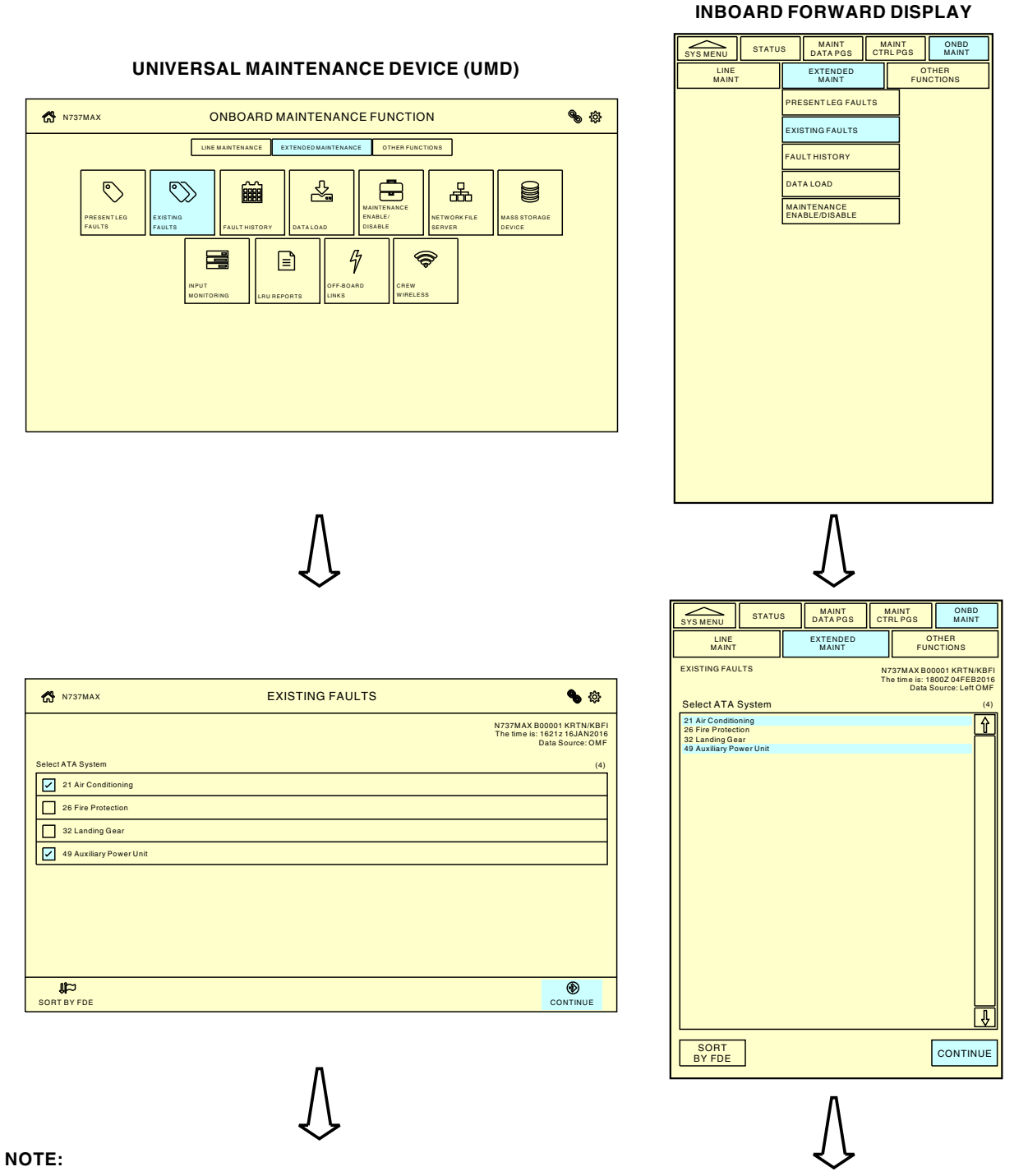
(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

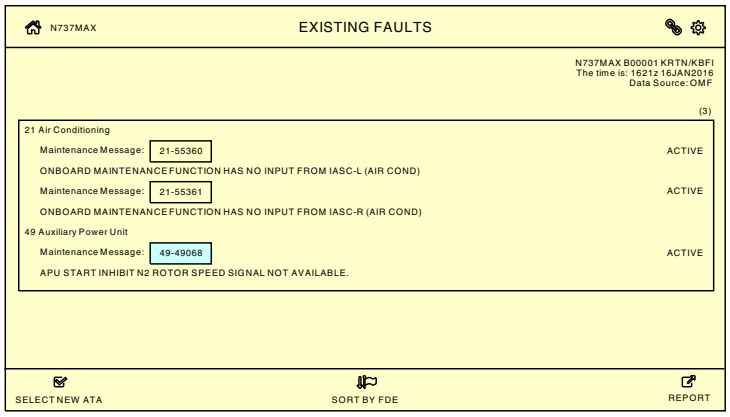
THE DATA ON THE DISPLAY IS AN EXAMPLE.

2484061 S0000584107_V3

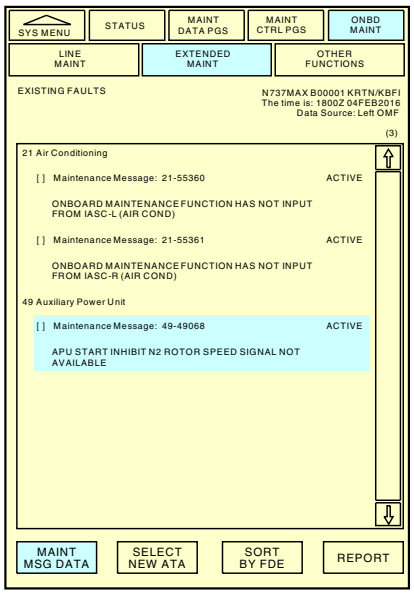
Existing Faults Check
Figure 206/46-13-02-990-808 (Sheet 1 of 2)

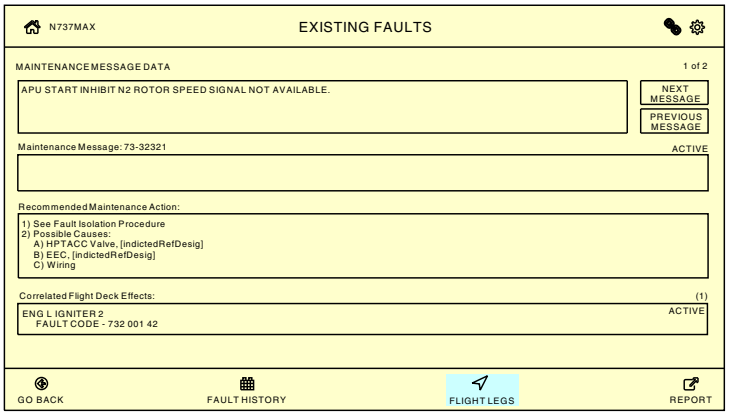
ECCN 9E991 BOEING PROPRIETARY - See title page for details

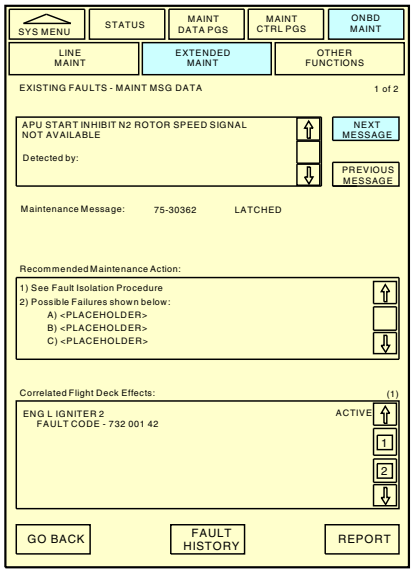
UNIVERSAL MAINTENANCE DEVICE (UMD)



INBOARD FORWARD DISPLAY







NOTE:

THE DATA ON THE DISPLAY IS AN EXAMPLE.

2484087 S0000584108_V3

Existing Faults Check
Figure 206/46-13-02-990-808 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-02-700-806

8. Fault History Check

A. General

(1) This procedure gives the instructions to do a check of fault history.

NOTE: If performance on the Fault History page is slow, this may be due to a large amount of maintenance message instances. To fix this issue, delete the fault history (TASK 46-13-02-700-806). If no action is taken, all functionality on the Fault History page will still be available but performance may be slower.

(2) A fault history check shows the information that follows:

- Fault history and Flight Deck Effects (FDE) sorted by ATA.
- Fault History and FDE sorted by Flight Leg.
- Maintenance messages with details.
- Faults recorded for each flight leg, flight phase, and fault type.
- Recommended maintenance actions.

(3) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
05-00-00-750-801	Scheduled Maintenance Task (SMT) - Repair Time Limits (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Procedure

SUBTASK 46-13-02-480-027

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Procedure

SUBTASK 46-13-02-700-004

(1) Do the Fault History check as follows:

(a) For the UMD, select EXTENDED MAINTENANCE > Fault History.

(b) For the MFD, select EXTENDED MAINT > FAULT HISTORY.

(c) Make sure that the “Select ATA System” menu shows.

(d) From the Select ATA System menu, make your ATA selection.

NOTE: It is possible to select more than one ATA at the same time.

(e) Select the CONTINUE option.

(f) Select a maintenance message to show details of when the fault occurred and recommended maintenance action.

NOTE: When selecting a single maintenance message, do not navigate away from the selection until the maintenance message has fully loaded.

(g) Select the maintenance message again to show recommended maintenance action.

(h) Make the LEG selection to show a menu of flight legs.

(i) Make a leg selection and then select CONTINUE.

NOTE: This shows all faults recorded during the selected leg.

(j) If applicable, select the IMPORT/EXPORT option and follow the steps on the IMPORT/EXPORT pop-up window and select CONTINUE.

NOTE: The maintenance messages cannot be imported from/exported to the MFD. This function exists only on the UMD.

1) When the Windows pop-up shows, follow your airline procedures for file-naming convention and file location.

(k) If applicable, refer to the IFIM and do the applicable procedure(s) to troubleshoot and correct the faults that show.

(l) For Scheduled Maintenance Tasks (SMT), go to Scheduled Maintenance Task (SMT) - Repair Time Limits, TASK 05-00-00-750-801.

(m) To delete all fault history, select the DELETE ALL button, then CONTINUE, then CONFIRM.

NOTE: It is not possible to delete one fault, or to select a group of faults. It is possible to delete only the complete fault history.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-860-049

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-02-860-801

9. Maintenance Enable/Disable Operation

A. General

(1) This procedure gives the instructions to do the Maintenance Enable/Disable operation.

(2) The Maintenance Enable/Disable operation has two functions:

- Flight Leg Transition (Leg Enable or Leg Disable).
- Maintenance Phase (Phase Enable or Phase Disable).

(3) To stop the Onboard Maintenance Function (OMF) from rolling a flight leg every time an engine starts, there is a “flight leg enable” function. If the leg is enabled, starting an engine will roll a leg and disable future leg rolls. During normal operation, the flight leg will be re-enabled automatically when the airplane hits 80 kts; setting the airplane up for the next leg after it hits the ground again. The flight leg enable/disable menu will allow the operator to override that enable setting.

(4) When you enable the Maintenance Phase, the OMF stops storing faults in fault history while the phase is active. For example, enabling the maintenance phase allows the mechanic to work an air crew reported fault but not create additional faults on the ground when opening circuit breakers and disconnecting connectors while troubleshooting the original fault. If the Maintenance Phase is left enabled accidentally, the OMF will end the maintenance phase and revert back to normal phases at takeoff.

(5) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Procedure

SUBTASK 46-13-02-480-030

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321 is required for access.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Flight Leg Transition

SUBTASK 46-13-02-700-005

(1) Do a check of the Flight Leg Transition status:

(a) For the UMD, select EXTENDED MAINTENANCE > Maintenance Enable/Disable.

(b) For the MFD, select EXTENDED MAINT > MAINTENANCE ENABLE/DISABLE.

(c) Make a note of the Flight Leg Transition status.

1) If the Flight Leg Transition status is Disabled, make the LEG ENABLE selection.

NOTE: Flight Leg will become enabled when LEG ENABLE is selected or upon take-off.

2) If the Flight Leg Transition status is Enabled, make the LEG DISABLE selection.

NOTE: A new flight leg will begin at the next engine start.

G. Maintenance Phase

SUBTASK 46-13-02-700-006

(1) Do a check of the Maintenance Phase status:

(a) For the UMD, select EXTENDED MAINTENANCE > Maintenance Enable/Disable.

(b) For the MFD, select EXTENDED MAINT > MAINTENANCE ENABLE/DISABLE.

(c) Make a note of the Maintenance Phase status.

1) If the Maintenance Phase is Disabled, make the PHASE ENABLE selection.

NOTE: Maintenance Phase will be enabled until Phase Disable is selected or at take-off.

2) If the Maintenance Phase is Enabled, make the PHASE DISABLE selection.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-860-052

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

END OF TASK

TASK 46-13-02-860-802

10. Special Functions

A. General

(1) This task gives a general procedure to operate the special functions feature.

(2) This task is used in this list of Special Functions.

(3) Refer to the applicable ATA system when and where to use the Special Functions feature:

- PSEU - Air/Ground Override
- PSEU - Tailskid Extend
- PSEU - Reset Latches
- Landing Gear – Brake Release Function

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Landing Gear – Wheel Speed Transducer Spin Function
- Landing Gear – Memory Clear Function
- Landing Gear – Autobrake Application Function
- AACU - Brake 1 Release Function
- AACU - Brake 2 Release Function
- AACU - Brake 3 Release Function
- AACU - Brake 4 Release Function
- AACU - Wheel 1 Speed Transducer Spin Function
- AACU - Wheel 2 Speed Transducer Spin Function
- AACU - Wheel 3 Speed Transducer Spin Function
- AACU - Wheel 4 Speed Transducer Spin Function
- AACU - Memory Clear Function
- AACU - Autobrake Application Function
- PSEU - Air/Ground Override
- PSEU - Tailskid Extend
- PSEU - Reset Latches
- APU - Reset Hours since Installation
- APU - Ice Break Logic
- APU - Limited Restart
- Engine - L (R) - Fuel System Preservation
- Engine - L (R) - Water Wash
- Engine - L (R) - Engine Serial Number Entry
- Engine - L (R) - Menu Mode Hardware Code Entry
- Engine - L (R) - Erase All Channel A Faults
- Engine - L (R) - Erase All Channel B Faults
- Engine - L (R) - Debris Monitoring System Reset Channel B
- Engine - L (R) - Debris Monitoring System Reset
- Engine - L (R) - Thrust Reverser Cycle
- Engine - L (R) - Engine Running Simulation
- Engine - L (R) - Lamp Check Out
- Engine - L (R) - Starter Air Valve Manual Start.

(4) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD)
- Other type of approved Universal Maintenance Device (UMD).

B. References

Reference	Title
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
32-09-10-750-801	Proximity Switch Electronics Unit Special Functions (P/B 501)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)

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(Continued)

Reference	Title
46-13-00-480-803	Multi Function Display Connection (P/B 201)
49-11-00-400-801	APU Power Plant Installation (Selection) (P/B 401)
71-00-00-700-801-G00	Test No. 25 - Fuel System Preservation Special Functions Test (P/B 501)
71-00-00-700-802-G00	Test No. 26 - Debris Monitoring System Channel B Special Functions Test (P/B 501)
71-00-00-700-805-G00	Test No. 29 - Water Wash Special Functions Test (P/B 501)
71-00-00-700-806-G00	Test No. 30 - Engine Running Simulation Special Functions Test (P/B 501)
71-00-00-700-809-G00	Test No. 33 - Lamp Check Out Special Functions Test (P/B 501)
71-00-00-700-810-G00	Test No. 34 - Menu Mode Hardware Code Entry Special Functions Test (P/B 501)
71-00-00-700-811-G00	Test No. 35 - Erase All Channel A Faults Special Functions Test (P/B 501)
71-00-00-700-812-G00	Test No. 36 - Erase All Channel B Faults Special Functions Test (P/B 501)
71-00-00-700-813-G00	Test No. 37 - Engine Serial Number Entry Special Functions Test (P/B 501)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)
71-00-00-700-818-G00	Test No. 39 - RBS Blower Activation Test (P/B 501)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. List of Procedures

SUBTASK 46-13-02-860-015

(1) These AMM tasks use the Onboard Maintenance Function (OMF) Special Function:

(a) PSEU - Air/Ground Override - Proximity Switch Electronics Unit Special Functions, TASK 32-09-10-750-801

(b) PSEU - Tailskid Extend - Proximity Switch Electronics Unit Special Functions, TASK 32-09-10-750-801

(c) PSEU - Reset Latches - Proximity Switch Electronics Unit Special Functions, TASK 32-09-10-750-801

(d) APU - Reset Hours since Installation - APU Power Plant Installation (Selection), TASK 49-11-00-400-801

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(e) Engine - L (R) - Fuel System Preservation - Test No. 25 - Fuel System Preservation Special Functions Test, TASK 71-00-00-700-801-G00

(f) Engine - L (R) - Water Wash - Test No. 29 - Water Wash Special Functions Test, TASK 71-00-00-700-805-G00

(g) Engine - L (R) - Engine Serial Number Entry - Test No. 37 - Engine Serial Number Entry Special Functions Test, TASK 71-00-00-700-813-G00

(h) Engine - L (R) - Fan Accelerometer Selection - Test No. 34 - Menu Mode Hardware Code Entry Special Functions Test, TASK 71-00-00-700-810-G00

(i) Engine - L (R) - Erase All Channel A Faults - Test No. 35 - Erase All Channel A Faults Special Functions Test, TASK 71-00-00-700-811-G00

(j) Engine - L (R) - Erase All Channel B Faults - Test No. 36 - Erase All Channel B Faults Special Functions Test, TASK 71-00-00-700-812-G00

(k) Engine - L (R) - Debris Monitoring System Reset Channel B - Test No. 26 - Debris Monitoring System Channel B Special Functions Test, TASK 71-00-00-700-802-G00

(l) Engine - L (R) - Thrust Reverser Cycle - Test No. 38 - Thrust Reverser Cycle Special Functions, TASK 71-00-00-700-816-G00

(m) Engine - L (R) - Engine Running Simulation - Test No. 30 - Engine Running Simulation Special Functions Test, TASK 71-00-00-700-806-G00

(n) Engine - L (R) - Lamp Check Out - Test No. 33 - Lamp Check Out Special Functions Test, TASK 71-00-00-700-809-G00

(o) Engine - L (R) - Engine Starter Air Valve Manual Start - Power Plant Test Reference Table, TASK 71-00-00-800-804-G00

(p) Engine - L (R) - RBS Blower Activation Test - Test No. 39 - RBS Blower Activation Test, TASK 71-00-00-700-818-G00.

F. Prepare for the Procedure

SUBTASK 46-13-02-860-019

(1) Make sure that no other MAX Display System (MDS) maintenance test function is active.

SUBTASK 46-13-02-860-020

(2) Make sure that the air/ground system is in ground mode (TASK 32-09-00-860-802).

SUBTASK 46-13-02-480-033

(3) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801 or Multi Function Display Connection, TASK 46-13-00-480-803.

NOTE: A universal maintenance device, SPL-11321, is required for access.

G. Special Functions Procedure

SUBTASK 46-13-02-860-060

(1) For the UMD, select OTHER FUNCTIONS > Special Functions.

SUBTASK 46-13-02-860-076

(2) For the MFD, select OTHER FUNCTIONS > SPECIAL FUNCTIONS.

SUBTASK 46-13-02-860-061

(3) Select the applicable ATA system.

NOTE: The page shows these systems under 'Select ATA System':

- 32 - Landing Gear
- 32 - Proximity Switch Electronic Unit

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- 49 - Auxiliary Power Unit
- 71 - Left Engine
- 71 - Right Engine.

SUBTASK 46-13-02-860-062

(4) Select the applicable System Function.

NOTE: The page shows these systems under 'Select System Function':

- Landing Gear - Brake 1 (2,3,4) Release Function
- Landing Gear - Wheel 1 (2,3,4) Speed Transducer Spin Function
- Landing Gear – Memory Clear Function
- Landing Gear – Autobrake Application Function
- PSEU - Air/Ground Override
- PSEU - Tailskid Extend
- PSEU - Reset Latches
- APU - Reset Hours since Installation
- APU - Ice Break Logic
- APU - Limited Restart
- Engine - L (R) - Fuel System Preservation
- Engine - L (R) - Water Wash
- Engine - L (R) - Engine Serial Number Entry
- Engine - L (R) - Fan Accelerometer Selection
- Engine - L (R) - Erase All Channel A Faults
- Engine - L (R) - Erase All Channel B Faults
- Engine - L (R) - Debris Monitoring System Reset Channel B
- Engine - L (R) - Thrust Reverser Cycle
- Engine - L (R) - Engine Running Simulation
- Engine - L (R) - Lamp Check Out
- Engine - L (R) - Starter Air Valve Manual Start
- Engine - L (R) - RBS Blower Activation Test.

SUBTASK 46-13-02-860-063

(5) Select CONTINUE.

SUBTASK 46-13-02-860-064

(6) Do the instructions that show on the PRECONDITIONS.

(a) Make sure that you check and follow the inhibited conditions instructions closely.

NOTE: If one of the inhibited conditions is not met, the special function will fail.

SUBTASK 46-13-02-860-065

(7) Select CONTINUE.

SUBTASK 46-13-02-860-066

(8) Set the GND TEST ENABLE SWITCH, on the P61-4 Maintenance Built-In-Test Equipment Panel (MBITEP), to the ENABLE position.

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SUBTASK 46-13-02-860-067

(9) Select the START FUNCTION screen button under FUNCTION.

(a) Follow any additional instructions for that special function.

SUBTASK 46-13-02-860-068

(10) Select STOP FUNCTION to stop the function.

SUBTASK 46-13-02-860-069

(11) Select the OPERATE FUNCTION AGAIN button to operate the function again.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-860-043

(1) Set the GND TEST ENABLE SWITCH, on the P61-4 MBITEP, to the NORM position.

SUBTASK 46-13-02-080-006

(2) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(3) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-02-970-802

11. Download Manager

(Figure 207)

A. General

(1) This task has the steps to get access to and use the download manager function.

(2) Download Manager lets the mechanic select files (such as system logs, QAR data, and engine data) to transfer from ONS to the UMD. You can also delete the files from the ONS. Files can be filtered by application and file type.

(3) Do this task when you get access to the Onboard Maintenance Function (OMF) with the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

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E. Prepare for the Download Manager

SUBTASK 46-13-02-480-009

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

F. Download Manager Procedure

SUBTASK 46-13-02-860-070

(1) From the ONS main menu, make the selection: LINE MAINTENANCE > Download Manager.

(a) Select the Application and FILE TYPE filter by clicking the box(es) next to your selections.

1) Make sure that a green checkmark appears in the selected box(es).

(b) Select CONTINUE.

(c) Select GET CURRENT (if active) on the page footer.

NOTE: A popup window with GETTING CURRENT FILES and a progress indicator shows.

(d) Select the files

NOTE: To select all files not previously transferred, select UNTRANSFERRED on the page footer.

1) Make sure that a green check mark shows adjacent to your selection.

NOTE: If a check mark is displayed in the TRANSFERRED column, then the files have already been downloaded.

NOTE: If you select DETAILS, you can see the number and size of the files.

(e) To delete the files, select the DELETE icon.

NOTE: Delete the files only when directed by your engineering department.

(f) To download the files, select the REPORT icon.

(g) Make sure that destination “Maintenance Device” is selected.

(h) Select the SEND button.

(i) When the Windows popup shows, make your location selection on the UMD where the data will be downloaded and save the file.

NOTE: The Boeing Company recommends that you use the auto-created file name. The file name is created automatically. The naming convention is airplane identification/tail ID, filter type (ADR-QAR, for example), date and time.

(j) Wait until you see the message “COMPLETE”.

1) If “COMPLETE” does not show after 5 minutes, refer to the IFIM and do the applicable procedure(s).

NOTE: The REPORT popup communicates STARTING, percentage complete, and FAILED.

(k) Wait ten (10) seconds after the “Complete” message shows.

(l) Select the CLOSE button.

(m) If an external media is used, eject the media in Windows prior to removing media from the UMD and wait for Windows to indicate “Safe To Remove”.

NOTE: Failure to eject the media using Windows can result in an incomplete download of the files.

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-080-001

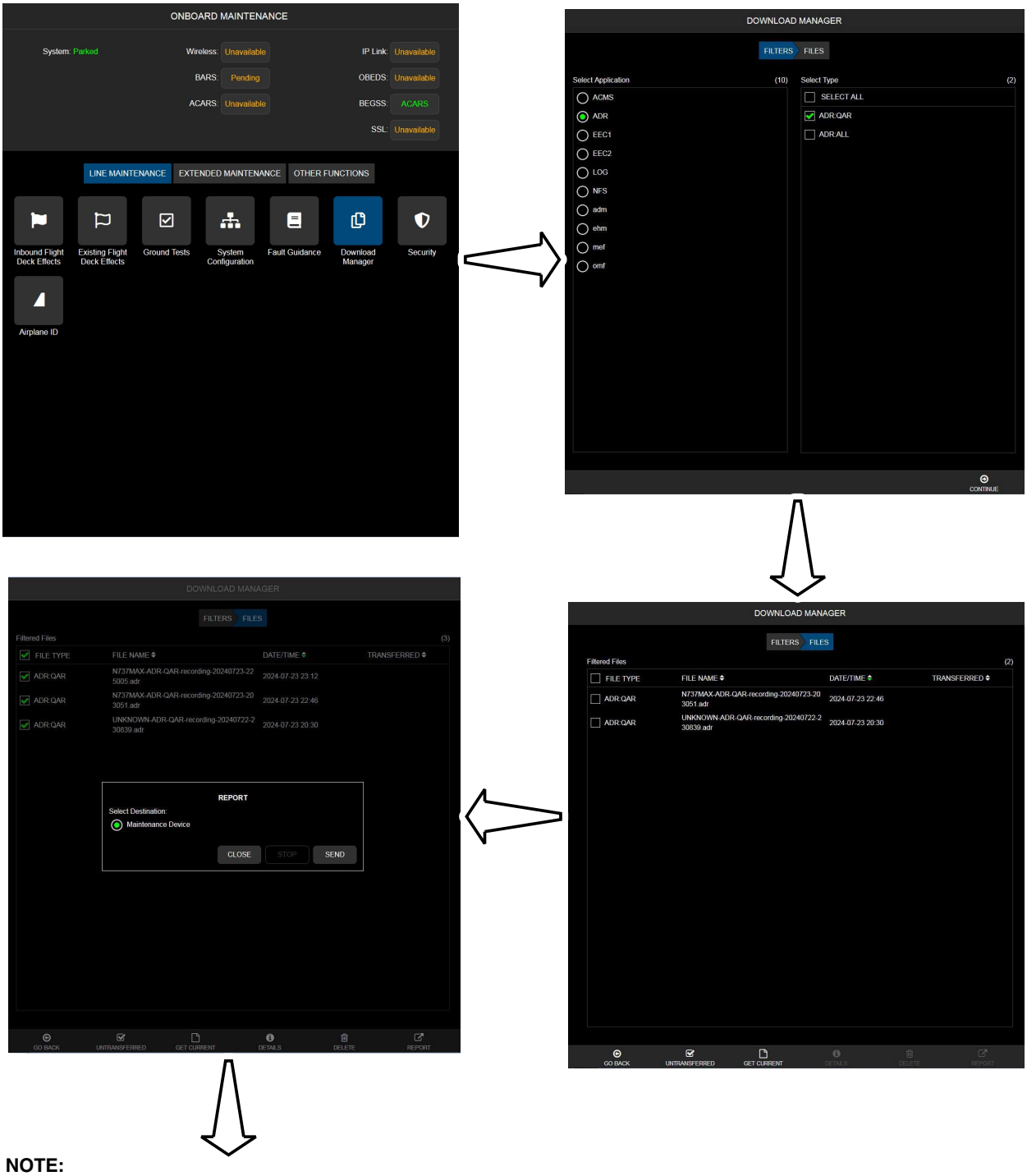
(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, stop the Onboard Network System (ONS) maintenance browser session and disconnect the maintenance device.

——— END OF TASK ———

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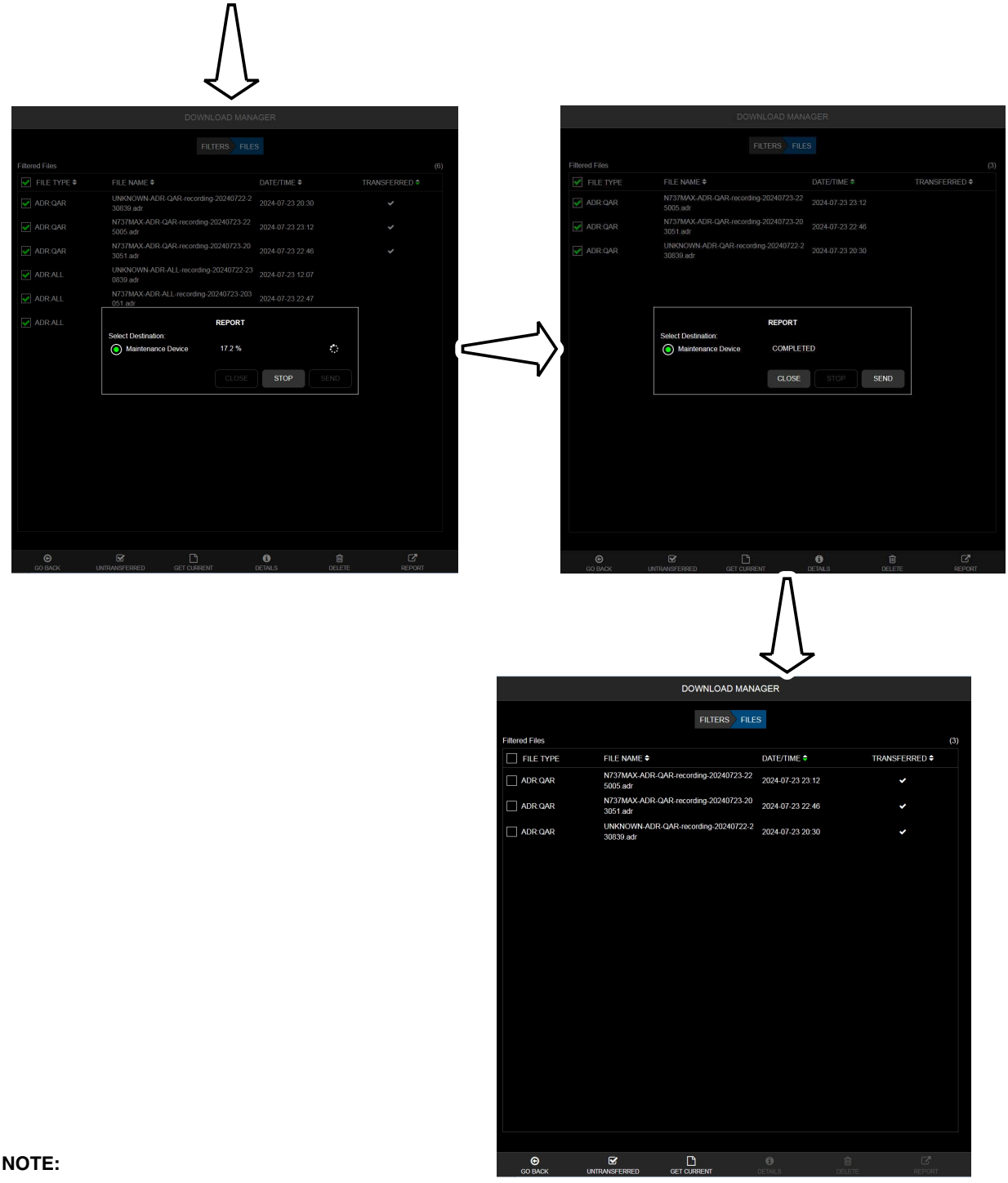


NOTE:
THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

2479773 S0000582248_V2

Download Manager
Figure 207/46-13-02-990-805 (Sheet 1 of 2)

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NOTE:
THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

2479851 S0000582250_V2

Download Manager
Figure 207/46-13-02-990-805 (Sheet 2 of 2)

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TASK 46-13-02-970-803

12. LRU Reports

A. General

(1) This procedure gives the instructions to create an LRU Report.

(2) This procedure is applicable to the units that follow:

- Flight Control Computer (FCC)
- Forward Attendant Control Panel (ACP)
- Display Processing Computer (DPC)
- Flight Management Computer (FMC)
- Spoiler Control Electronics (SCE)

(3) LRU Reports show the data that follows:

- Hardware serial number.
- Hardware part number.
- Software part numbers.

(4) When a report is generated, it can be read and then deleted by navigating to another page or it can be downloaded.

(5) To do this task, the DATA REPORTS parts for the applicable LRU must be installed on the Onboard Network System (ONS) Mass Storage Device (MSD).

(6) This task is available when the ONS is accessed with the Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-470-802	Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD) (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-860-804	Mass Storage Device Software Parts Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Task

SUBTASK 46-13-02-480-010

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

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SUBTASK 46-13-02-470-001

(2) Make sure that the applicable DATA REPORTS parts are installed in the MSD (TASK 46-13-00-860-804).

(a) If it is necessary, do this task: Stage Software Parts on a Mass Storage Device via Universal Maintenance Device (UMD), TASK 46-13-00-470-802.

F. LRU Reports Procedure

SUBTASK 46-13-02-700-007

(1) Do the LRU Reports procedure as follows:

(a) Select EXTENDED MAINTENANCE > LRU REPORTS.

(b) On the ATA SYSTEM tab, scroll down until you see the applicable ATA.

(c) Select the applicable ATA.

NOTE: It is not possible to select more than one ATA at the same time.

(d) If it is necessary, scroll down the LRU list and then select the applicable LRU.

NOTE: It is not possible to select more than one LRU at the same time.

(e) Select the CONTINUE icon.

(f) Make sure that the next screen that shows is the PRECONDITIONS tab.

(g) Do the steps and instructions that show on the PRECONDITIONS tab.

NOTE: If there are no preconditions necessary, this message shows: “No LRU Precondition message was reported.”

(h) Select the CONTINUE icon.

(i) Make sure that the “RETRIEVING REPORT TYPES” popup window shows with a progress indicator.

(j) Make sure that the REPORT TYPES tab shows and a report type list with nomenclature shows.

1) If the REPORT TYPES tab does not show a report type list, make sure that the applicable DATA REPORTS part is installed in the MSD (TASK 46-13-00-860-804).

(k) From the REPORT TYPES tab, select the applicable report.

(l) Select CONTINUE.

(m) Make sure that the DATA REPORTING popup window shows with a progress indicator.

(n) Make sure that the RETRIEVED REPORTS tab shows and status shows the word “COMPLETED” adjacent to the report name.

(o) Make sure that all pop ups have been closed before navigating to other tabs on the LRU Reports page.

NOTE: After a report is generated, options become available to the user that will generate popups on the RETRIEVED REPORTS page.

(p) Select the DETAILS icon to see the report summary with report size.

(q) To delete the report, do the steps that follow:

1) Select the DELETE icon.

2) Select CONTINUE.

(r) To download the report, do the steps that follow:

1) Select the REPORT icon.

2) Select the destination (Maintenance Device or Download Manager).

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3) Select SEND.

4) When the Windows Explorer window shows, select the location on the maintenance device where the data will be downloaded.

NOTE: The Boeing Company recommends that you use the auto-created file name. The file name is created automatically.

5) Wait until you see the message “Complete”.

6) Select the CLOSE button.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-860-051

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-02-860-803

13. Off-Board Links

A. General

(1) This procedure gives the instructions for Offboard Communication.

(2) The Off-Board Communications screens allow the user to view the status of off-board links to the ONS, such as the WWU or AID. The status indicates if the interface is active.

NOTE: This function requires a connectivity option, such as the WWU or AID, to be installed in the airplane.

(a) To determine whether the optional WWU is installed, run a system configuration for ATA “46 – Information Systems”. “Wireless WAN Unit” will show on the system configuration if WWU is installed. Do this task: System Configuration Check, TASK 46-13-00-710-802.

(b) To determine whether the optional AID is installed, run a system configuration for ATA “46 – Information Systems”. “Aircraft Interface Device – Forward (AID-FWD)” will show on the system configuration if AID is installed. Do this task: System Configuration Check, TASK 46-13-00-710-802 .

(3) This function also lets the user to do a test of the link.

(4) There are these tabs in the off-board links function:

(a) LINKS

(b) WIRELESS

(c) CELLULAR

(d) ACARS

(e) MESSAGING

(5) This task is available when the Onboard Network System (ONS) is accessed with the Universal Maintenance Device (UMD).

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B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Procedure

SUBTASK 46-13-02-480-011

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321 is required for access.

F. Default Link Status Test

SUBTASK 46-13-02-700-018

(1) Do a status test as follows:

(a) Select EXTENDED MAINTENANCE > Offboard Communications.

(b) On the LINKS tab, do a test of the default link status as follows:

(c) Make a record of which link has the green check mark in the DEFAULT column.

1) If no link has a green check mark in the DEFAULT column, refer to the IFIM and do the applicable procedure(s).

(d) Select the TEST DEFAULT icon.

(e) Make sure that the TEST DEFAULT LINK window shows with a progress indicator.

(f) When the test is complete, make sure that a green check mark shows in the TEST column.

NOTE: The green check mark now shows in the same row for the same link as the green check mark in the DEFAULT column.

G. Cellular Link Test

SUBTASK 46-13-02-700-011

(1) Do a test of the cellular link as follows:

(a) Select EXTENDED MAINTENANCE > Offboard Communications.

(b) Select the CELLULAR tab.

(c) Select DIAGNOSTICS.

(d) Make sure that the Link State shows the following:

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LINK STATE	DESCRIPTION
Home Network	Cellular connected to home network
Home Network-Default	Cellular connected to home network and is the default route
Roaming	Cellular connected but not to home network
Roaming-Default	Cellular connected but not to home network and is the default route

1) If one of the links above is not shown, then refer to the IFIM and do the applicable procedure(s).

——— END OF TASK ———

TASK 46-13-02-700-807

14. Shop Faults

A. General

(1) This procedure gives the instructions to do a Shop Faults check.

(2) The purpose of the OMF Shop Faults function is to provide a means for maintenance personnel to display and report shop fault data stored in member systems. That shop fault data could be used directly by the maintenance personnel or sent (upon request) to either Boeing or an LRU supplier that is performing troubleshooting.

(3) Present Leg Faults, Inbound Flight Deck Effects, Existing Faults and other fault and flight deck effect applications show airplane faults at the system level. Mechanics use these applications to access the Interactive Fault Isolation Manual (IFIM) for maintenance and troubleshooting guidance. Shop Faults however, give a deeper level of troubleshooting down to the specific LRU level.

(a) The Boeing Company does not determine the Shop Fault content.

(b) The member system's supplier defines what information shows.

(c) Shop fault data is a combination of standard LRU maintenance information and failure information detected by member system BITE.

(d) The Onboard Maintenance Function (OMF) provides access and outputs to shop fault data when you use the Universal Maintenance Device (UMD).

(4) Use this task when Shop Fault content is sent to Boeing or LRU suppliers for specific LRU troubleshooting purposes.

(5) This task is available when the Onboard Network System (ONS) is accessed with the UMD.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

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D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Task

SUBTASK 46-13-02-480-012

(1) To access the ONS, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

F. Shop Faults Check Procedure

SUBTASK 46-13-02-700-012

(1) Do the Shop Faults Check procedure as follows:

(a) Select OTHER FUNCTIONS > Shop Faults.

(b) On the ATA SYSTEM tab, scroll down until you see the applicable ATA.

(c) Select the applicable ATA.

- 32 - Landing Gear
- 49 - Airborne Auxiliary Power Unit
- 71 - Engine-1
- 71 - Engine-2

NOTE: It is not possible to select more than one ATA at the same time.

(d) If necessary, scroll down the LRU list and then select the applicable LRU.

NOTE: It is not possible to select more than one LRU at the same time.

(e) Select the CONTINUE icon.

(f) Make sure that the “Retrieving Fault Data” popup shows with a progress indicator.

(g) Make sure that the fault data shows for the ATA and the LRU.

NOTE: The fault data includes flight leg information, UTC time and date when the fault occurred, airplane registration, and flight phase. The fault data also shows fault ID (or fault code), the type of fault and number of occurrences, fault description and fault ATA chapter-section.

(h) If necessary, select the NEXT PAGE icon to show more fault data.

(i) If applicable, select the REPORT icon.

NOTE: The REPORT icon will only create a report of the current page of fault data being viewed on the screen. To report all data for the ATA, each page must be individually selected by using the PREVIOUS PAGE or NEXT PAGE button and then selecting REPORT on each page.

(j) Make sure that the REPORT page shows with “SELECT DESTINATION” options.

(k) Select the applicable destination:

1) Maintenance Device.
2) Download Manager.
3) Flight Deck Printer.

(l) Select SEND.

(m) If applicable, follow the onscreen instructions.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-02-860-050

(1) Verify that there are no active maintenance messages for ATA “46 - Information Systems". Do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the IFIM and do the applicable procedure(s).

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

END OF TASK

TASK 46-13-02-860-805

15. Maintenance Data Pages

Figure 208

A. General

(1) This tasks gives the steps to access the Maintenance Data pages for system status of the different airplane systems.

B. Location Zones

Zone	Area
211	Flight Compartment - Left

C. Prepare for Maintenance Data Page Access

SUBTASK 46-13-02-860-054

(1) If you use one of the large inboard displays (Captain's or First Officer's MFD) to do the ground test, do the steps that follow:

(a) Push the MFD SYS button on the P9-82 panel, MULTIFUNCTION DISPLAY (MFD) panel.

NOTE: The Synoptic page will display with the MAINT button highlighted.

(b) Push the SEL button on the knob.

(c) Make sure that you see the main Maintenance Menu on the MFD.

SUBTASK 46-13-02-860-073

(2) Select MAINT DATA PGS.

SUBTASK 46-13-02-860-074

(3) Push the SEL button on the knob.

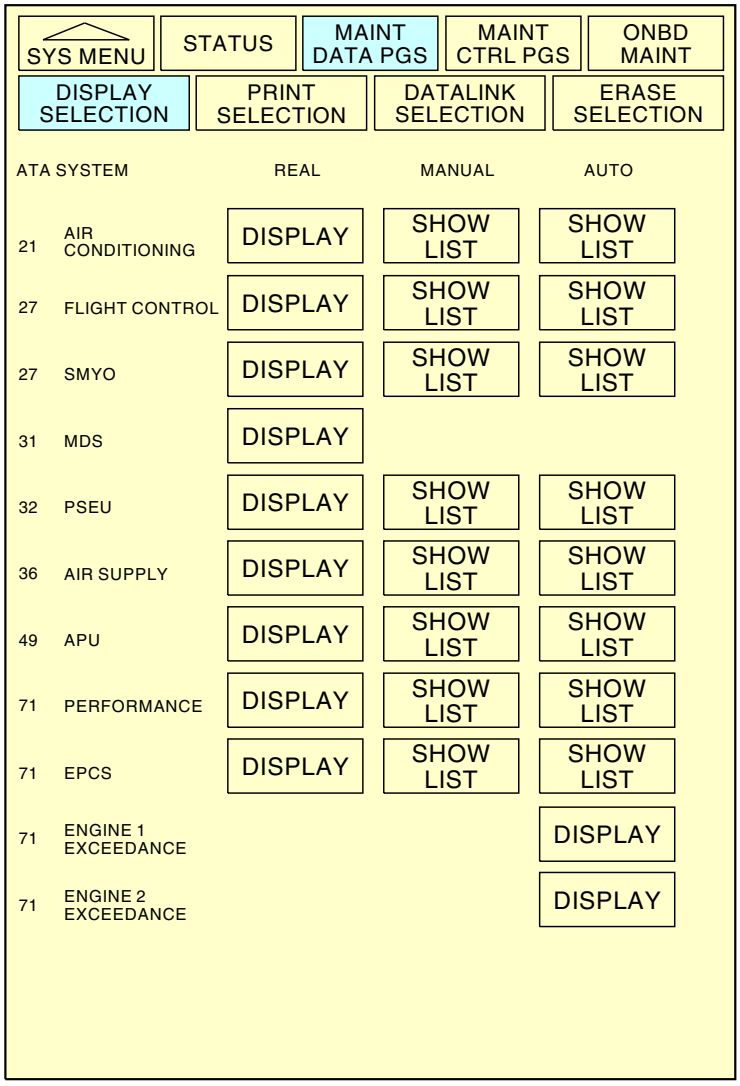
SUBTASK 46-13-02-860-075

(4) Select desired ATA DISPLAY button to view system data.

NOTE: Some ATA systems have more than one data page. To navigate through the other pages, use the NEXT PAGE or the PREV PAGE button.

END OF TASK

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2478060 S0000581532_V2

ATA System Page
Figure 208/46-13-02-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-02-860-806

16. Erasing a Latched Status Message - Verification Check

A. General

(1) This procedure contains the steps to follow after erasing a latched STATUS message with the inoperative Network File Server (NFS).

B. References

Reference	Title
12-13-31-200-801	APU Oil Level Inspection (P/B 301)
21-51-00-700-802	Pack Flow Control Valve Operational Test (P/B 501)
21-61-22 P/B 401	INTEGRATED AIR SYSTEMS CONTROLLER - REMOVAL/INSTALLATION
27-32-00-740-801	Stall Management Yaw Damper (SMYD) BITE Test - Existing Faults (P/B 501)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-10 P/B 401	PROXIMITY SWITCH ELECTRONICS UNIT (PSEU) - REMOVAL/INSTALLATION
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
34-61-00-730-801	Flight Management Computer System - System Test (P/B 501)
49-15-00-710-801	APU Air Inlet Door Test (P/B 501)
52-71-00-730-801	Door Warning System Test (P/B 501)
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-00-00-910-803-G00	Start the Engine (Normal Start) (P/B 201)
71-00-00-910-806-G00	Stop the Engine (Usual Engine Stop) (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Erasing a Latched Status Message - Verification Check Procedure

SUBTASK 46-13-02-860-078

(1) Do the verification steps after erasing a latched STATUS message.

(a) Use the Table 201 if the STATUS message latched by the MAX Display System (MDS).

(b) Use the Table 202 if the STATUS message latched by other Line Replaceable Unit (LRU).

Table 201/46-13-02-993-801 Latched STATUS messages from MDS

Status Text	Dispatch Relief	Alternate Reset Procedure
APU OIL QTY	N/A	1) Do this task: APU Oil Level Inspection, TASK 12-13-31-200-801.
APU DOOR	Y	1) Do this task: APU Air Inlet Door Test, TASK 49-15-00-710-801.

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Table 201/46-13-02-993-801 Latched STATUS messages from MDS (Continued)

Status Text	Dispatch Relief	Alternate Reset Procedure
ENG 1 EEC OVERHEAT ENG 2 EEC OVERHEAT ENG 1 FUEL FILTER ENG 2 FUEL FILTER ENG 1 FUEL STRAINER ENG 2 FUEL STRAINER ENG 1 FUEL MAIN HX ENG 2 FUEL MAIN HX ENG 1 PSS OVERHEAT ENG 2 PSS OVERHEAT ENG 1 OIL PRESS SENSOR ENG 2 OIL PRESS SENSOR ENG 1 OIL DEBRIS ENG 2 OIL DEBRIS ENG 1 OIL TEMP SENSOR ENG 2 OIL TEMP SENSOR	N	1) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.
ENG 1 FAN COMP OVHT ENG 2 FAN COMP OVHT ENG 1 FAN COMP OVHT SENSOR ENG 2 FAN COMP OVHT SENSOR	N	1) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.
ENG 1 CONTROL ENG 2 CONTROL ENG 1 OVERSPEED PROTECT ENG 2 OVERSPEED PROTECT ENG 1 LPTACC VALVE ENG 2 LPTACC VALVE	N	1) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00. 2) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00.
ENG 1 FUEL FILTER SENSOR ENG 2 FUEL FILTER SENSOR ENG 1 FUEL STRAINER SENSOR ENG 2 FUEL STRAINER SENSOR ENG 1 FUEL MAIN HX SENSOR ENG 2 FUEL MAIN HX SENSOR ENG 1 OIL FILTER SENSOR ENG 2 OIL FILTER SENSOR ENG 1 OIL DEBRIS SENSOR ENG 2 OIL DEBRIS SENSOR	Y	1) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.
ENG 1 MTC VALVE ENG 2 MTC VALVE ENG 1 HPTACC VALVE ENG 2 HPTACC VALVE ENG 1 VIBRATION MONITOR ENG 2 VIBRATION MONITOR	Y	1) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00. 2) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00.

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Table 201/46-13-02-993-801 Latched STATUS messages from MDS (Continued)

Status Text	Dispatch Relief	Alternate Reset Procedure
ENG 1 REVERSER ENG 2 REVERSER	Y	Cycle the thrust reversers with the engines on. Do these steps: 1) Do this task: Start the Engine (Normal Start), TASK 71-00-00-910-803-G00. 2) Move the applicable reverse thrust lever up and aft to the extend position. 3) Make sure that the green REV indication shows when the thrust reverser fully deployed. 4) Wait a minimum of 20 seconds. 5) Move the applicable reverse thrust lever forward and down to the retract (stow) position. 6) Do this task: Stop the Engine (Usual Engine Stop), TASK 71-00-00-910-806-G00.
SPOILER LOAD ALLEVIATION	Y	1) Refer to the observed fault code 276 105 00 in the Interactive Fault Isolation Manual (IFIM).
STALL WARNING SYS L STALL WARNING SYS R	N	1) Do this task: Stall Management Yaw Damper (SMYD) BITE Test - Existing Faults, TASK 27-32-00-740-801.
TAT DISAGREE	N	1) Do this task: Flight Management Computer System - System Test, TASK 34-61-00-730-801.

Table 202/46-13-02-993-802 Latched STATUS Messages from non-MDS LRUs

Status Text	Dispatch Relief	Alternate Reset Procedure
AIR/GND SYS MLG L SENSOR MLG R SENSOR NLG SENSOR	N	1) Open and close the circuit breakers for the Proximity Switch Electronic Unit (PSEU), M2061 (PAGEBLOCK 32-09-10/401). 2) Put the airplane in the air mode with deactuators (alternate method) to do the operational check of the landing gear sensors (TASK 32-09-00-860-801).
ALTN NOSE STEERING	N	1) Open and close the circuit breakers for the PSEU, M2061 (PAGEBLOCK 32-09-10/401). 2) Do the test of the alternate steering system (TASK 32-51-00-700-801).
AUTOSLAT FAIL SMYD 1 AUTOSLAT FAIL SMYD 2 YAW DAMPER SYS	Y	1) Do this task: Stall Management Yaw Damper (SMYD) BITE Test - Existing Faults, TASK 27-32-00-740-801.
AUTOSLAT FAIL	N	1) Do this task: Stall Management Yaw Damper (SMYD) BITE Test - Existing Faults, TASK 27-32-00-740-801.

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Table 202/46-13-02-993-802 Latched STATUS Messages from non-MDS LRUs (Continued)

Status Text	Dispatch Relief	Alternate Reset Procedure
DOOR AFT CARGO DOOR AFT ENTRY DOOR AFT SERVICE DOOR EE ACCESS DOOR EQUIP IND DOOR FWD CARGO DOOR FWD ENTRY DOOR FWD SERVICE DOOR L MID CABIN FLTLOCK DOOR L OVERWING FLTLOCK DOOR L OVERWING DOOR R MID CABIN FLTLOCK DOOR R OVERWING FLTLOCK DOOR R OVERWING	Y	1) Open and close the circuit breakers for the PSEU, M2061 (PAGEBLOCK 32-09-10/401). 2) Do this task: Door Warning System Test, TASK 52-71-00-730-801.
PACK FLOW CTRL SENSOR L PACK FLOW CTRL VALVE L PACK FLOW PRIMARY L PACK TEMP CTRL PRIMARY L	Y	1) Open and close the circuit breakers for the Left Integrated Air System Controller (IASC), M2939 (PAGEBLOCK 21-61-22/401). 2) Do this task: Pack Flow Control Valve Operational Test, TASK 21-51-00-700-802.
PACK FLOW CTRL SENSOR R PACK FLOW CTRL VALVE R PACK FLOW PRIMARY R PACK TEMP CTRL PRIMARY R	Y	1) Open and close the circuit breakers for the Right IASC, M2957 (PAGEBLOCK 21-61-22/401). 2) Do this task: Pack Flow Control Valve Operational Test, TASK 21-51-00-700-802.

——— END OF TASK ———

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AIRCRAFT INTERFACE DEVICE - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Aircraft Interface Device activation.

(2) Aircraft Interface Device deactivation.

(3) Aircraft Interface Device power cycle.

(4) Aircraft Interface Device coaxial cable WWU antenna verification.

(5) Aircraft Interface Device software verification.

(6) Aircraft Interface Device software installation.

(7) Aircraft Interface Device SIM replacement.

TASK 46-13-52-440-801

2. Aircraft Interface Device (AID) Activation

A. General

(1) This task gives two ways to operate the Aircraft Interface Device (AID) radio enable discrete.

(a) The Cellular RF Status ON function shows on the CELLULAR DISCRETES page.

(b) The Wi-Fi RF Status ON function shows on the WIRELESS DISCRETES page.

(2) When the Radio Frequency (RF) Status functions are set to off, or disabled, the AID remains energized.

(a) To remove electrical power from the AID, you must open the correct circuit breaker.

(3) The aircraft must be on the ground to do this task.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Prepare for the Activation

SUBTASK 46-13-52-480-001

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

E. Wireless Discrete Activation

SUBTASK 46-13-52-440-001

(1) On the Onboard Network System (ONS) maintenance browser:

(a) Select EXTENDED MAINTENANCE > Offboard Communication.

(b) Select WIRELESS on the top of the page.

(c) Select DISCRETES.

(d) Below the words DISCRETE & PROFILE CONTROL, examine the Wi-Fi RF Status.

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(e) If the Wi-Fi RF Status shows OFF, select the ON/OFF button to turn the Wi-Fi RF Status to ON.

NOTE: The label of the button changes from OFF to ON.

1) If the Wi-Fi RF Status button is grayed out, do this task: Client Credentials Check, TASK 46-13-00-860-803.

F. Cellular Discrete Activation

SUBTASK 46-13-52-440-003

(1) On the ONS maintenance browser:

NOTE: If necessary, make sure that SIM card 1 (J10) and/or SIM card 2 (J9) are installed.

(a) Select EXTENDED MAINTENANCE > Offboard Communication.

(b) Select CELLULAR on the top of the page.

(c) Select DISCRETES.

(d) Below the words DISCRETE & PROFILE CONTROL, examine the Cellular RF Status.

(e) If the Cellular RF Status shows OFF, select the ON/OFF button to turn the Cellular RF Status to ON.

NOTE: The label of the button changes from OFF to ON.

1) If the Cellular RF Status button is grayed out, do this task: Client Credentials Check, TASK 46-13-00-860-803.

——— END OF TASK ———

TASK 46-13-52-040-801

3. Aircraft Interface Device (AID) Deactivation

A. General

(1) This task gives two ways to operate the Aircraft Interface Device (AID) radio disable discrete.

(a) The Cellular RF Status OFF function shows on the CELLULAR DISCRETES page.

(b) The Wi-Fi RF Status OFF function shows on the WIRELESS DISCRETES page.

(2) When the Radio Frequency (RF) discrete status function is set to off, or disabled, the AID remains energized.

(a) To remove electrical power from the AID, you must open the correct circuit breaker.

(3) The aircraft must be on the ground to do this task.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Prepare for the Deactivation

SUBTASK 46-13-52-480-002

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

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E. Wireless Discrete Deactivation

SUBTASK 46-13-52-040-001

(1) On the Onboard Network System (ONS) maintenance browser:

(a) Select EXTENDED MAINTENANCE > Offboard Communication.

(b) Select WIRELESS on the top of the page.

(c) Select DISCRETES.

(d) Below the words DISCRETE & PROFILE CONTROL, examine the Wi-Fi RF Status.

(e) If the Wi-Fi RF Status shows ON, select the ON/OFF button to turn the Wi-Fi RF Status to OFF.

NOTE: The label of the button changes from ON to OFF.

1) If the Wi-Fi RF Status button is grayed out, do this task: Client Credentials Check, TASK 46-13-00-860-803.

F. Cellular Discrete Deactivation

SUBTASK 46-13-52-040-003

(1) On the ONS maintenance browser:

(a) Select EXTENDED MAINTENANCE > Offboard Communication.

(b) Select CELLULAR on the top of the page.

(c) Select DISCRETES.

(d) Below the words DISCRETE & PROFILE CONTROL, examine the Cellular RF Status.

(e) If the Cellular RF Status shows ON, select the ON/OFF button to turn the Cellular RF Status to OFF.

NOTE: The label of the button changes from ON to OFF.

1) If the Cellular RF Status button is grayed out, do this task: Client Credentials Check, TASK 46-13-00-860-803.

——— END OF TASK ———

TASK 46-13-52-440-802

4. Aircraft Interface Device (AID) Power Cycle

A. General

(1) This task contains the procedure to power cycle the Aircraft Interface Device (AID).

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Procedure

SUBTASK 46-13-52-440-006

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

(2) Wait for 20 seconds.

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(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

——— END OF TASK ———

TASK 46-13-52-200-801

5. Aircraft Interface Device (AID) Coaxial Cable WWU Antenna Verification

A. General

(1) This task gives the steps to perform the Aircraft Interface Device (AID) coaxial cables Wireless WAN Unit (WWU) antenna verification on the aircraft, using a FDR, TDR, or Multimeter.

(2) The AID, M3501 is installed at STA 467, WL 305, BL R005.

(3) The Antenna Disconnect Bracket, AB0488 is installed at STA 488, WL 304, BL R009.

(4) You must have access to the interior crown of the fuselage to do this task.

B. References

Reference	Title
20-10-72-210-801	Coaxial Cable Inspection with a Time Domain Reflectometer (P/B 201)
20-10-72-211-801	Coaxial Cable and Antenna System Inspection with a Frequency Domain Reflectometer (P/B 201)
46-13-52-000-801	Aircraft Interface Device Removal (P/B 401)
46-13-52-400-801	Aircraft Interface Device Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-5187	Reflectometer - Time Domain (TDR) Part #: 1205CXB Supplier: 93929 Part #: 6021-5154 Supplier: 1UW81 Part #: W1101 Supplier: FB3T5 Opt Part #: 1205CXA Supplier: 93929 Opt Part #: 1270A Supplier: 93929 Opt Part #: 900AST Supplier: 53387
COM-15185	Reflectometer - Frequency Domain (FDR) Part #: FH-AV-KIT Supplier: 70998 Part #: W4101 Supplier: FB3T5 Opt Part #: 7003A001-4 Supplier: 21844

D. Location Zones

Zone	Area
234	Area Above Passenger Compartment Ceiling, Right - Section 43

E. Prepare for the Aircraft Interface Device (AID) Coaxial Cable WWU Antenna Verification

SUBTASK 46-13-52-040-009

(1) Deactivate the AID (TASK 46-13-52-040-801).

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(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

F. Aircraft Interface Device (AID) Coaxial Cable WWU Antenna Verification - Using an FDR or TDR

SUBTASK 46-13-52-200-001

(1) Boeing recommends that you use tool frequency domain reflectometer, COM-15185 or time domain reflectometer, COM-5187 to do the coaxial cable check if available.

(a) These tasks tell you how to use the Frequency Domain Reflectometer and Time Domain Reflectometer:

- Coaxial Cable and Antenna System Inspection with a Frequency Domain Reflectometer, TASK 20-10-72-211-801
- Coaxial Cable Inspection with a Time Domain Reflectometer, TASK 20-10-72-210-801

(2) Remove the antenna coaxial cables from the AID (TASK 46-13-52-000-801).

(a) Do a check of the coaxial cable between the AID and the antennas.

Coaxial Cable Wire Number	AID, M3501
W4923-002CX-BMS 13-65-0G	D17863
W4923-003CX-BMS 13-65-0G	D17864
W4923-004CX-BMS 13-65-0G	D17861
W4923-005CX-BMS 13-65-0G	D17862

(b) Re-connect the antenna coaxial cables to the AID (TASK 46-13-52-400-801).

G. Aircraft Interface Device (AID) Coaxial Cable WWU Antenna Verification - Using a Multimeter

SUBTASK 46-13-52-200-002

(1) Get access to the AID (TASK 46-13-52-000-801).

(a) Disconnect these connectors at the AID:

- D17861 (J3 port)
- D17862 (J4 port)
- D17863 (J5 port)
- D17864 (J6 port)

(2) Get access to the Antenna Disconnect Bracket, AB0488, from inside the aircraft at approximately STA 488, WL 304, BL R009.

(a) Disconnect these connectors at the Antenna Disconnect Bracket:

- D16890
- D16891
- D16892
- D16893

(3) Do a wiring check between these pins:

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Coaxial Cable Wire Number	AID, M3501	Bracket, AB0488
W4923-002CX-BMS 13-65-0G	D17863, Pin A1	D16890, Pin A1
W4923-003CX-BMS 13-65-0G	D17864, Pin A1	D16891, Pin A1
W4923-004CX-BMS 13-65-0G	D17861, Pin A1	D16892, Pin A1
W4923-005CX-BMS 13-65-0G	D17862, Pin A1	D16893, Pin A1

(4) Re-connect the connectors at the Antenna Disconnect Bracket.

(5) Re-connect the connectors at the AID.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-52-840-006

(1) Make sure that the AID is installed correctly (TASK 46-13-52-400-801).

——— END OF TASK ———

TASK 46-13-52-470-801

6. Aircraft Interface Device (AID) Software Verification

A. General

(1) This task gives the steps to make sure that the Operational Program Software (OPS) and Operational Program Configuration (OPC) software installed in the Aircraft Interface Device (AID).

(2) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD),
- Universal Maintenance Device (UMD).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-00-710-802	System Configuration Check (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Aircraft Interface Device (AID) Software Verification

SUBTASK 46-13-52-840-001

(1) Access the ONS via these methods:

(a) To access the ONS via the UMD, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

(b) To access the ONS via the MFD, do this task: Multi Function Display Connection, TASK 46-13-00-480-803.

SUBTASK 46-13-52-840-005

(2) Do a check of the AID software (TASK 46-13-00-710-802).

(a) Record and make sure that the software part numbers are correct.

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1) If the software part number is not correct, do this task: Aircraft Interface Device (AID) Software Installation, TASK 46-13-52-470-802.

2) If there are still problems with the AID software, or if the AID System Config does not show, then refer to the IFIM and do the applicable procedure(s).

——— END OF TASK ———

TASK 46-13-52-470-802

7. Aircraft Interface Device (AID) Software Installation

A. General

(1) This task gives the steps to install the Operational Program Software (OPS) and Operational Program Configuration (OPC) into the Aircraft Interface Device (AID) from the Mass Storage Device (MSD) to the AID.

(2) If the software parts are not already staged to the MSD, then gather all of the software parts you need on removable media or the Universal Maintenance Device (UMD) before going to the airplane.

(3) This task is available when the Onboard Network System (ONS) is accessed with the methods that follow:

- Multi Function Display (MFD)
- UMD.

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-480-803	Multi Function Display Connection (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Aircraft Interface Device (AID) Software Installation

SUBTASK 46-13-52-470-007

(1) Access the ONS via these methods:

(a) To access the ONS via the UMD, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

(b) To access the ONS via the MFD, do this task: Multi Function Display Connection, TASK 46-13-00-480-803.

SUBTASK 46-13-52-470-005

(2) To install the applicable AID software, do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

NOTE: If the AID does not contain software (factory state), all software parts can be added to the load queue for loading. Both the AID OPC and OPS must be loaded at the same time. If data loading one part at a time, you must data load the OPC first and then the OPS. Otherwise, data load will fail.

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(a) If the AID software load is failed, then refer to the IFIM and do the applicable procedure(s).

NOTE: If AID Data-Load fails and prompts with a message "System Busy Generating", try another Data-Load.

SUBTASK 46-13-52-470-006

(3) To make and install new client credentials for the AID, do this task: Client Credentials Check, TASK 46-13-00-860-803.

NOTE: It is necessary to install new credentials after the AID software is installed on a new AID unit.

NOTE: For the LRUs in the Table 201, wait the minimum time specified before you continue.

Table 201/46-13-52-993-804 Minimum Reboot Time for the LRUs

LRU	Minimum Reboot Time (after successful Data Load Message)
AID	9 minutes

NOTE: These LRUs must reboot fully before you continue.

——— END OF TASK ———

TASK 46-13-52-480-801

8. Aircraft Interface Device (AID) SIM Card Replacement

(Figure 201)

A. General

(1) This task contains the procedures to replace the Subscriber Identity Module (SIM) cards on the Aircraft Interface Device (AID).

(2) The AID is installed in the ceiling crown at STA 467, WL 305, RBL 5.

(3) Up to two SIM cards can be installed in the AID.

- SIM card 1 is installed at slot J10.
- SIM card 2 is installed at slot J9.

B. References

Reference	Title
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)
25-80-00-000-801	Insulation Blanket Removal (P/B 401)
25-80-00-400-801	Insulation Blanket Installation (P/B 401)
46-13-52-710-801	Aircraft Interface Device (AID) Operational Test - Cellular Connection (P/B 501)

C. Location Zones

Zone	Area
234	Area Above Passenger Compartment Ceiling, Right - Section 43

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D. Aircraft Interface Device (AID) SIM Card Replacement

SUBTASK 46-13-52-840-004

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-52-010-008

WARNING THE AIRCRAFT INTERFACE DEVICE MUST BE OFF IF YOU ARE LESS THAN 8 INCHES (20 CMS) FROM THE RADIO FREQUENCY ANTENNAS. IF YOU DO NOT OBEY, INJURY FROM ELECTROMAGNETIC RADIATION CAN OCCUR.

(2) Get access to the AID [1]:

(a) If it is necessary, remove the applicable ceiling panels (TASK 25-21-45-000-801).

(b) Remove the applicable insulation (TASK 25-80-00-000-801).

SUBTASK 46-13-52-020-003

(3) If there is a SIM CONFIGURATION label on the access door [2], remove and discard the label.

SUBTASK 46-13-52-420-012

(4) Loosen the captive screws [3] on the access door [2].

SUBTASK 46-13-52-420-013

(5) Open the access door [2].

SUBTASK 46-13-52-420-014

(6) For the SIM cards [5], do these steps:

- SIM card 1 is installed at slot J10.
- SIM card 2 is installed at slot J9.

(a) To remove the card(s), carefully push on the applicable card to release it.

(b) To install the card(s), carefully push on the applicable card until it latches.

1) Make sure that the cut corner is in the correct position (Figure 201).

SUBTASK 46-13-52-420-016

(7) Close the access door [2].

SUBTASK 46-13-52-420-017

(8) Tighten the captive screws [3].

SUBTASK 46-13-52-440-005

(9) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

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E. AID SIM Replacement Test

SUBTASK 46-13-52-710-003

(1) Do this task: Aircraft Interface Device (AID) Operational Test - Cellular Connection, TASK 46-13-52-710-801.

NOTE: To do the operational test, you must have access to an active wireless connection from a remote access terminal, or airport messaging terminal.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-52-860-005

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-52-410-003

WARNING

THE AIRCRAFT INTERFACE DEVICE MUST BE OFF IF YOU ARE LESS THAN 8 INCHES (20 CMS) FROM THE RADIO FREQUENCY ANTENNAS. IF YOU DO NOT OBEY, INJURY FROM ELECTROMAGNETIC RADIATION CAN OCCUR.

(2) Close access to the AID [1]:

(a) Install the applicable insulation (TASK 25-80-00-400-801).

(b) If removed, install the applicable ceiling panels (TASK 25-21-45-400-801).

SUBTASK 46-13-52-860-006

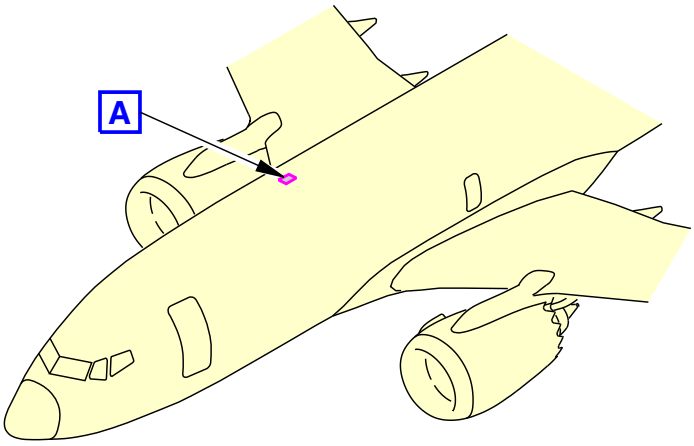
(3) Remove the safety tag and close this circuit breaker:

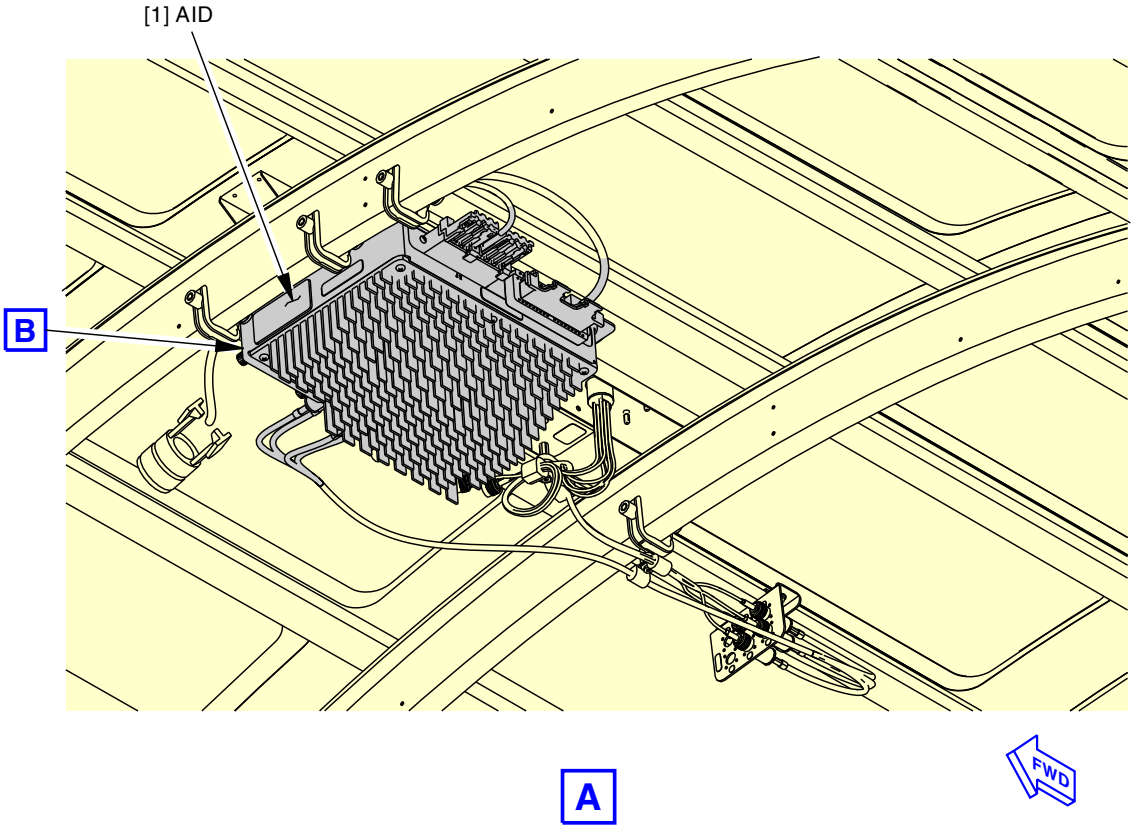
F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

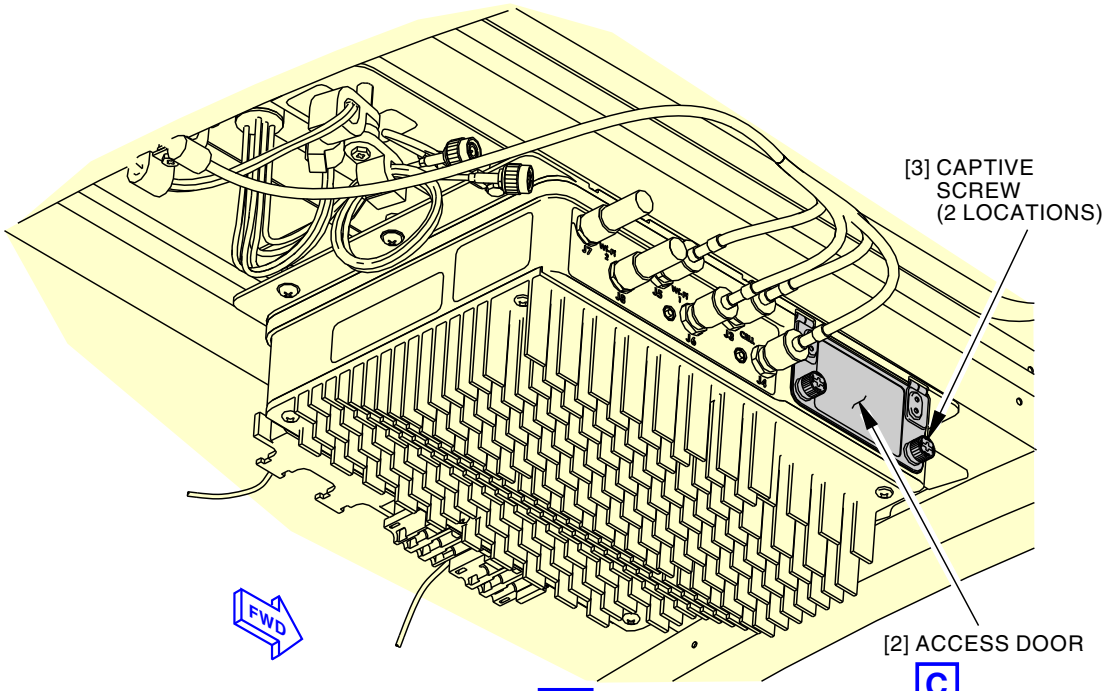


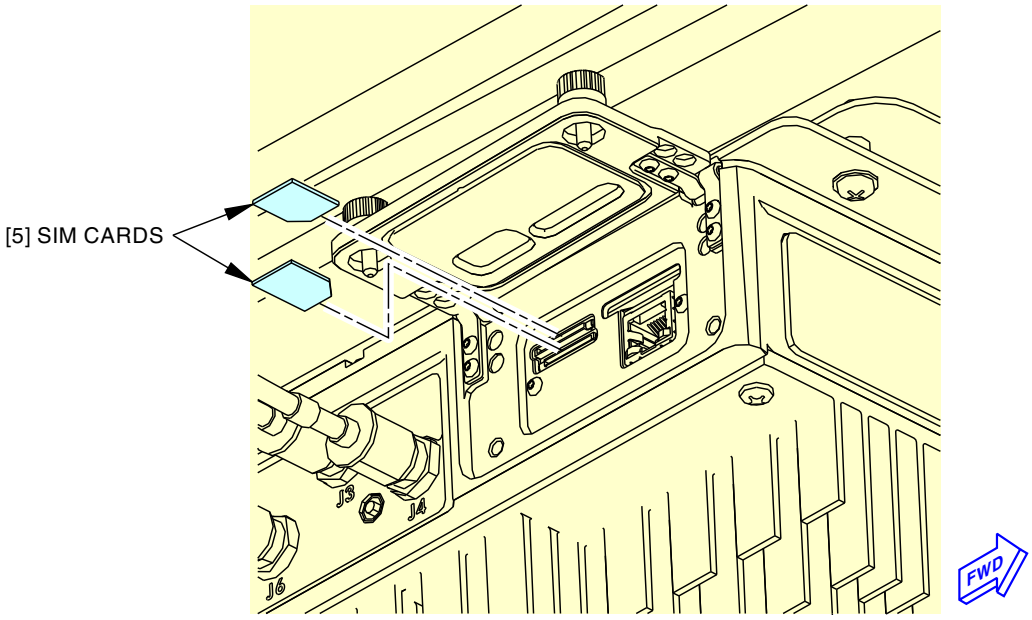


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Aircraft Interface Device (AID) SIM Card Replacement
Figure 201/46-13-52-990-801 (Sheet 1 of 2)

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Aircraft Interface Device (AID) SIM Card Replacement
Figure 201/46-13-52-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIRPLANE INTERFACE DEVICE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Aircraft Interface Device Removal

(2) Aircraft Interface Device Installation.

TASK 46-13-52-000-801

2. Aircraft Interface Device Removal

(Figure 401)

A. General

(1) This task gives the steps to remove the Aircraft Interface Device (AID).

(2) The AID is installed in the ceiling crown at STA 467, WL 305, RBL 5.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-803	Conductive Dust Cap and Connector Cover Installation (P/B 201)
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)
25-80-00-000-801	Insulation Blanket Removal (P/B 401)
46-13-52-480-801	Aircraft Interface Device (AID) SIM Card Replacement (P/B 201)

C. Location Zones

Zone	Area
234	Area Above Passenger Compartment Ceiling, Right - Section 43

D. Prepare for the Removal

SUBTASK 46-13-52-040-008

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-52-010-006

WARNING THE AIRCRAFT INTERFACE DEVICE MUST BE OFF IF YOU ARE LESS THAN 8 INCHES (20 CMS) FROM THE RADIO FREQUENCY ANTENNAS. IF YOU DO NOT OBEY, INJURY FROM ELECTROMAGNETIC RADIATION CAN OCCUR.

(2) Get access to the AID [1]:

(a) If it is necessary, remove the applicable ceiling panels (TASK 25-21-45-000-801).

(b) Remove the applicable insulation (TASK 25-80-00-000-801).

E. AID Removal

SUBTASK 46-13-52-000-001

(1) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

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SUBTASK 46-13-52-020-002

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(2) Remove the AID [1] as follows:

(a) Disconnect the electrical connector [10] and electrical connector [11] from the AID [1].

(b) Disconnect the WIFI connectors [13] and the cellular connectors [12] from the AID [1].

(c) Put the protective covers on the AID connectors (TASK 20-40-12-400-803).

(d) Put the protective covers on all other connectors.

(e) Remove the bolt [5], washers [3], spacer [7], and bonding jumper [8] from the AID [1]

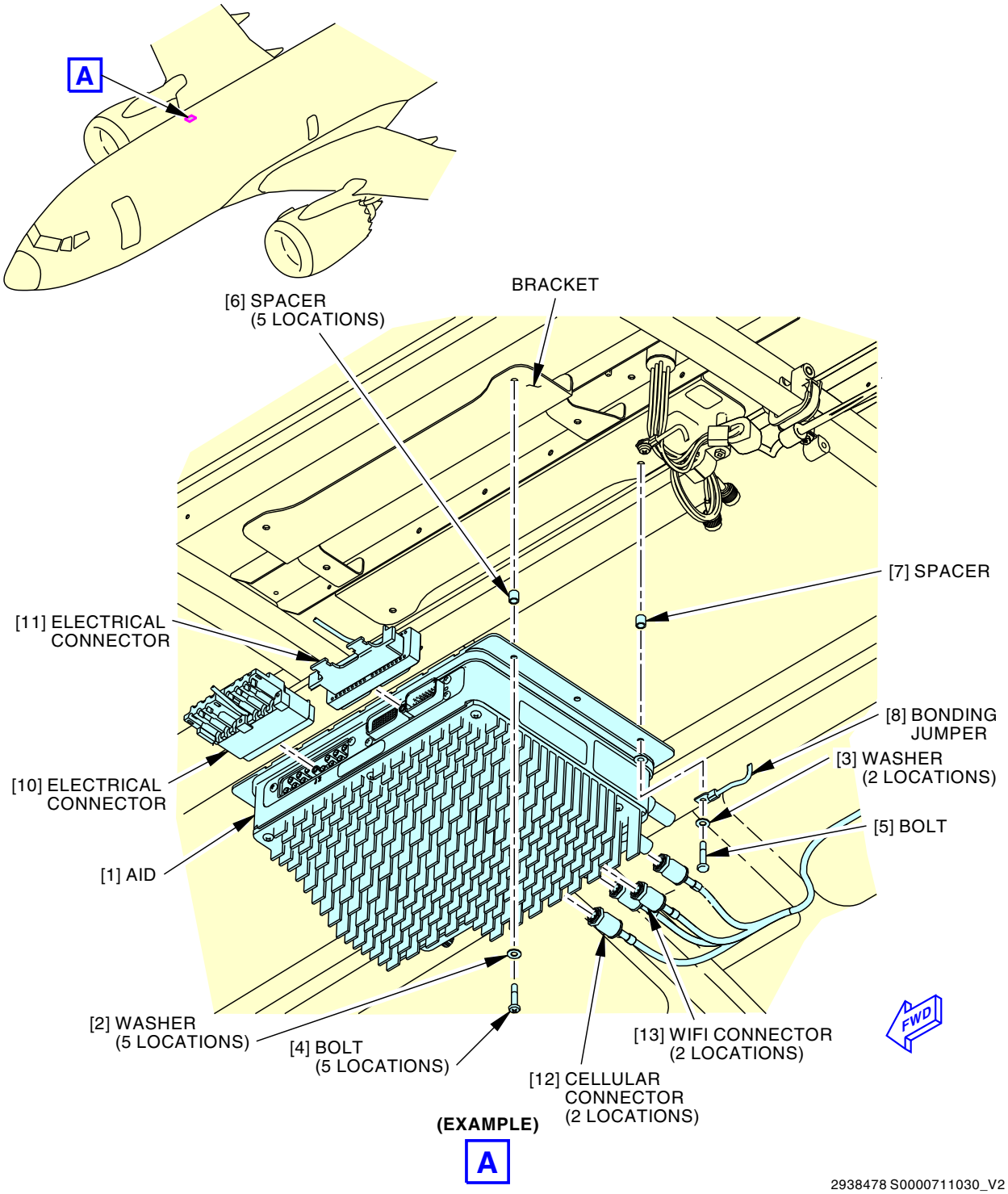
(f) Remove the bolts [4], washers [2], and spacers [6] from the AID [1].

(g) Remove the AID [1] from the overhead location.

(h) If it is necessary, remove the AID Subscriber Identity Module (SIM) card and Secure Digital (SD) card (TASK 46-13-52-480-801).

——— END OF TASK ———

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Aircraft Interface Device (AID) - Installation
Figure 401/46-13-52-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-52-400-801

3. Aircraft Interface Device Installation

(Figure 401)

A. General

(1) This task gives the steps to install the Aircraft Interface Device (AID).

(2) The AID is installed in the ceiling crown at STA 467, WL 305, RBL 5.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)
25-80-00-400-801	Insulation Blanket Installation (P/B 401)
46-13-52-470-801	Aircraft Interface Device (AID) Software Verification (P/B 201)
46-13-52-470-802	Aircraft Interface Device (AID) Software Installation (P/B 201)
46-13-52-480-801	Aircraft Interface Device (AID) SIM Card Replacement (P/B 201)
46-13-52-710-801	Aircraft Interface Device (AID) Operational Test - Cellular Connection (P/B 501)
46-13-52-710-802	Aircraft Interface Device (AID) Operational Test - Wi-Fi Connection (P/B 501)
46-13-52-730-801	Aircraft Interface Device (AID) System Test (P/B 501)
51-21-00-160-801	Surface Preparation Of Aluminum Bonding Surfaces For Antenna Installations (P/B 701)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
51-21-72-370-804	BMS10-11 Primer - Application (P/B 701)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)

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(Continued)

Reference	Description	Specification
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	AID	46-13-52-01-045	PTF ALL

F. Location Zones

Zone	Area
234	Area Above Passenger Compartment Ceiling, Right - Section 43

G. Prepare for the Installation

SUBTASK 46-13-52-040-007

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

H. AID Installation

SUBTASK 46-13-52-840-003

(1) Abrasive clean the area around the hole for bolt [5] on the AID [1] (TASK 51-21-00-160-801).

SUBTASK 46-13-52-420-018

(2) Apply a layer of primer, C00259, on the unpainted surface around the holes for bolts [4] (TASK 51-21-72-370-804).

SUBTASK 46-13-52-420-004

(3) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 46-13-52-420-005

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(4) Install the AID [1]:

(a) Put the AID [1] in position on the bracket.

(b) Install the AID [1] with the bolts [4], washers [2], and spacers [6].

(c) Apply sealant, A00247, to the surfaces that follow:

1) The threads and the shank of the bolt [5].

2) The washers [3], spacer [7], and the terminal of the bonding jumper [8].

(d) Install the bolt [5], washers [3], bonding jumper [8], and spacer [7].

(e) Tighten the bolts [4] and bolt [5] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(f) Use an intrinsically safe approved bonding meter, COM-1550, to measure the resistance between the terminal of the bonding jumper [8] and the flange of the AID [1].

1) Make sure that the resistance is 0.001 ohms (1 milliohm) or less.

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WARNING

DO NOT GET CHEMICAL CONVERSION COATING SOLUTION IN YOUR MOUTH, IN YOUR EYES, ON YOUR SKIN, OR ON YOUR CLOTHES. DO NOT BREATHE THE FUMES FROM THIS MATERIAL. AIR MUST FLOW FREELY THROUGH THE WORK AREA. USE THE NECESSARY RESPIRATORY PROTECTION. KEEP THIS MATERIAL AWAY FROM SPARKS, FLAME, AND HEAT. THIS MATERIAL IS POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(g) Manually apply coating, C00064, on the bolt [5] and adjacent bare metal surfaces (TASK 51-21-41-370-802).

(h) Apply a layer of primer, C00259 on the bolt [5] and adjacent bare metal surfaces (TASK 51-21-72-370-804).

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(i) Remove the protective covers from the AID electrical connectors.

(j) Make sure that the electrical connectors do not have bent or broken pins, dirt, or damage.

(k) Connect the electrical connector [10] and electrical connector [11] to the AID [1].

(l) Connect the WIFI connectors [13] to the AID [1].

(m) Connect the cellular connectors [12] to the AID [1].

SUBTASK 46-13-52-420-006

(5) If it is necessary, install the AID Subscriber Identity Module (SIM) card and Secure Digital (SD) card (TASK 46-13-52-480-801).

I. AID Installation Test

SUBTASK 46-13-52-470-004

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-52-470-008

(2) Dataload the OPS and OPC software to the AID: Aircraft Interface Device (AID) Software Installation, TASK 46-13-52-470-802.

SUBTASK 46-13-52-470-002

(3) Verify that the OPS and OPC software is installed in the AID: Aircraft Interface Device (AID) Software Verification, TASK 46-13-52-470-801.

SUBTASK 46-13-52-750-001

(4) Do this task: Aircraft Interface Device (AID) System Test, TASK 46-13-52-730-801.

SUBTASK 46-13-52-710-004

(5) Do this task: Aircraft Interface Device (AID) Operational Test - Wi-Fi Connection, TASK 46-13-52-710-802.

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SUBTASK 46-13-52-710-002

(6) Do this task: Aircraft Interface Device (AID) Operational Test - Cellular Connection, TASK 46-13-52-710-801.

NOTE: To do the operational test, you must have access to an active wireless connection from a remote access terminal, or airport messaging terminal.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-52-860-003

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-52-410-004

WARNING

THE AIRCRAFT INTERFACE DEVICE MUST BE OFF IF YOU ARE LESS THAN 8 INCHES (20 CMS) FROM THE RADIO FREQUENCY ANTENNAS. IF YOU DO NOT OBEY, INJURY FROM ELECTROMAGNETIC RADIATION CAN OCCUR.

(2) Close access to the AID [1]:

(a) Install the applicable insulation (TASK 25-80-00-400-801).

(b) If removed, install the applicable ceiling panels (TASK 25-21-45-400-801).

SUBTASK 46-13-52-860-004

(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

END OF TASK

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AIRCRAFT INTERFACE DEVICE - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Aircraft interface device operational test - cellular connection.

(2) Aircraft interface device operational test - Wi-Fi connection.

(3) Aircraft interface device system test.

(4) Aircraft interface device cellular scan test.

(5) Aircraft interface device Wi-Fi scan test.

TASK 46-13-52-710-801

2. Aircraft Interface Device (AID) Operational Test - Cellular Connection

A. General

(1) This task makes sure that the Aircraft Interface Device (AID) cellular function is serviceable. This task should only be completed if the aircraft is configured for cellular function.

(2) The aircraft must be on the ground to do this task.

(3) UMS parts must be installed to the Network File Server (NFS) to do this task.

(4) A SIM card must be installed to the AID to do this task.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)
46-13-52-470-801	Aircraft Interface Device (AID) Software Verification (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. AID Operational Test - Cellular Connection

SUBTASK 46-13-52-730-001

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

SUBTASK 46-13-52-730-010

(2) If not already completed, then verify that the OPS and OPC software is installed in the AID (Aircraft Interface Device (AID) Software Verification, TASK 46-13-52-470-801).

SUBTASK 46-13-52-730-011

(3) If not already completed, then verify that the AID has a client credential (Client Credentials Check, TASK 46-13-00-860-803).

SUBTASK 46-13-52-730-004

(4) On the Onboard Network System (ONS) maintenance browser:

(a) Select EXTENDED MAINTENANCE > Offboard Communication.

(b) Select CELLULAR on the top of the page.

(c) Select DISCRETES.

(d) If the Aircraft Location field is empty, select the arrow (>) for this field.

1) Select the applicable airplane location.

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2) Select Back to Cellular Discretes.

(e) If the Active Profile field is empty, select the arrow (>) for this field.

1) Select an available profile.

2) Select Back to Cellular Discretes.

(f) Select DIAGNOSTICS.

1) Make sure that a sim serial number populates next to “Sim Serial Number”.

NOTE: If necessary, use the scroll bar to find “Sim Serial Number”.

2) Make sure that the applicable Link State is shown.

NOTE: It may take a few minutes for the Link State to populate.

LINK STATE	DESCRIPTION
Home Network	Cellular connected to home network
Home Network-Default	Cellular to home network and is the default route
Roaming	Cellular connected but not to home network
Roaming-Default	Cellular connected but not to home network and is the default route

(g) Select TEST LINK.

1) Make sure that the link test says “Success”.

NOTE: It may take a few minutes for the link test to complete.

(h) Select MESSAGING.

1) Verify that an active link shows for the service “OBEDS”.

NOTE: An active link will show in green text. An inactive link will show in orange text.

(i) Select the three-line icon on the menu bar at the upper left-hand corner.

(j) Select HOME.

(k) Select EXTENDED MAINTENANCE > Offboard Communication.

(l) Select the CELLULAR radio button under “LINK”.

(m) If the SET DEFAULT button is not grayed out, then do the steps that follow:

1) Select SET DEFAULT.

2) Select TEST DEFAULT.

a) Verify that a green check mark appears under “TEST” for the CELLULAR link.

(n) If the SET DEFAULT button is grayed out, then do the steps that follow:

1) Select TEST DEFAULT.

a) Verify that a green check mark appears under “TEST” for the CELLULAR link.

2) Select the CELLULAR tab in the horizontal menu at the top of the page.

3) Select CONDITION.

a) Make sure that the Active Profile and Associated Network Speciation fields are not empty.

b) Make sure that PARAMETERS show the values that follow:

- the WWU RF Status is On.

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- The WWU WOW Status is Ground.
- The Controller State is Running.

——— END OF TASK ———

TASK 46-13-52-710-802

3. Aircraft Interface Device (AID) Operational Test - Wi-Fi Connection

A. General

(1) This task makes sure that the Aircraft Interface Device (AID) Wi-Fi function is servicable. This task should only be completed if the aircraft is configured for Wi-Fi function.

(2) The aircraft must be on the ground to do this task.

(3) UMS parts must be installed to the Network File Server (NFS) to do this task.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)
46-13-52-470-801	Aircraft Interface Device (AID) Software Verification (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. AID Operational Test - Wi-Fi Connection

SUBTASK 46-13-52-730-012

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

SUBTASK 46-13-52-730-013

(2) If not already completed, then verify that the OPS and OPC software is installed in the AID (Aircraft Interface Device (AID) Software Verification, TASK 46-13-52-470-801).

SUBTASK 46-13-52-730-014

(3) If not already completed, then verify that the AID has a client credential (Client Credentials Check, TASK 46-13-00-860-803).

SUBTASK 46-13-52-730-015

(4) On the Onboard Network System (ONS) maintenance browser:

(a) Select EXTENDED MAINTENANCE > Offboard Communication.

(b) Select WIRELESS on the top of the page.

(c) Select DISCRETES.

(d) If the Aircraft Location field is empty, select the arrow (>) for this field.

1) Select the applicable airplane location.

2) Select Back to Wireless Discretes.

(e) If the Active Profile field is empty, select the arrow (>) for this field.

1) Select an available profile.

2) Select Back to Wireless Discretes.

(f) Select DIAGNOSTICS.

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1) Make sure that the applicable Link State is shown.

NOTE: It may take a few minutes for the Link State to populate.

LINK STATE	DESCRIPTION
Connected	Offboard Wi-Fi is connected
Connected - Default	Offboard Wi-Fi is connected and is the default route

(g) Select TEST LINK.

1) Make sure that the link test says “Success”.

NOTE: It may take a few minutes for the link test to complete.

(h) Select MESSAGING.

1) Verify that an active link shows for the service “OBEDS”.

NOTE: An active link will show in green text. An inactive link will show in orange text.

(i) Select the three-line icon on the menu bar at the upper left-hand corner.

(j) Select HOME.

(k) Select EXTENDED MAINTENANCE > Offboard Communication.

(l) Select the TWLU radio button under “LINK”.

(m) If the SET DEFAULT button is not grayed out, then do the steps that follow:

1) Select SET DEFAULT.

2) Select TEST DEFAULT.

a) Verify that a green check mark appears under “TEST” for the Wi-Fi link.

(n) If the SET DEFAULT button is grayed out, then do the steps that follow:

1) Select TEST DEFAULT.

a) Verify that a green check mark appears under “TEST” for the Wi-Fi link.

2) Select the WIRELESS tab in the horizontal menu at the top of the page.

3) Select CONDITION.

a) Make sure that the Active Profile and Associated SSID fields are not empty.

b) Make sure that PARAMETERS show the values that follow:

- the WWU RF Status is On.
- The WWU WOW Status is Ground.
- The Controller State is Running.

——— END OF TASK ———

TASK 46-13-52-730-801

4. Aircraft Interface Device (AID) System Test

A. General

(1) This task makes sure that the Aircraft Interface Device (AID) is serviceable.

(2) The aircraft must be on the ground to do this task.

(3) No 46 NFS UMS parts are needed to be installed to the Network File Server (NFS) to complete this task.

(4) No sim card needs to be installed to the AID to complete this task.

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B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-860-803	Client Credentials Check (P/B 201)
46-13-52-470-801	Aircraft Interface Device (AID) Software Verification (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. AID System Test

SUBTASK 46-13-52-480-003

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

SUBTASK 46-13-52-730-016

(2) If not already completed, then make sure that the Operational Program Software (OPS) and Operational Program Configuration (OPC) software is installed in the AID (TASK 46-13-52-470-801).

SUBTASK 46-13-52-730-017

(3) If not already completed, then verify that the AID has a client credential (TASK 46-13-00-860-803).

SUBTASK 46-13-52-730-008

(4) On the Onboard Network System (ONS) maintenance browser:

(a) Select EXTENDED MAINTENANCE > Input Monitoring.

(b) Select NFS.

1) Make sure that the END CONNECTION for PORT 4 is AID-FWD.

2) Make sure that the STATUS for PORT 4 is Up.

(c) Select INPUT DISCRETES.

1) Make sure that the STATUS for W-Air/Ground Relay is Ground.

(d) Select the three-line icon on the menu bar at the upper left-hand corner.

(e) Select HOME.

(f) Select EXTENDED MAINTENANCE > OFFBOARD COMMUNICATION.

(g) Select WIRELESS button on the top of the page.

(h) If the WIRELESS functionality is enabled in the software configuration, then do these steps:

NOTE: If the WIRELESS functionality is disabled in the software configuration, the message will appear on the screen.

1) From the list of functions on the WIRELESS page footer, select DISCRETES.

2) If the outline of the WWU RF Discrete Status toggle is white and operable:

a) Make sure that the WWU RF Discrete Status is ON.

<1> If the WWU RF Discrete Status is OFF, then select the WWU RF Discrete Status button from OFF to ON.

b) From the list of functions on the WIRELESS page footer, select CONDITION.

<1> Make sure that the WWU RF Status is ON.

c) From the list of functions on the WIRELESS page footer, select DISCRETES.

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d) Set the WWU RF Discrete Status button from ON to OFF.

e) From the list of functions on the WIRELESS page footer, select CONDITION.

<1> Make sure that the WWU RF Status is OFF.

(i) Select CELLULAR at the top of the page.

(j) Select DISCRETES.

(k) If the outline of the Cellular RF Status toggle is white and operable:

1) Make sure that the Cellular RF Status is ON.

a) If the Cellular RF Status is OFF, then select the Cellular RF Status button from OFF to ON.

2) Select CONDITION.

a) Make sure that the WWU RF Status is ON.

3) Select DISCRETES.

4) Set the Cellular RF Status button from ON to OFF.

5) Select CONDITION.

a) Make sure that the WWU RF Status is OFF.

SUBTASK 46-13-52-730-018

(5) Do the steps that follow to set the PSEU SYS 2 to IN AIR using the Onboard Maintenance Function (OMF) functionality in the ONS, and to make sure that the state of the Weight-On-Wheels (WOW) discrete from the PSEU to AID is set to AIR:

(a) Select the SYS button on the P9-82 Multi Function Display (MFD) panel.

(b) Select MAINT > ONBD MAINT > OTHER FUNCTIONS > SPECIAL FUNCTIONS.

(c) Select the 32-Proximity Switch Electronic Unit under “Select ATA System”.

(d) Select PSEU-Air/Ground Override under “Select System Function”.

(e) Select CONTINUE.

(f) Verify the preconditions are met and then select CONTINUE.

(g) Select START FUNCTION under “FUNCTION”.

(h) Select Set SYS 2 only IN AIR.

(i) Select CONTINUE.

1) Make sure that the following is shown on the MAX Display System (MDS).

- Air/Gnd Override is ACTIVE
- System 1: GND
- System 2: AIR

(j) Select the menu icon in the upper left corner of the ONS maintenance screen on the Universal Maintenance Device (UMD).

(k) Select HOME > EXTENDED MAINTENANCE > Offboard Communication > CELLULAR > CONDITION.

1) Make sure that the WWU WOW Status is Air.

(l) Select DISCRETES.

1) Make sure that the Cellular RF Status is OFF and cannot be toggled ON.

(m) Select WIRELESS > CONDITION.

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1) Make sure that the WWU WOW Status is Air.

(n) Select DISCRETES.

1) Make sure that the Wi-Fi RF Status is OFF and cannot be toggled ON.

SUBTASK 46-13-52-730-019

(6) Do the steps that follow to set the PSEU SYS 2 to ON GND using the OMF functionality in the ONS, and to make sure that the state of the WOW discrete from the PSEU to AID is set to GND:

(a) Select the SYS button on the P9-82 MFD panel.

(b) Select MAINT > ONBD MAINT > OTHER FUNCTIONS > SPECIAL FUNCTIONS.

(c) Select the 32-Proximity Switch Electronic Unit under “Select ATA System”.

(d) Select PSEU-Air/Ground Override under “Select System Function”.

(e) Select CONTINUE.

(f) Verify the preconditions are met and then select CONTINUE.

(g) Select START FUNCTION under “FUNCTION”.

(h) Select Set SYS 2 only ON GND.

(i) Select CONTINUE.

1) Make sure that the following is shown on the MDS.

- Air/Gnd Override is NOT ACTIVE
- System 1: GND
- System 2: GND

(j) Select the menu icon in the upper left corner of the ONS maintenance screen on the UMD.

(k) Select HOME > EXTENDED MAINTENANCE > Offboard Communication.

(l) Select WIRELESS > DISCRETES.

1) If the outline of the Wi-Fi RF Status toggle is white and operable:

a) Make sure that the Wi-Fi RF Status is ON.

b) If the Wi-Fi RF Status is OFF, then select the Wi-Fi RF Status button from OFF to ON.

(m) Select CELLULAR > DISCRETES.

1) If the outline of the Cellular RF Status toggle is white and operable:

a) Make sure that the Cellular RF Status is ON.

b) If the Cellular RF Status is OFF, then select the Cellular RF Status button from OFF to ON.

SUBTASK 46-13-52-730-020

(7) Do the steps that follow to do a check of the AID antenna and coax connections:

(a) Select the menu icon in the upper left-hand corner of the browser client window.

(b) Select HOME > EXTENDED MAINTENANCE > Offboard Communication.

(c) Select WIRELESS > DISCRETES.

(d) If the outline of the Wi-Fi RF Status toggle is white and operable:

1) Make sure that the Wi-Fi RF Status is ON.

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2) If the Wi-Fi RF Status is OFF, then select the Wi-Fi RF Status button from OFF to ON.

3) Select SCAN.

4) Select CONFIRM.

5) Verify that the “Scan Results” table populates with at least one result.

(e) Select CELLULAR > DISCRETES.

(f) If the outline of the Cellular RF Status toggle is white and operable:

1) Make sure that the Cellular RF Status is ON.

2) If the Cellular RF Status is OFF, then select the Cellular RF Status button from OFF to ON.

3) Select SCAN.

4) Select CONFIRM.

5) Verify that the “Scan Results” table populates with at least one result.

SUBTASK 46-13-52-730-021

(8) Do the steps that follow to do a check for existing faults of the AID.

(a) Select the three-line icon on the menu bar at the upper left-hand corner.

(b) Select EXTENDED MAINTENANCE > Existing Faults.

(c) Select 46 - Information Systems.

NOTE: If there are no fault messages for ATA 46 - Information System, then the selection will not be displayed.

(d) Select CONTINUE.

1) Make sure that there are no active AID fault messages.

2) If there are active AID fault messages, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

——— END OF TASK ———

TASK 46-13-52-720-801

5. Aircraft Interface Device (AID) – Cellular Scan Test

A. General

(1) This task gives the steps to perform the Aircraft Interface Device (AID) cellular scan test.

(2) To do this task, the Weight-On-Wheels (WOW) discrete status must be in the ground condition.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Prepare for the Cellular Scan Test

SUBTASK 46-13-52-860-007

(1) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-52-480-004

(2) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

F. Aircraft Interface Device Cellular Scan Test

SUBTASK 46-13-52-720-001

(1) From the Onboard Network System (ONS) maintenance browser main menu, select EXTENDED MAINTENANCE.

SUBTASK 46-13-52-720-002

(2) Select Offboard Communication.

SUBTASK 46-13-52-720-003

(3) Select CELLULAR on the top of the page.

SUBTASK 46-13-52-720-004

(4) Select DISCRETES on the bottom of the page.

SUBTASK 46-13-52-720-005

(5) Make sure that the “Cellular RF Status” toggle is set to ON.

SUBTASK 46-13-52-720-006

(6) Select SCAN on the bottom of the page.

SUBTASK 46-13-52-720-007

(7) Make sure that the following message is displayed: “Initiating a cellular scan may take up to 2 minutes and will cause an active cellular connection to disconnect”.

SUBTASK 46-13-52-720-008

(8) Select CONFIRM.

SUBTASK 46-13-52-720-009

(9) Wait for the “CELLULAR SCAN” modal to disappear.

SUBTASK 46-13-52-720-010

(10) Make sure that a list of cellular networks is displayed.

SUBTASK 46-13-52-720-011

(11) Make sure that each cellular network in the list has values listed for the following headers:

(a) NETWORK NAME

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(b) NETWORK SPECIFICATION

(c) STATUS

(d) NETWORK TYPE.

NOTE: The values under the NETWORK NAME and NETWORK SPECIFICATION columns are managed by the cellular service provider.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-52-700-002

(1) Make sure that there are no active maintenance messages for ATA “46 - Information Systems", do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

SUBTASK 46-13-52-720-012

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

TASK 46-13-52-720-802

6. Aircraft Interface Device (AID) - Wi-Fi Scan Test

A. General

(1) This task gives the steps to perform the Aircraft Interface Device (AID) Wi-Fi scan test.

(2) To do this task, the Weight-On-Wheels (WOW) discrete status must be in the Ground condition.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
212	Flight Compartment - Right

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E. Prepare for the Wi-Fi Scan Test

SUBTASK 46-13-52-860-008

(1) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-52-480-005

(2) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

NOTE: A universal maintenance device, SPL-11321, is required for access.

F. Aircraft Interface Device Wi-Fi Scan Test

SUBTASK 46-13-52-720-013

(1) From the Onboard Network System (ONS) maintenance browser main menu, select EXTENDED MAINTENANCE.

SUBTASK 46-13-52-720-014

(2) Select Offboard Communication.

SUBTASK 46-13-52-720-015

(3) Select WIRELESS on the top of the page.

SUBTASK 46-13-52-720-016

(4) Select DISCRETES on the bottom of the page.

SUBTASK 46-13-52-720-017

(5) Make sure that the “Wi-Fi RF Status” toggle is set to ON.

SUBTASK 46-13-52-720-018

(6) Select SCAN on the bottom of the page.

SUBTASK 46-13-52-720-019

(7) Wait for the “WIRELESS SCAN IN PROGRESS” modal to disappear.

NOTE: A Wi-Fi scan may take up to 2 minutes.

SUBTASK 46-13-52-720-020

(8) Make sure that a list of available Wi-Fi Networks’ is displayed.

SUBTASK 46-13-52-720-021

(9) Make sure that each Wi-Fi Network in the list has values listed for the following headers:

(a) MAC ADDRESS
(b) SSID
(c) MODE
(d) CHANNEL
(e) SIGNAL
(f) NOISE
(g) SNR
(h) INFO.

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-52-700-003

(1) Make sure that there are no active maintenance messages for ATA “46 - Information Systems", do this task: Existing Faults Check, TASK 46-13-02-700-805.

(a) If you find an active maintenance message, then refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

SUBTASK 46-13-52-720-022

(2) If the universal maintenance device, SPL-11321, is not necessary, close the ONS maintenance browser and disconnect the maintenance device.

——— END OF TASK ———

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AIRCRAFT INTERFACE DEVICE ANTENNA - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Aircraft Interface Device Antenna Removal

(2) Aircraft Interface Device Antenna Installation.

TASK 46-13-53-000-801

2. Aircraft Interface Device Antenna Removal

(Figure 401)

A. General

(1) This task gives the steps to remove the Aircraft Interface Device (AID) antenna from the aircraft fuselage.

(2) The antenna is installed externally at approximately STA 477, WL 306, RBL 5.

(3) You must have access to the top and the interior crown of the fuselage to do this task.

B. References

Reference	Title
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)
25-80-00-000-801	Insulation Blanket Removal (P/B 401)
46-13-52-040-801	Aircraft Interface Device (AID) Deactivation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975

D. Location Zones

Zone	Area
234	Area Above Passenger Compartment Ceiling, Right - Section 43

E. Prepare for the Removal

SUBTASK 46-13-53-040-002

(1) Do this task: Aircraft Interface Device (AID) Deactivation, TASK 46-13-52-040-801.

SUBTASK 46-13-53-860-005

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-53-040-001

WARNING THE AIRCRAFT INTERFACE DEVICE MUST BE OFF IF YOU ARE LESS THAN 8 INCHES (20 CMS) FROM THE RADIO FREQUENCY ANTENNAS. IF YOU DO NOT OBEY, INJURY FROM ELECTROMAGNETIC RADIATION CAN OCCUR.

(3) Get access to the AID antenna [1]:

(a) If it is necessary, remove the applicable ceiling panels (TASK 25-21-45-000-801).

(b) Remove the applicable insulation (TASK 25-80-00-000-801).

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F. AID Antenna Removal

SUBTASK 46-13-53-020-001

(1) At the interior crown of the aircraft (at approximately STA 488):

(a) Disconnect the coaxial connectors [4] from the coaxial receptacles [5].

(b) Remove the screw [10], screw [9], spacer [12] from the clamp [8], clamp [7], and clamp [6].

(c) Put the protective covers on the coaxial connectors [4] and coaxial receptacles [5].

SUBTASK 46-13-53-020-002

(2) On the top of the aircraft:

CAUTION

BE CAREFUL WHEN YOU REMOVE THE SEAL WITH THE SEALANT REMOVAL TOOL. DAMAGE TO THE AIRPLANE SKIN CAN OCCUR.

(a) Remove the sealant, with the sealant removal tool, COM-2481.

1) Remove the sealant, from the heads of the bolts [2].

2) Remove the aerodynamic fillet seal around the base of the AID antenna [1].

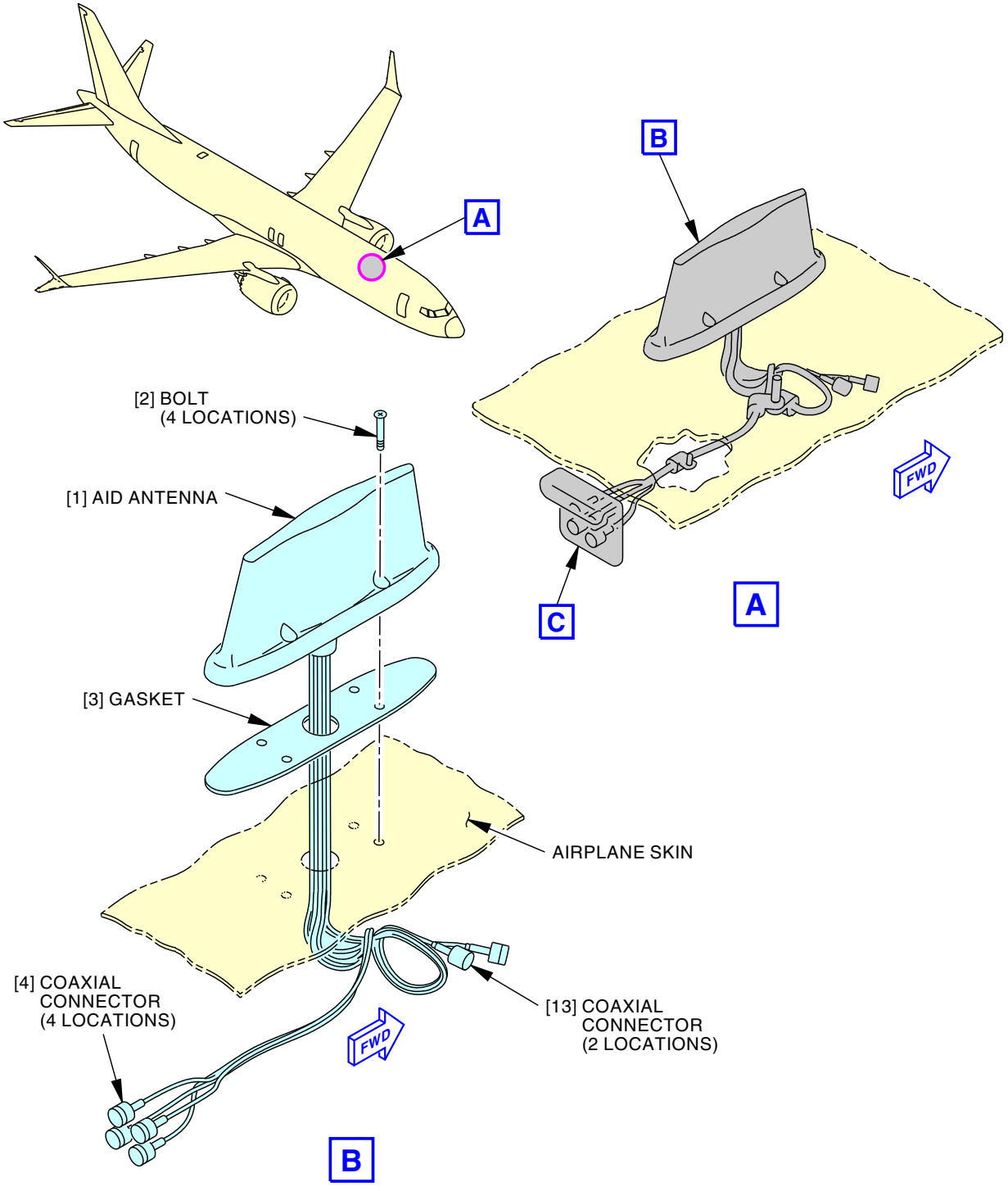
(b) Remove the bolts [2] from the AID antenna [1].

(c) Remove the AID antenna [1].

(d) Remove and discard the gasket [3].

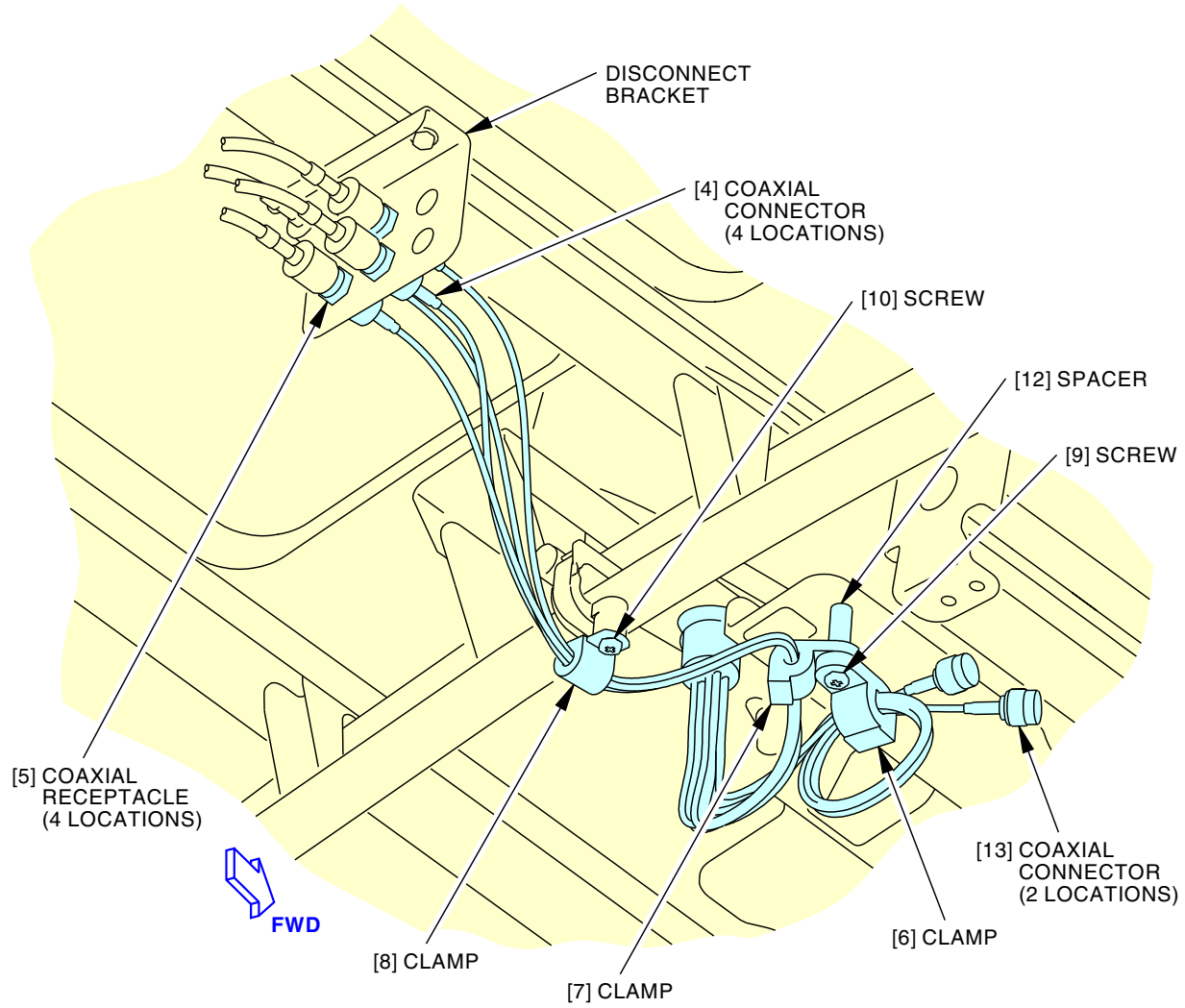
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AID Antenna Installation
Figure 401/46-13-53-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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AID Antenna Installation
Figure 401/46-13-53-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 46-13-53-400-801

3. Aircraft Interface Device Antenna Installation

(Figure 401)

A. General

(1) This task gives the steps to install the Aircraft Interface Device (AID) antenna to the aircraft fuselage.

(2) The AID antenna is installed externally at approximately STA 477, WL 306, RBL 5.

(3) You must have access to the top and the interior crown of the fuselage to do this task.

B. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)
25-80-00-400-801	Insulation Blanket Installation (P/B 401)
46-13-52-440-801	Aircraft Interface Device (AID) Activation (P/B 201)
46-13-52-710-801	Aircraft Interface Device (AID) Operational Test - Cellular Connection (P/B 501)
51-21-31-350-801	Removal and Control of Corrosion for Aluminum and Aluminum Alloys (P/B 701)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
51-22-00-390-809	Aerodynamic Smoother Application (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-765	Scraper - Plastic

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7

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(Continued)

Reference	Description	Specification
C50033	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	AID antenna	46-13-52-05-065	PTF ALL

F. Location Zones

Zone	Area
234	Area Above Passenger Compartment Ceiling, Right - Section 43

G. Prepare for the Installation

SUBTASK 46-13-53-860-002

(1) Make sure that this circuit breaker is open:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

H. AID Antenna Installation

SUBTASK 46-13-53-390-001

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS) AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(1) Solvent clean the mating surfaces between the AID antenna [1] and the aircraft skin (TASK 20-10-34-110-802).

SUBTASK 46-13-53-840-001

(2) If there is corrosion on the airplane skin below the AID antenna [1], do the steps that follow to prepare the airplane skin:

(a) To remove the corrosion, do this task: Removal and Control of Corrosion for Aluminum and Aluminum Alloys, TASK 51-21-31-350-801.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS) AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(b) Clean the airplane skin in the area of the corrosion with solvent, B00062 or solvent, B00184.

(c) Apply a layer of Bonderite M-CR 1200S Aero coating, C50033, to the airplane skin in the area where the AID antenna [1] touches the skin (TASK 51-21-41-370-802).

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SUBTASK 46-13-53-020-007

(3) If a new AID antenna [1] will be installed, discard the O-ring supplied with the AID antenna [1].

SUBTASK 46-13-53-420-003

(4) On the top of the aircraft:

(a) Make sure that the coaxial connectors [4] do not have bent or broken pins, dirt, or damage.

(b) Remove the protective film from the AID antenna [1] side of the gasket [3].

(c) Align the gasket [3] to the fastener and the connector holes of the AID antenna [1].

(d) Press the gasket [3] firmly onto the AID antenna [1] base.

(e) Remove the protective film from the airplane side of the gasket [3].

(f) Put the coaxial connectors [4] and coaxial connectors [13] through the airplane skin connector hole.

(g) Put the AID antenna [1] on the airplane skin.

(h) Apply the sealant, A00247, to the threads and shank of the bolts [2].

(i) Install the AID antenna [1] with the bolts [2].

(j) Tighten the bolts [2] to 30.0 in-lb (3.4 N·m).

(k) Wait 15 minutes.

NOTE: The gasket gel material can compress during this time.

(l) Tighten the bolts [2] to 35.0 in-lb (4.0 N·m).

(m) Use an intrinsically safe approved bonding meter, COM-1550 to measure the resistance between the base of the AID antenna [1] and the airplane skin.

1) Make sure that the electrical bonding resistance is 0.001 ohms (1.0 milliohms) or less.

(n) Remove the unwanted sealant, A00247 from the bolts [2].

(o) Remove the unwanted gel squeeze out from the gasket [3] with a plastic scraper, STD-765.

1) If it is necessary, use a clean cotton wiper, G00034, moist (not soaked) with alcohol, B00130, to help remove the unwanted gel.

(p) Apply an aerodynamic fillet seal around the base of the AID antenna [1] with sealant, A00247 (TASK 51-22-00-390-809).

1) Make sure that there are no gaps between the AID antenna [1] base and the aircraft skin.

SUBTASK 46-13-53-410-002

(5) At the interior crown of the aircraft (at approximately STA 488)

(a) Connect the coaxial connectors [4] to the coaxial receptacles [5] at the disconnect bracket.

(b) Attach the coaxial connectors [4] and coaxial connectors [13] as follows:

1) Install the clamp [6], clamp [7], clamp [8], spacer [12], screw [9], and screw [10].

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I. AID Antenna Installation Test

SUBTASK 46-13-53-440-001

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-53-710-002

(2) Do this task: Aircraft Interface Device (AID) Activation, TASK 46-13-52-440-801.

SUBTASK 46-13-53-710-001

(3) Do this task: Aircraft Interface Device (AID) Operational Test - Cellular Connection, TASK 46-13-52-710-801.

NOTE: To do the operational test, you must have access to an active wireless connection from a remote access terminal, or airport messaging terminal.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 46-13-53-860-006

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

SUBTASK 46-13-53-410-001

WARNING THE AIRCRAFT INTERFACE DEVICE MUST BE OFF IF YOU ARE LESS THAN 8 INCHES (20 CMS) FROM THE RADIO FREQUENCY ANTENNAS. IF YOU DO NOT OBEY, INJURY FROM ELECTROMAGNETIC RADIATION CAN OCCUR.

(2) Close access to the AID antenna [1]:

(a) Install the applicable insulation (TASK 25-80-00-400-801).

(b) If removed, install the applicable ceiling panels (TASK 25-21-45-400-801).

SUBTASK 46-13-53-860-007

(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	12	C04113	AID PWR C/B

END OF TASK

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